

PART B – ELECTRONICS

UNIT – IV ELECTRONICS CIRCUITS

V-I Characteristics of diode – Half wave rectifier and Full-wave rectifier – with and without capacitor filter – Transistor – Construction & working – Input and Output characteristics of CB and CE configuration – Transistor as an Amplifier – Principle and working of Hartley oscillator and RC phase shift oscillator – Construction and working of JFET & MOSFET.

UNIT – V DIGITAL ELECTRONICS

Boolean algebra – Reduction of Boolean expressions – De-Morgan's theorem – Logic gates – Implementation of Boolean expressions – Flip flops – RS, JK, T and D. Combinational logic – Half adder, Full adder and Subtractors, Sequential logic – Ripple counters and shift registers.

UNIT – VI COMMUNICATION AND COMPUTER SYSTEMS

Model of communication system – Analog and digital – wired and wireless channel. Block diagram of various communication systems – Microwave, satellite, optical fibre and cellular mobile system. Network model – PAN, LAN, MAN and WAN – Circuit and packet switching – Overview of ISDN.

4.10 THE PN JUNCTION DIODE [April 2015, 17, 19, JAN-2016, 18]

Definition : The *PN* junction forms a popular semiconductor device called *PN* junction diode. The *PN* junction has two terminals called electrodes, one from *P* region and other from *N* region. Due to the two electrodes it is called diode (*di* + electrode).

Symbol of a Diode

The symbol of *PN* junction diode is shown in Fig. 4.16. The *P* region acts as anode while the *N*- region acts as cathode. The arrowhead in the symbol indicates the direction of the conventional current, which can flow when an external voltage is connected in a specific manner across the diode.

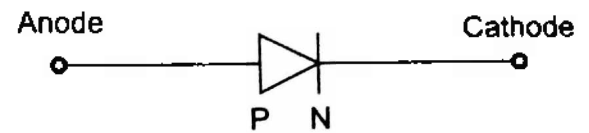


Fig. 4.16 Symbol of a diode

4.11 BIASING OF PN JUNCTION DIODE

Definition : Applying external *d.c* voltage to any electronic device is called biasing. As seen, there is no current in the unbiased *PN* junction at equilibrium.

Types of Biasing

Depending upon the polarity of the *d.c.* voltage externally applied to it, biasing can be classified into two types

1. Forward biasing
2. Reverse biasing

4.11.1 Forward Biasing of *PN* Junction Diode

Definition : If an external *d.c.* voltage is connected in such a way that the *P*- region terminal (anode) is connected to the positive of the *d.c.* voltage and the *N*- region (cathode) is connected to the negative of the *d.c* voltage, the biasing condition is called forward biasing. In this condition the *PN* junction is said to be forward biased.

Circuit diagram

Figure 4.17 (a) shows the connection of forward biasing of a *PN* junction diode Fig. 4.17. To limit the current, practically a current limiting resistor (*R*) is connected in series with the *PN* junction diode. Fig. 4.17 (b) shows the symbolic representation of a forward biased diode.

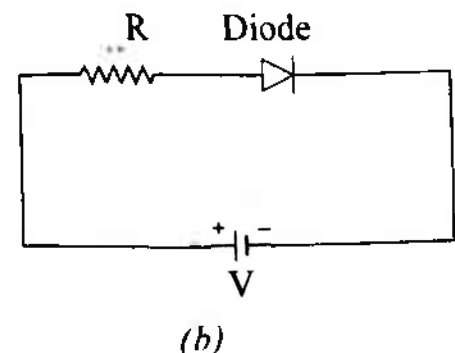
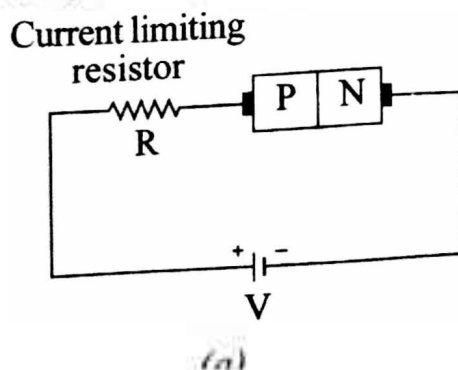


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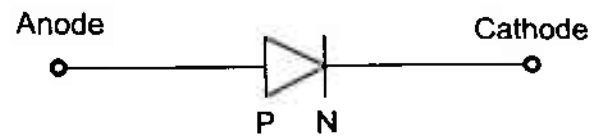


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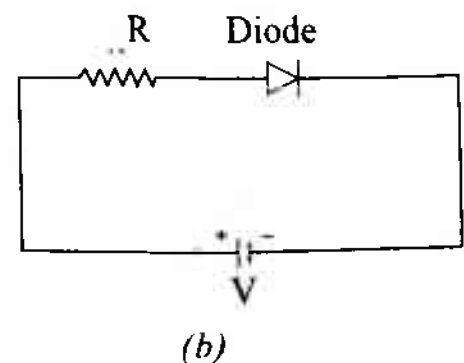
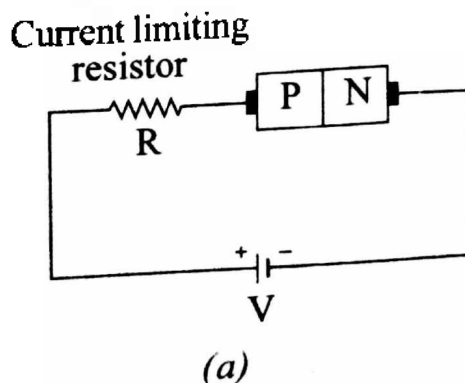


Fig. 4.17

Operation of forward Biased Diode

When the PN junction is biased, as long as the applied voltage is less than the barrier potential there cannot be any conduction. When the applied voltage becomes more than the barrier potential the negative terminal of battery forces the free electrons against barrier potential from N to P region. Similarly positive terminal pushes the holes from P to N region. Thus the holes get repelled by positive terminal and cross the junction against barrier potential. Thus the applied voltage overcomes the potential. This reduces the width of depletion region.

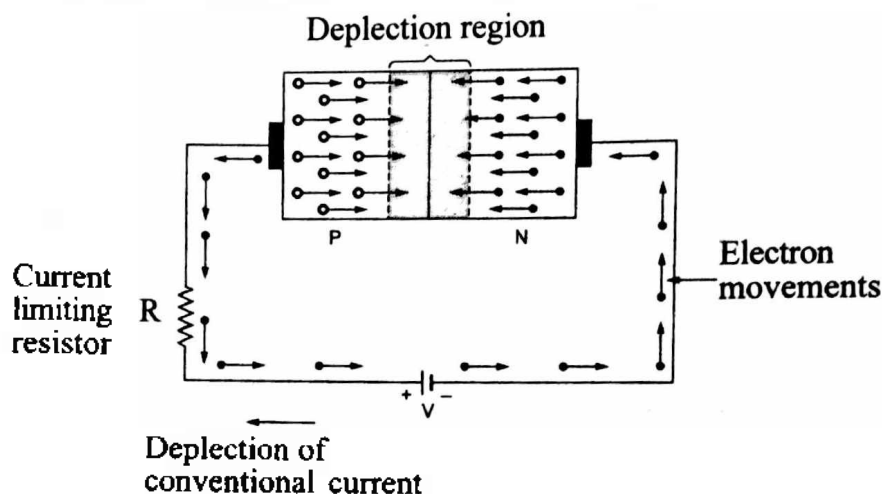


Fig. 4.18 Forward biased PN junction

As forward voltage is increased, at a particular value the depletion region becomes very much narrow such that large number of majority charge carriers can cross the junction. Hence the overall forward current is due to the majority charge carriers. This action is shown in the Fig. 4.18

These majority carriers can travel around the closed circuit and a relatively large current flows. The direction of flow of electrons is from negative to positive of the battery. While the direction of the conventional current is from positive to negative of the battery as shown in Fig. 4.18

Effect on the Depletion Region

Due to the forward biased voltage, more electrons flow into the depletion region, which reduces the number of positive ions. Similarly flow of holes reduces the number of negative ions. This reduces the width of the depletion region as shown in Fig. 4.19

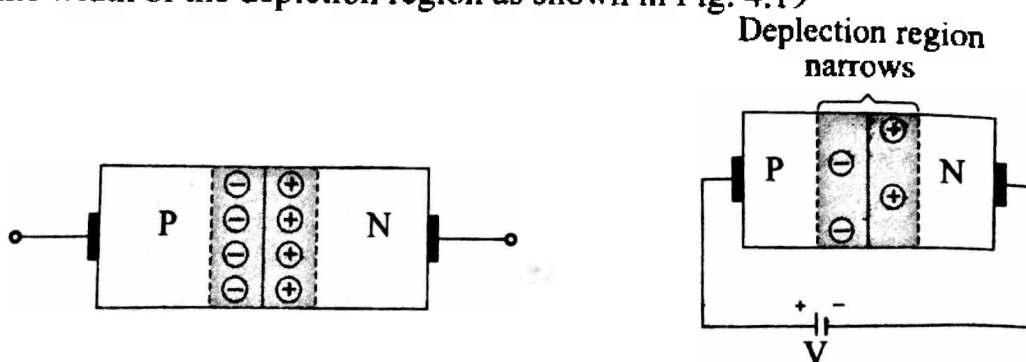


Fig. 4.19

4.11.2 Reverse Biasing of P-N Junction Diode

Definition : If an external d.c voltage is connected in such a way that the P region terminal of PN junction is connected to the negative of the battery and the N-region terminal of a PN junction is connected to the positive terminal of the battery. The biasing condition is called reverse biasing of a PN junction

Circuit diagram

Shows the connection of a reverse biasing Fig. 4.20 (a) of PN junction and shows the symbolic representation of a reverse biased diode. Fig. 4.20 (b)

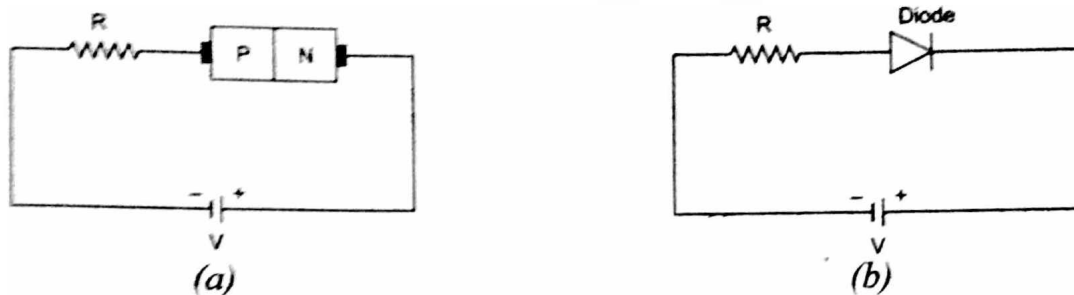


Fig. 4.20

Operation of Reverse Biased Diode

When the PN junction is reverse biased the negative terminal attracts the hole in the P-region, away from the junction. The positive terminal of battery attracts the free electrons in the N-region away from the junction. No charge carrier is able to cross the junction. As electrons and holes both move away from the junction, the depletion region widens. This creates more positive ions in the N-region and more negative ions in the P-region. This is because the applied voltage increase the barrier potential. This is shown in the Fig. 4.21

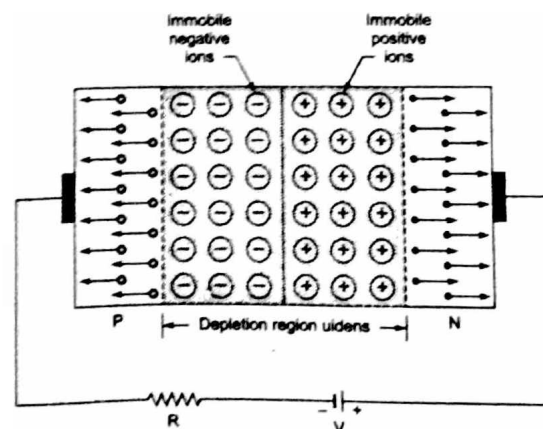


Fig. 4.21 Depletion region widens in reverse bias

As the depletion region indem and also increased barrier potential across the junction. However, this process cannot continue for a long time. In the steady state, majority current as holes and electrons stop moving away from the junction.

The polarities of barrier potential are as same as that of the applied voltage. Due to increased barrier potential, the positive side drags the electrons from P-region towards the positive of battery. Similarly negative side of barrier potential drags the holes from N-region towards the negative of battery. The electrons on P-side and holes on N- side are minority charge carriers, which constitute the current in reverse biased condition. Thus reverse conduction takes place.

The reverse current flows due to minority charge carriers which are small in number. Hence, the reverse current is always very small. Thus the reverse current depends on the temperature i.e., thermal generation and not on the reverse voltage applied.

4.12 VOLTAGE- AMPERE (V - I) CHARACTERISTICS OF A DIODE

Definition : The VI characteristics curve is a graph that shows how much current will pass through a typical junction diode when it is forward biased or reverse biased.

In order to know how a device responds when it is connected in an electrical circuit. VI characteristics of PN junction diode is analyzed. The VI characteristics in the forward biased and reverse biased is analyzed condition unit of the voltage across the diode against the diode current.

4.13 VI CHARACTERISTICS OF A PRACTICAL DIODE

4.13.1 Forward characteristics of a Practical PN junction diode

Figure 4.22 shows the forward biased diode. The applied voltage is V while the voltage across the diode is V_F . The current flowing in the circuit is the forward current I_F . The graph of forward current I_F against the forward voltage V_F across the diode is called forward characteristics of diode. The forward characteristics of a diode is shown in Fig. 4.22. This characteristic curve can be divided into two regions.

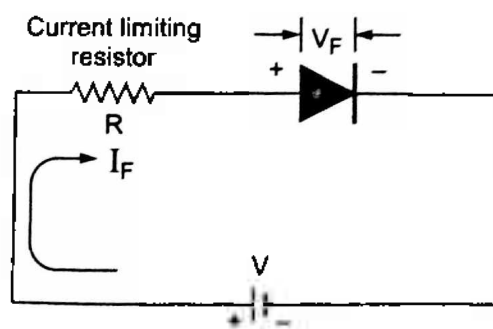


Fig. 4.22 Circuit diagram for forward bias

1. Region 0 to P

As long as V_F is less than cut in Voltage (V_γ), the current flowing is very small practically this current is assumed to be zero

2. Region P to Q and onwards

As V_F increases towards V_γ , the width of depletion region goes on reducing. When V_F exceeds V_γ i.e., cut in voltage, the depletion region becomes very thin and current I_F increases exponentially as shown in Fig. 4.23. The point P after which the forward current increases exponentially is called knee point of the curve.

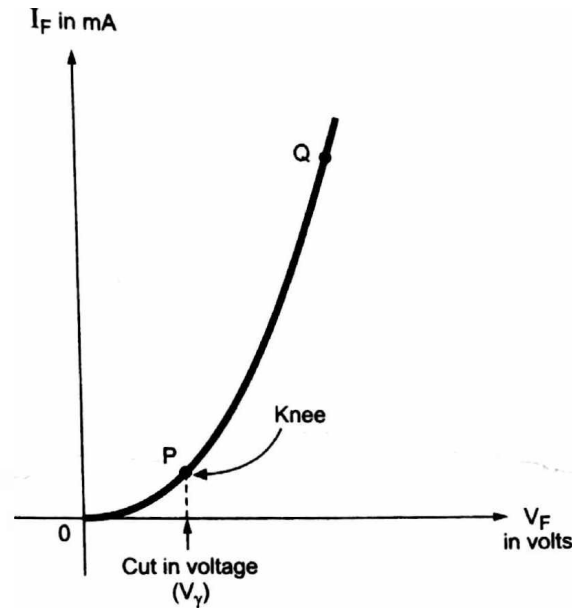


Fig. 4.23 Forward characteristics

4.13.2 Reverse characteristics of Practical PN junction diode

Shows the reverse biased diode. The reverse voltage across the diode is V_R while the current flowing through the circuit due to minority charge carriers is reverse current I_R .

The graph of I_R against V_R is called reverse characteristic of a diode. The polarity of reverse voltage applied is opposite to that of forward voltage. Hence in practice reverse voltage is taken

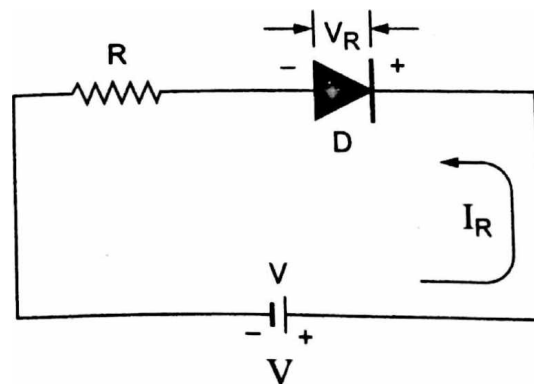


Fig. 4.24 Reverse biased diode

as negative. Similarly the reverse saturation current is due to minority carriers and is opposite to the forward current.

As reverse voltage is increased, reverse current increases initially but after a certain voltage, the current remains constant equal to reverse saturation current though reverse voltage is increased. The point A where breakdown occurs and reverse current increases rapidly is called knee of the reverse characteristic as shown in Fig. 4.25

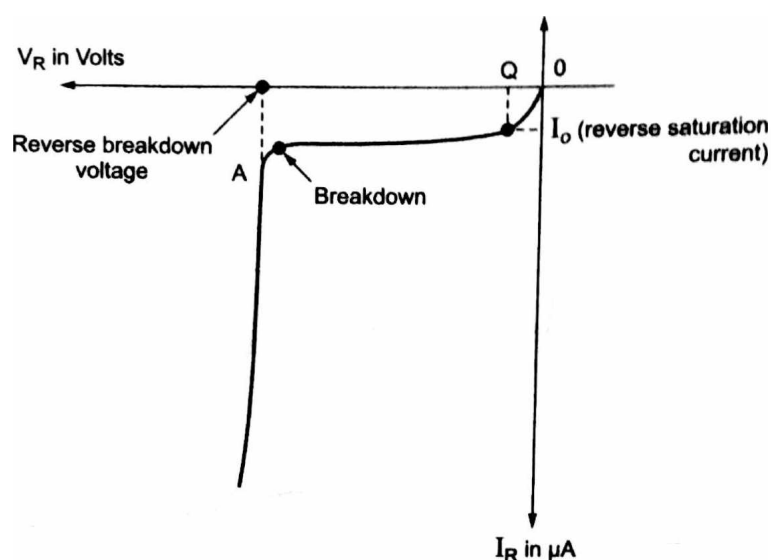


Fig. 4.25 Reverse characteristic of a diode

4.13.3 Complete VI characteristics of a Diode

The complete VI characteristics of a diode is the combination of its forward as well as reverse characteristics. This is shown in the Fig. 4.26

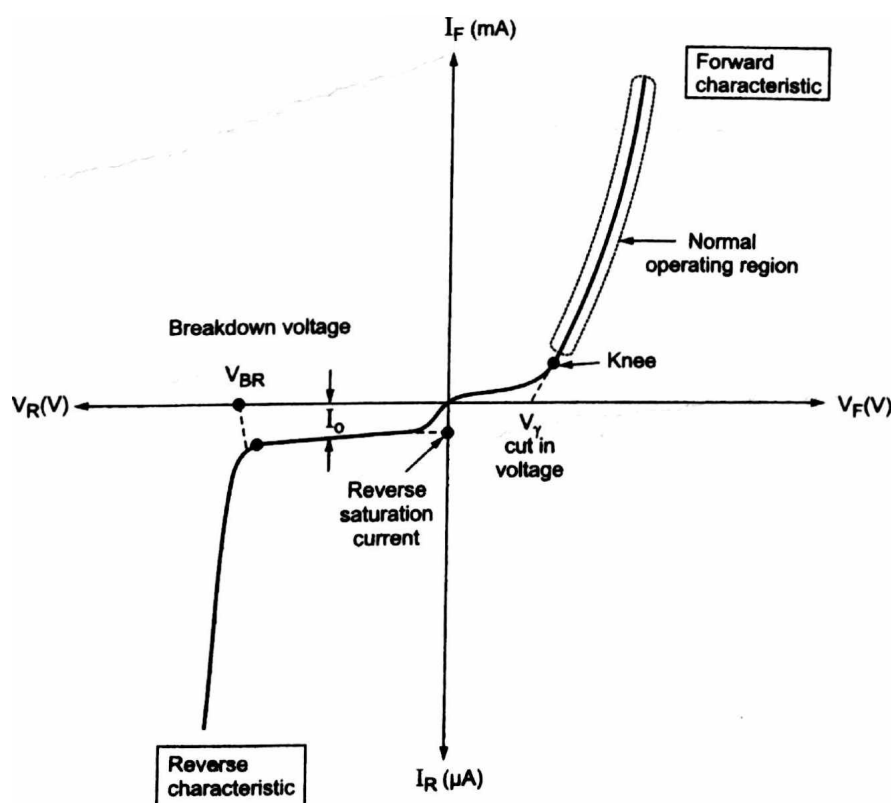


Fig. 4.26 Complete VI characteristics of a diode

1. Rectifiers

2. Clippers or limiters

3. Clampers

4. Voltage multipliers

4.15 RECTIFIERS

Definition : The process of a.c. to d.c conversion is known as **rectification**. A circuit which is used for rectification is called as **rectifier**.

A rectifier is a circuit which is used to convert a.c. voltage into pulsating d.c voltage. It can be classified as follows.

4.16 HALF-WAVE RECTIFIER [JAN 2016, 18]

Definition : In half-wave rectification, the rectifier conducts current only during positive half-cycles of input a.c. supply.

Circuit diagram

The basic circuit diagram of half-wave rectifier is shown in Fig. 4.31. In this circuit there is a step down transformer at the input of the circuit. The transformer couples an a.c. input voltage to the rectifier circuit. A diode D is connected between secondary of transformer and load resistor R_L .

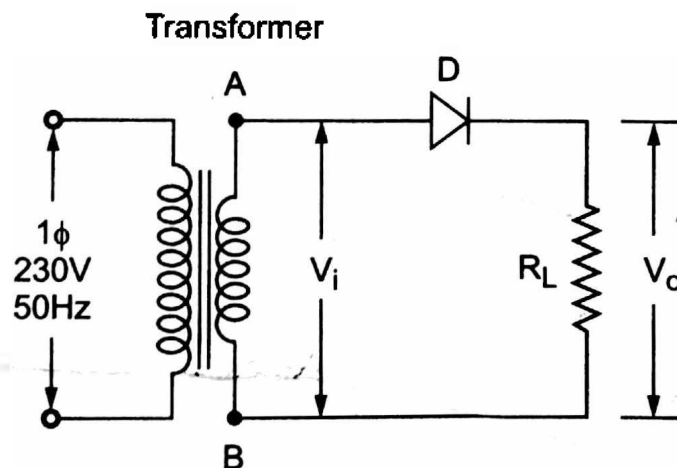


Fig. 4.28 Half-wave rectifier circuit

Operation of Half-wave rectifier

In an a.c. supply voltage, there are two half cycles, one is positive half cycle the other is negative half cycle. The operation of the circuit, can be understood well from Fig. 4.29 and waveforms generated by this circuit are also shown in the Fig. 4.30.

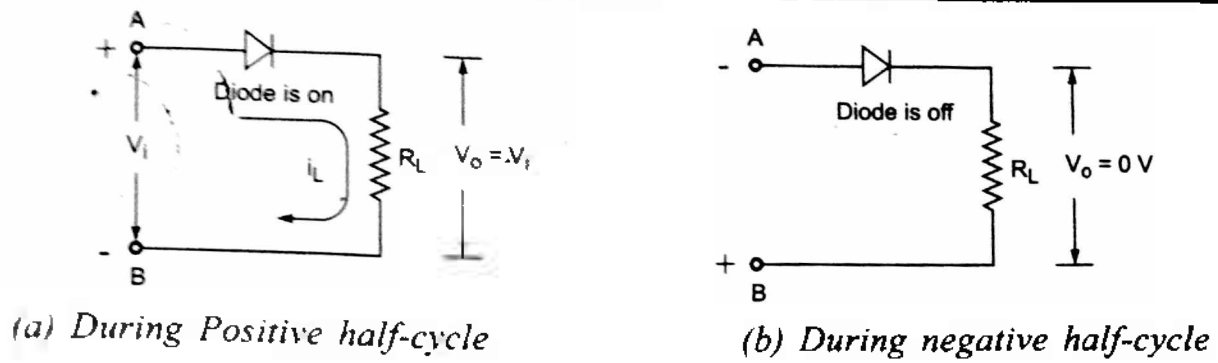


Fig. 4.29

During positive half-cycle

During the positive half cycle ($0 \leq \omega t \leq \pi$), the terminal A of secondary is positive w.r.t terminal B . The diode D gets forward biased and hence possesses a low resistance path (low resistance R_f) to the load current. The load current i_L flows through the load resistor R_L to produce output voltage V_o .

During Negative half-cycle

During negative half cycle ($\pi \leq \omega t \leq 2\pi$), the terminal B of secondary is positive w.r.t terminal A and diode D becomes reverse biased and hence acts as an open switch. Therefore, no current flows in the circuit ($i_L = 0$) and the output voltage across load resistor is also zero ($V_o = 0$ V).

It is clear from above discussion that diode D conducts only for one half cycle or in other words we get only one half cycle of the input across the load. The output voltage obtained from this circuit is not of constant. The magnitude but pulsating in nature.

4.16.1 Analysis of Half-Wave Rectifier Parameters

We assume that the input is sinusoidal, then the mathematical representation of sinusoidal signal is

$$v_i = V_m \sin \omega t \quad \dots (4.1)$$

If we neglect the drop across the diode D , we have output voltage in the form from Fig. 4.30

$$\begin{aligned} v_L &= v_i = V_m \sin \omega t & ; 0 \leq \omega t \leq \pi \\ v_L &= 0 & ; \pi \leq \omega t \leq 2\pi \end{aligned} \quad \dots (4.2)$$

Similarly, the current i is represented as

$$\begin{aligned} i_L &= I_m \sin \omega t & ; 0 \leq \omega t \leq \pi \\ i_L &= 0 & ; \pi \leq \omega t \leq 2\pi \end{aligned} \quad \dots (2.3)$$

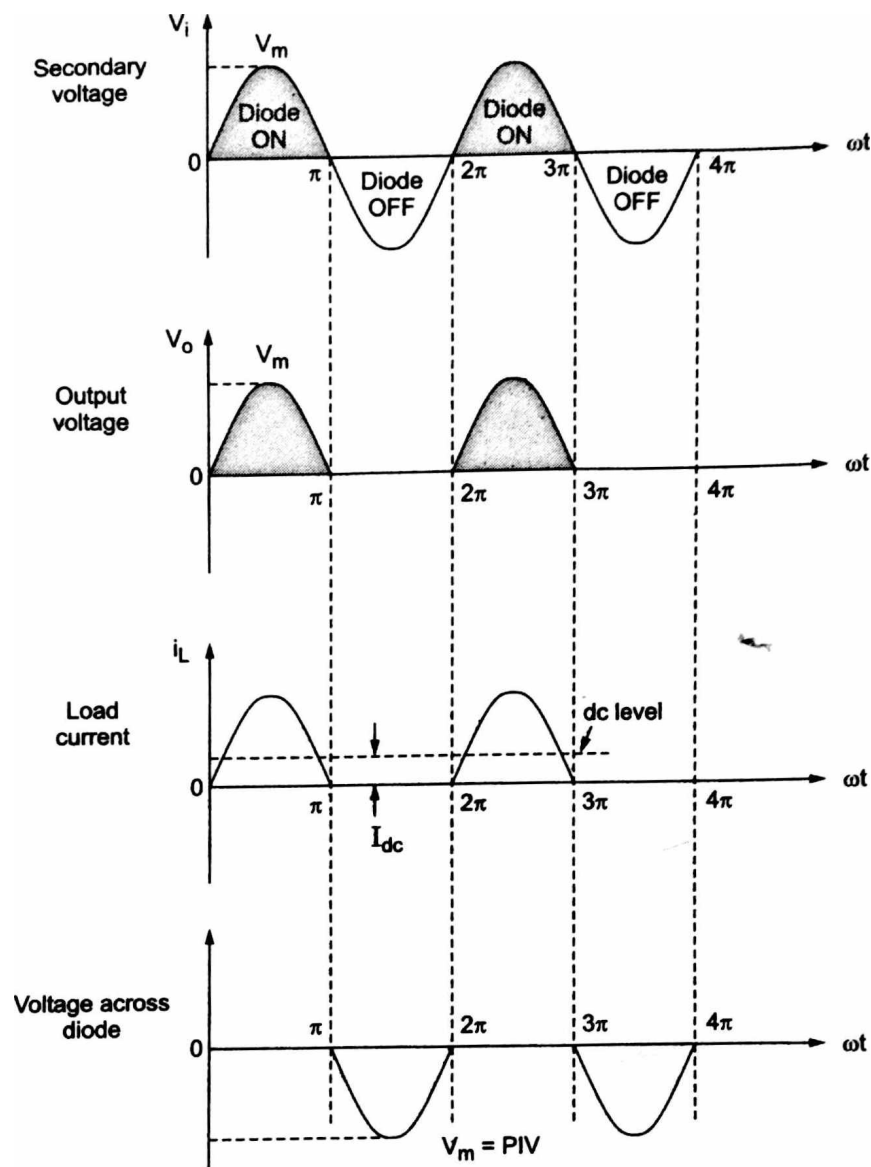


Fig. 4.30 Waveforms of the half-wave rectifier

where
$$I_m = \frac{V_m}{R_2 + R_F + R_L} \quad \dots(4.4)$$

The following parameters of rectifier are analyzed

- | | |
|---|--------------------------------|
| 1. Average (d.c.) load current | 2. R.M.S. (a.c.) load current |
| 3. Form factor | 4. Average (d.c.) load voltage |
| 5. R.M.S. (a.c.) load voltage | 6. Ripple factor |
| 7. Voltage regulation | 8. Efficiency |
| 9. Transformer utilization factor (TUF) | 10. Peak inverse voltage |

1. Average load current

Mathematically, average or mean value of any function $f(x)$ can be obtained between the limits l and u as

$$F_{\text{mean}} = \frac{1}{u-l} \int_l^u f(x) dx \quad \dots(4.5)$$

where l = lower limit

u = upper limit

Therefore from equation (4.5), we can calculate the average value of load current i_L for one cycle as

$$I_{\text{av}} \text{ (or) } I_{dc} = \frac{1}{2\pi} \int_0^{2\pi} i_L d(\omega t)$$

Substituting the value of i_L from equation (4.3) in the above equation

$$\begin{aligned} I_{dc} &= \frac{1}{2\pi} \left[\int_0^{2\pi} I_m \sin \omega t d(\omega t) \right] \\ &= \frac{1}{2\pi} \left[\int_0^{\pi} I_m \sin \omega t d\omega t + \int_{\pi}^{2\pi} I_m \sin \omega t d\omega t \right] \\ &= \frac{1}{2\pi} \left[\int_0^{\pi} I_m \sin \omega t d\omega t + 0 \right] \\ &= \frac{1}{2\pi} \int_0^{\pi} I_m \sin \omega t d\omega t \\ &= \frac{I_m}{2\pi} [-\cos(\omega t)]_0^{\pi} \\ &= \frac{I_m}{2\pi} [-\cos \pi - (-\cos 0)] \end{aligned} \quad \dots(4.6)$$

$$I_{dc} = \frac{I_m}{\pi} \quad \dots(4.7)$$

$$\text{(or)} \quad I_{dc} = 0.318 I_m$$

Substituting the value of I_m from equation (2.4) in equation (2.6)

$\dots(4.8)$

$$I_{dc} = \frac{V_m}{\pi (R_2 + R_F + R_L)}$$

where R_s = Transformer secondary resistance

R_f = Forward diode resistance

R_L = Load resistance

2. RMS Load current

Mathematically, Root Mean Square (RMS) value of a function $f(x)$ is defined as

$$F_{\text{rms}} = \sqrt{\frac{1}{u-l} \int_l^u [f(x)^2] dx} \quad \dots(4.9)$$

Using the equation (4.9), we can calculate the RMS value of the load current i_L

$$\begin{aligned} I_{\text{rms}} &= \left[\frac{1}{2\pi} \int_0^\pi i_L^2 d(\omega t) \right]^{1/2} \\ &= \left[\frac{1}{2\pi} \int_0^{2\pi} I_m^2 \sin^2 \omega t d\omega t \right]^{1/2} \\ &= \left[\frac{I_m^2}{2\pi} \int_0^\pi \frac{1 - \cos(2\omega t)}{2} d\omega t \right]^{1/2} \\ &= \left[\frac{I_m^2}{2\pi \times 2} \left(\omega t - \frac{\sin 2\omega t}{2} \right)_0^\pi \right]^{1/2} \\ &= \left[\frac{I_m^2}{4\pi} \times \pi \right]^{1/2} \\ I_{\text{rms}} &= \frac{I_m}{2} \quad \dots(4.10) \end{aligned}$$

$$\text{(or)} \quad I_{\text{rms}} = \frac{V_m}{2(R_s + R_f + R_L)} \quad \dots(4.11)$$

3. Form Factor (FF)

Form factor for any periodic signal is defined as the ratio of RMS value to average value

$$\text{FF} = \frac{\text{RMS value}}{\text{DC value}} \quad \text{or} \quad \text{Average value} \quad \dots(4.12)$$

Therefore form factor for the half wave rectifier is

$$FF = \frac{I_m / 2}{I_m / \pi} = \frac{\pi}{2}$$

$$FF = 1.57$$

4. Average Load voltage

We can determine the dc load voltage using the ohms law

$$\begin{aligned} V_{dc} &= I_{dc} \times R_L \\ &= \frac{V_m}{\pi (R_F + R_2 + R_L)} \times R_L \end{aligned}$$

If $R_L \gg (R_2 + R_F)$, the above equation becomes

$$V_{dc} = \frac{V_m}{\pi R_L} \times R_L$$

$$V_{dc} = \frac{V_m}{\pi} \quad \dots(4.13)$$

5. RMS Load voltage

Similarly, the RMS value of load voltage can be obtained as

$$\begin{aligned} V_{Lrms} &= I_{rms} \times R_L \\ &= \frac{V_m}{2 (R_F + R_2 + R_L)} \times R_L \end{aligned}$$

If we assume $R_L \gg R_2 + R_F$, then the above equation becomes

$$V_{Lrms} = \frac{V_m}{2 R_L} \times R_L$$

$$V_{Lrms} = \frac{V_m}{2} \quad \dots(4.14)$$

6. Ripple Factor

The rectifier output contains a.c as well as d.c component. The ripple factor measures the percentage of a.c component in the rectified output. The ideal value of ripple factor should be zero. i.e. output should be pure d.c. Ripple factor is defined as

$$\text{Ripple factor, } \gamma = \frac{\text{RMS value of the a.c component of output}}{\text{D.C. or Average value of output}}$$

As the instantaneous value of a.c fluctuations is measured with respect to the d.c level (i.e. the difference of instantaneous total value and the d.c level as shown in Fig. 2.3.). Thus the instantaneous a.c value is given as

$$i_{ac} = i_L - I_{dc} \quad \dots (4.15)$$

Therefore, the RMS value of a.c component is given as

$$\begin{aligned} I_{ac \text{ rms}} &= \left[\frac{1}{2\pi} \int_0^{2\pi} (i_L - I_{dc})^2 d\omega t \right]^{1/2} \\ &= \left[\frac{1}{2\pi} \int_0^{2\pi} (i_L^2 + I_{dc}^2 - 2i_L I_{dc}) d\omega t \right]^{1/2} \\ &= \left[\frac{1}{2\pi} \int_0^{2\pi} i_L^2 d\omega t + \frac{1}{2\pi} \int_0^{2\pi} I_{dc}^2 d\omega t - \frac{2I_{dc}}{2\pi} \int_0^{2\pi} i_L d\omega t \right]^{1/2} \\ &= [I_{rms}^2 + I_{dc}^2 - 2I_{dc}^2]^{1/2} \\ I_{ac \text{ rms}} &= [I_{rms}^2 - I_{dc}^2]^{1/2} \quad \dots(4.16) \end{aligned}$$

$$\begin{aligned} \therefore \text{The ripple factor } \gamma &= \frac{I_{ac \text{ rms}}}{I_{dc}} = \frac{[I_{rms}^2 - I_{dc}^2]^{1/2}}{I_{dc}} \\ &= \left[\left(\frac{I_{rms}}{I_{dc}} \right)^2 - 1 \right]^{1/2} \\ \gamma &= [(FF)^2 - 1]^{1/2} \quad \dots(4.17) \end{aligned}$$

For half wave rectifier, ripple factor can be calculated by substituting the value of form factor into equation (2.17) therefore

$$\gamma = [(1.57)^2 - 1]^{1/2}$$

$$\gamma = 1.21$$

7. Voltage Regulation

Under d.c conditions, the rectifier circuit can be modelled as shown in Fig. 4.31 (a) where R_f is the forward resistance of diode. R_2 is the d.c resistance of transformer's secondary winding and R_L is the load resistance.

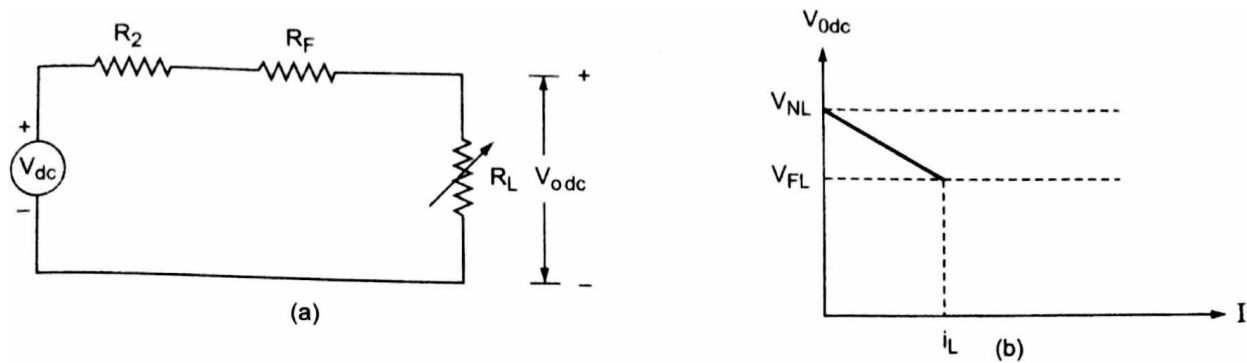


Fig. 4.31

The voltage at output terminals can be calculated by voltage divider rule as

$$V_{odc} = \frac{V_{dc} \times R_L}{R_2 + R_F + R_L} = I_{dc} R_L$$

The value of load current depends on the value of the load resistance *i.e* lower the value of R_L and higher will be the value of current (I_{dc}). Since the secondary voltage V_{dc} is constant, the output voltage V_{odc} will decrease with decreasing value of R_L . This variation in output voltage from no-load (V_{NL}) to full-load (V_{FL}) is measured by voltage regulation

$$\% \text{ Voltage regulation } V_R = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100 \quad \dots(4.18)$$

where

$$V_{NL} = \text{Voltage at no-load}$$

$$V_{FL} = \text{Voltage at full load}$$

Voltage regulation is also known as load regulation. The ideal value of load regulation should be zero.

8. Rectification or Power Conversion Efficiency

The efficiency of a rectifier can be defined as the ratio of output d.c power available at the load to the input a.c power from the mains and can be represented mathematically as

$$\text{Rectifier efficiency} = \frac{\text{output d.c. power available at the load (P}_{odc})}{\text{Input a.c. power from the mains (P}_{iac})}$$

$$\% \eta = \frac{P_{odc}}{P_{iac}} \times 100 \quad \dots(4.19)$$

The d.c output power (P_{odc}) at the load

$$P_{odc} = I_{dc}^2 R_L$$

$$= \left(\frac{I_m}{\pi} \right)^2 R_L \quad \dots(4.20)$$

And the input a.c power from the mains

$$\begin{aligned} P_{\text{ac}} &= I_{\text{rms}}^2 (R_L + R_F + R_2) \\ &= \frac{I_m^2}{4} (R_L + R_F + R_2) \end{aligned} \quad \dots(4.21)$$

Substituting equations (4.20) and (4.21), in equation (4.19), we get

$$\begin{aligned} \% \eta &= \frac{(I_m / \pi)^2 R_L}{(I_m / 2)^2 (R_L + R_2 + R_F)} \times 100 \\ &= \frac{4R_L}{\pi^2 (R_L + R_2 + R_F)} \times 100 \\ &= \frac{0.4053 \times 100}{1 + \left(\frac{R_F + R_2}{R_L} \right)} \\ &= \frac{40.53}{1 + \frac{R_F + R_2}{R_L}} \end{aligned}$$

If we assume $R_L \gg (R_2 + R_F)$, we have

$$\eta = 40.53\% \quad \dots(4.22)$$

It is clear from the equation (2.22) that maximum achievable rectification and conversion efficiency with the half wave rectifier circuit is approximately 41%. In practical case, where $(R_F + R_2)$ is comparable to the load resistance R_L , the efficiency is very less than this maximum value.

9. Transformer Utilization Factor (TUF)

Transformer utilization factor (TUF) can be defined as the ratio of output dc power delivered to the load and the a.c power rating of the transformer. Mathematically,

$$\text{TUF} = \frac{\text{Output d.c power delivered to the load}}{\text{a.c power rating of the transformer}}$$

The a.c power rating or VA (volt-ampere) rating of a transformer can be calculated with the RMS voltage developed across the winding and RMS current flowing through the winding
a.c power rating of transformer = $V_{\text{rms}} \times I_{\text{rms}}$

As the voltage developed across the secondary winding is full sinusoidal cycle but the current flows only for the half cycle so

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}}$$



$$I_{\text{rms}} = \frac{I_m}{2}$$

$$\text{a.c. power rating of transformer} = \frac{V_m}{\sqrt{2}} \times \frac{I_m}{2} = \frac{V_m I_m}{2\sqrt{2}}$$

$$\text{but we know that } I_m = \frac{V_m}{R_L + R_F + R_2}$$

$$\text{Therefore a.c. power rating of transformer} = \frac{I_m^2 R_L}{\pi^2} \times \frac{2\sqrt{2}}{I_m^2 (R_L + R_F + R_2)} \quad \dots(4.23)$$

Hence, using equations (4.20) and (4.23), we can calculate the TUF as

$$\begin{aligned} \text{TUF} &= \frac{\text{Output d.c power delivered to the load}}{\text{a.c power rating of the transformer}} \\ &= \frac{2\sqrt{2}}{\pi^2} \times \frac{1}{1 + \left(\frac{R_F + R_2}{R_L} \right)} \end{aligned}$$

if $(R_F + R_2) \ll R_L$, we get

$$\text{TUF} = \frac{2\sqrt{2}}{\pi^2}$$

$$\text{TUF} = 28.7\% \quad \dots(4.24)$$

10. Peak Inverse Voltage (PIV)

This is the maximum voltage, which appears across the diode during non-conducting or off state. Figure 4.35 shows a rectifier during negative half cycle. As shown in Fig. 4.32 PIV appears across the diode when the negative peak occurs at the secondary of transformer

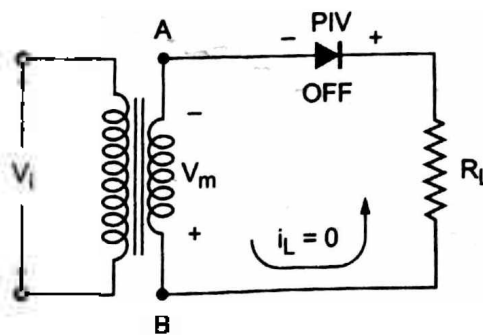


Fig. 4.32

Apply KVL in this closed loop of circuit shown in Fig. 4.32

$$V_m - \text{PIV} = 0$$

$$\text{PIV} = V_m \quad \dots(4.25)$$

This parameter is useful in selecting a particular diode for the rectifier. The diode must be selected for any rectifier circuit such that the PIV does not exceed the diodes maximum reverse voltage withstand capability, otherwise the diode will be damaged due to breakdown.

11. Ripple frequency

The frequency of the pulsating load voltage is known as ripple frequency. It is equal to the frequency of supply voltage in the halfwave rectifier

$$f = f_s \quad \dots(4.26)$$

4.16.2 Disadvantages of Half-Wave Rectifier

1. Low efficiency (41%)
2. High ripple factor (121%)
3. Low transformer utilization factor (28%)
4. Low d.c. output voltage and current
5. The d.c. current flowing through the secondary winding of the transformer may cause d.c. saturation of the core of the transformer. To minimise saturation, the transformer size has to be increased accordingly. This increases the cost.

Example 4.1 For a half wave rectifier circuit, the transformer used has a turn ratio $N_2 : N_1 = 4 : 1$ and load resistance R_L is 1000W. Calculate the average current and load voltage, RMS current and voltage, efficiency, TUF, and ripple factor if the input to primary is $V_i = 10 \sin \omega t$, assume ideal diode.

Given data: For a given half wave rectifier

$$N_2 : N_1 = 4 : 1$$

$$R_L = 1000 \, \Omega$$

$$V_i = 10 \sin \omega t$$

diode $\Rightarrow$ ideal

To find

$$I_{av}, V_{av}, V_{rms}, I_{rms}, \eta, \text{TUF and } \gamma$$

(i) As we have for a transformer

$$\frac{N_2}{N_1} = \frac{V_2}{V_1}$$

Therefore induced voltage at the secondary winding when V_i is applied.

$$V_2 = \frac{N_2}{N_1} \times V_1 \quad \because (V_i = V_1)$$

$$= \frac{4}{1} \times 10 \sin \omega t$$

$$V_2 = 40 \sin \omega t$$

From above equation $V_m = 40$ volts

(ii) Average load voltage

$$V_{L_{av}} = \frac{V_m}{\pi} = \frac{40}{\pi}$$

$$V_{dc} = V_{L_{av}} = 12.73 \text{ volts}$$

$$\text{Average load current } I_{av} = I_{dc} = \frac{I_m}{\pi}$$

We know that

$$I_{dc} = \frac{V_m}{R_L \pi} = \frac{12.73}{1000}$$

$$I_{dc} = 12.73 \text{ mA}$$

(iii) RMS load voltage

$$V_{L_{rms}} = \frac{V_m}{2} = \frac{40}{2}$$

$$V_{L_{rms}} = 20 \text{ volts}$$

RMS load current

$$I_{rms} = \frac{V_{L_{rms}}}{R_L}$$

$$= \frac{20}{1000}$$

$$I_{rms} = 20 \text{ mA}$$

$$\text{(iv) Efficiency } \% \eta = \frac{P_{odc}}{P_{iac}} = \frac{I_{Ldc}^2 R_L}{I_{rms}^2 R_L} \times 100$$

$$= \frac{(12.73)^2}{(20)^2} \times 100$$

$$\eta = 40.51\%$$

(v) Transformer utilization factor (TUF)

$$\text{TUF} = \frac{P_{odc}}{\text{a.c power rating of transformer}}$$

$$= \frac{I_{dc}^2}{V_{2rms} \times I_{2rms}} = \left[\begin{array}{l} V_{2rms} \Rightarrow \text{RMS voltage at secondary} \\ I_{2rms} \Rightarrow \text{RMS current in secondary} \end{array} \right]$$

$$= \frac{(12.73 \times 10^3)^2}{\left(\frac{40}{\sqrt{2}}\right) \times 20 \times 10^{-3}}$$

$$\text{TUF} = 28.65\%$$

(vi) Ripple factor (γ)

$$\gamma = \frac{[V_{Lrms}^2 - V_{Ldc}^2]}{V_{Ldc}}$$

Substituting values we have

$$\gamma = \frac{[(20)^2 - (12.73)^2]^{1/2}}{12.73} \times 100$$

$$\gamma = 121\%$$

4.17 FULL-WAVE RECTIFIER (FWR) [JAN 2015, 17]

Definition : In full wave rectifier circuit current flows through the load in the same direction for both half-cycles (full wave) of input a.c voltage

Types of full wave rectifier

The following two circuits are commonly used for full- wave rectification.

1. Full wave rectifier with centre tapped transformer
2. Full wave bridge rectifier.

4.18 FULL WAVE RECTIFIER WITH CENTRE TAPPED TRANSFORMER

Circuit diagram

The circuit diagram for the FWR with centre tapped transformer is shown in Fig. 4.33. In this circuit, two diodes D_1 and D_2 are used as the switching elements, one diode acts for one half cycle and second diode for another half cycle. A centre tapped transformer is used in this circuit to step down the input voltage.

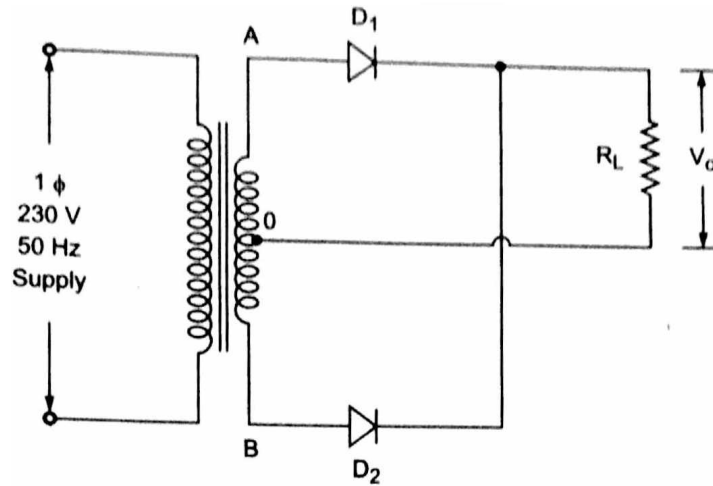


Fig. 4.33 FWR with centre tapped transformer

Operation of the circuit

During Positive half cycle

In the positive half cycle of the a.c. supply, the polarities of a secondary voltage are shown in the Fig. 4.33. Now terminal A of secondary winding is at a positive potential than the center terminal O and the terminal B of secondary is more negative with respect to terminal O .

This makes the diode D_1 forward biased and diode D_2 is reverse biased. Therefore in this positive half cycle diode D_1 conducts and diode D_2 remains OFF. Thus load current flows through the diode D_1 and load R_L producing a load voltage. The path of current is shown in Figure 4.37.

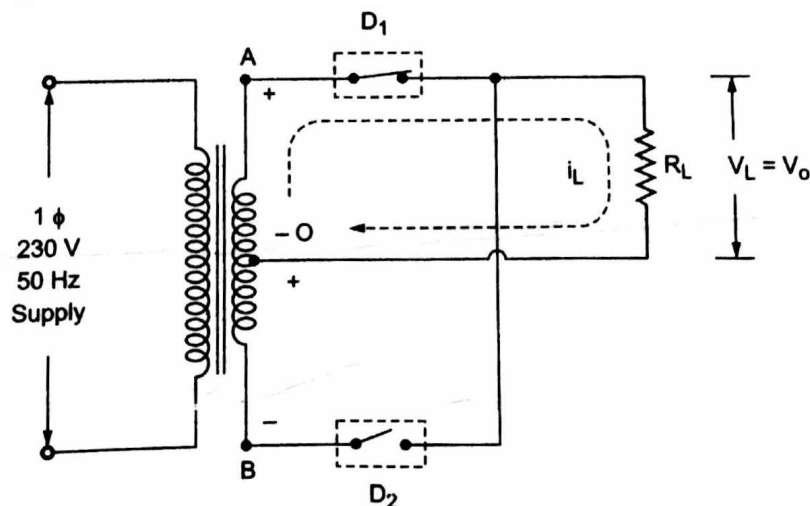


Fig. 4.34 (a) During positive half-cycle

During Negative half cycle

In the negative half cycle ($\pi \leq \omega t \leq 2\pi$), terminal B of transformer secondary winding becomes positive w.r.t terminal O and A becomes negative with respect to the terminal O . Therefore diode D_2 conducts for this duration due to forward bias and D_1 is OFF as shown in Fig. 4.34 (b).

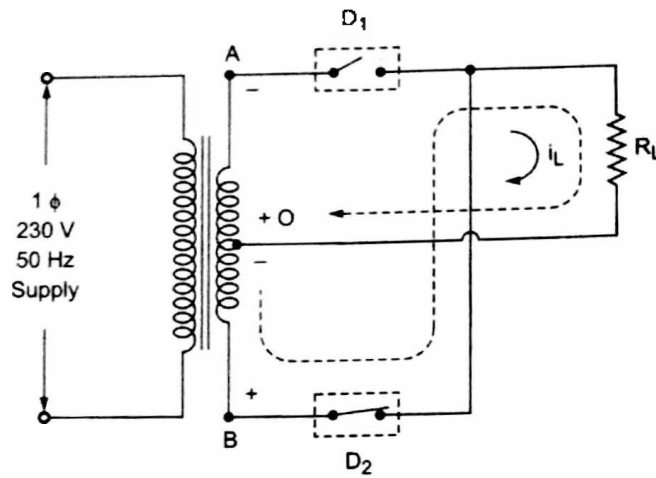


Fig. 4.34 (b) During negative half-cycle

Now, the load current flows through the diode D_2 and load resistance R_L . It is to be noted here that for both half cycles of a.c supply voltage, load current flows in the same direction, therefore producing an equidirectional pulsating d.c voltage. The waveforms for diode currents, load current and load voltage are shown in Fig. 4.34(a).

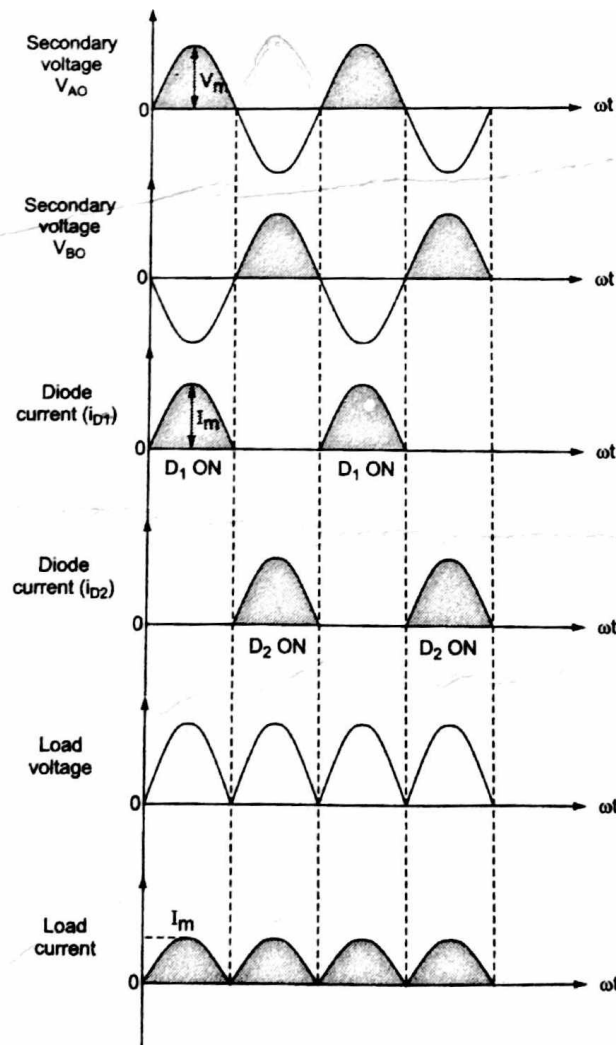


Fig. 4.35(a) Waveforms for FWR with centre tapped transformer

4.18.1 Analysis of centre tapped FWR parameters

Assume the sinusoidal input

$$V_i = V_m \sin \omega t$$

Now, if we neglect the diode drop in forward bias we have load voltage

$$\text{Similarly load current } V_L = \begin{cases} V_m \sin \omega t & ; \text{ for } 0 \leq \omega t \leq \pi \\ -V_m \sin \omega t & ; \text{ for } \pi \leq \omega t \leq 2\pi \end{cases} \quad \dots(4.27)$$

$$i_L = \begin{cases} I_m \sin \omega t & ; \text{ for } 0 \leq \omega t \leq \pi \\ -I_m \sin \omega t & ; \text{ for } \pi \leq \omega t \leq 2\pi \end{cases} \quad \dots(4.28)$$

$$\text{where } I_m = \frac{V_m}{R_2 + R_F + R_L} \quad \dots(4.29)$$

where R_F is the forward resistance of diode and R_2 is the d.c resistance of secondary winding.

1. Average load current

Average current for full wave rectifier is obtained as follows

$$\begin{aligned} I_{av.} &= I_{dc} = \frac{1}{2\pi} \left[\int_0^{2\pi} i_L d\omega t \right] \\ &= \frac{1}{2\pi} \left[\int_0^{\pi} I_m \sin \omega t d\omega t + \int_{\pi}^{2\pi} (-I_m \sin \omega t) d\omega t \right] \\ &= \frac{I_m}{2\pi} [(-\cos \omega t)_0^{\pi} + (\cos \omega t)_{\pi}^{2\pi}] \\ &= \frac{I_m}{2\pi} [-\cos \pi + \cos 0 + \cos 2\pi - \cos \pi] \\ I_{dc} &= \frac{2I_m}{\pi} \quad \dots(4.30) \end{aligned}$$

$$\text{(or)} \quad I_{dc} = \frac{2}{\pi} \frac{V_m}{(R_F + R_2 + R_L)}$$

2. RMS Load current

Substituting the value of current i_L from equation (2.28) into equation (2.9), we get

$$I_{rms} = \left[\frac{1}{2\pi} \left\{ \int_0^{\pi} I_m^2 \sin^2 \omega t d\omega t + \int_{\pi}^{2\pi} I_m^2 \sin^2 \omega t d\omega t \right\} \right]^{1/2}$$

$$= \left[\frac{I_m^2}{2\pi} \right]^{1/2} \left[\int_0^\pi \left(\frac{1 - \cos 2\omega t}{2} \right) d\omega t + \int_\pi^{2\pi} \left(\frac{1 - \cos 2\omega t}{2} \right) d\omega t \right]^{1/2}$$

After simplifying the above equation, we get

$$I_{rms} = \frac{I_m}{\sqrt{2}} \quad \dots(4.31)$$

$$(or) \quad I_{rms} = \frac{1}{\sqrt{2}} \frac{V_m}{(R_F + R_2 + R_L)}$$

3. Form Factor (F.F)

$$\begin{aligned} \text{Form Factor FF} &= \frac{\text{RMS value}}{\text{Average value}} \\ &= \frac{I_m / \sqrt{2}}{2I_m / \pi} \end{aligned}$$

$$= \frac{\pi}{2\sqrt{2}}$$

$$FF = 1.11 \quad \dots(4.32)$$

4. Average Load voltage

$$\text{Average d.c. load voltage } V_{Ldc} = V_{dc} = I_{dc} \times R_L$$

$$\begin{aligned} V_{Ldc} &= \frac{2I_m}{\pi} \times R_L \\ &= \frac{2}{\pi} \times \frac{V_m}{R_F + R_2 + R_L} \times R_L \\ &= \frac{2}{\pi} \times \frac{V_m}{1 + \left(\frac{R_F + R_2}{R_L} \right)} \end{aligned}$$

If $R_L \gg (R_F + R_2)$ then

$$V_{Ldc} = \frac{2V_m}{\pi} \quad \dots(4.33)$$

5. RMS Load voltage

The RMS value of load voltage is given by

$$V_{Lrms} = I_{rms} \times R_L$$

Substituting the value of I_{rms} , we get

$$V_{Lrms} = \frac{I_m}{\sqrt{2}} \times R_L$$

$$= \frac{1}{\sqrt{2}} \times \frac{V_m}{R_2 + R_F + R_L} \times R_L$$

$$= \frac{1}{\sqrt{2}} \times \frac{V_m}{1 + \frac{R_2 + R_F}{R_L}}$$

Assuming $(R_F + R_2) \ll R_L$

$$V_{Lrms} = \frac{V_m}{\sqrt{2}} \quad \dots(4.34)$$

6. Ripple Factor (γ)

From equation (2.17), we have

$$\gamma = [(FF)^2 - 1]^{1/2}$$

Substituting the value of form factor for the full wave rectifier, we get

$$\gamma = [(1.11)^2 - 1]^{1/2}$$

$$= 0.48$$

$$\gamma = 48\%$$

In the output of a full wave rectifier, the d.c component is more than the a.c component. Therefore, the quality of the d.c voltage obtained from full wave rectifier is better than that of half wave rectifier, because the ripples present in the rectified output of full wave rectifiers are just 48% which are much less than the 121% of half wave rectifier.

7. Rectification Efficiency

Rectification efficiency is defined as

$$\% \eta = \frac{P_{odc}}{P_{iac}} \times 100 \quad \dots(4.35)$$

where

$$P_{odc} = I_{dc}^2 R_L$$

$$P_{odc} = \frac{4I_m^2}{\pi^2} \times R_L \quad \dots(4.36)$$

and

$$P_{iac} = I_{rms}^2 \times (R_F + R_2 + R_L)$$

$$= \frac{I_m^2}{2} (R_F + R_2 + R_L) \quad \dots(4.37)$$

Therefore, substituting equations (4.36) and (4.37) into equation (4.35) we have

$$\% \eta = \frac{(4I_m^2 / \pi^2) R_L}{(I_m^2 / 2) (R_L + R_F + R_2)} \times 100$$

$$\begin{aligned}
&= \frac{4I_m^2}{\pi^2} \times \frac{2}{I_m^2} \times \frac{R_L}{R_L + R_F + R_2} \times 100 \\
&= \frac{8}{\pi^2} \times \frac{1}{1 + \frac{R_F + R_2}{R_L}} \quad \dots(4.38)
\end{aligned}$$

If we assume $(R_F + R_2) \ll R_L$, then equation (2.38) reduces to

$$\% \eta = 81.06$$

8. Transformer Utilization Factor (TUF)

The method to calculate the TUF for full wave rectifier with centre tapped transformer is slightly different to that for half wave rectifier. For this circuit the TUF for primary and secondary windings is calculated separately, and the average of these two is the TUF for the circuit. This is because the half of secondary supplies the power to the load only for the half cycle duration, while the current flows in the the primary winding for the full cycle of input voltage. Now

$$\begin{aligned}
\text{For secondary winding: } TUF &= \frac{\text{dc output power (P}_{odc})}{\text{a.c power rating of transformer}} \\
&= \frac{I_{dc} \times V_{Ldc}}{I_{rms} \times V_{rms}} \\
&= \frac{(2I_m / \pi) \times (2V_m / \pi)}{(I_m / \sqrt{2}) \times (V_m / \sqrt{2})} \\
&= \frac{8}{\pi^2}
\end{aligned}$$

$$\text{Secondary TUF} = 81.1\% \quad \dots(4.39)$$

and TUF for primary winding i.e.,

$$\begin{aligned}
\text{For primary winding TUF} &= 2 \times \text{TUF of HWR} \\
&= 2 \times 0.287 \text{ (from HWR)}
\end{aligned}$$

$$\text{Primary TUF} = 0.574$$

Thus, the TUF for a full wave rectifier with centre tapped transformer is given as

$$\begin{aligned}
TUF &= \\
&= \frac{0.811 + 0.574}{2} \\
TUF &= 69.25\% \quad \dots(4.40)
\end{aligned}$$

9. Peak Inverse Voltage (PIV)

Suppose V_m is the maximum voltage across the half secondary winding. Fig. 4.38(b) shows the circuit during positive half cycle, where diode D_2 is reverse biased at the time of peak of the input voltage and D_1 conducts, therefore the maximum voltage V_m also appears across the load resistance R_L .

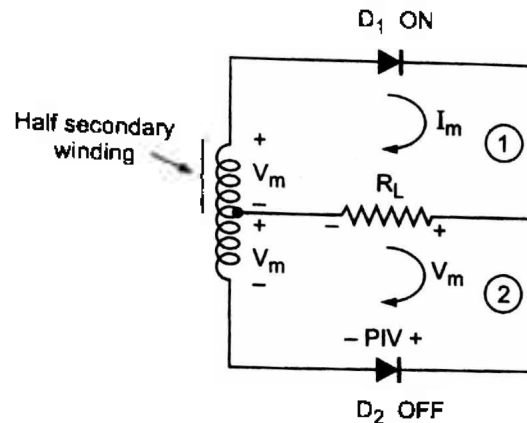


Fig. 4.35(b) Calculation of PIV

Now applying the KVL in loop 2, we have

$$V_m + V_m - \text{PIV} = 0$$

$$\text{PIV} = 2V_m \quad \dots(4.41)$$

10. Ripple Frequency

The ripple frequency in a full-wave rectifier is twice that of the supply frequency

$$f_r = 2f_s \quad \dots(4.42)$$

4.18.2 Advantages of FWR with Centre Tapped Transformer

1. Low ripples as compared to half wave rectifier
2. Higher rectification efficiency
3. Better TUF
4. Higher value of d.c. load voltage and current
5. No possibility of transformer core saturation

4.18.3 Disadvantages of FWR with Centre Tapped Transformer

1. The output maximum voltage is half that of maximum induced voltage at secondary winding.
2. PIV is twice that of HWR, therefore diodes are required higher rating
3. Increased cost due to centre tapped transformer

4.23 BASIC CONSTRUCTION OF A TRANSISTOR

Definition : Transistor consists of two PN junctions formed by sandwiching either P type or N type semiconductor between a pair of opposite types.

Transistor types

A transistor can be classified in two types based on its construction

1. NPN Transistor
2. PNP Transistor

A transistor in which two layers of N type semiconductor are separated by a thin layer of P type semiconductor is known as **NPN transistor**, it is shown in Fig. 4.44(a).

A transistor in which two layers of P type semiconductor are separated by thin layer of N type semiconductor is known as **PNP transistor**, it is shown in Fig. 4.44(b).

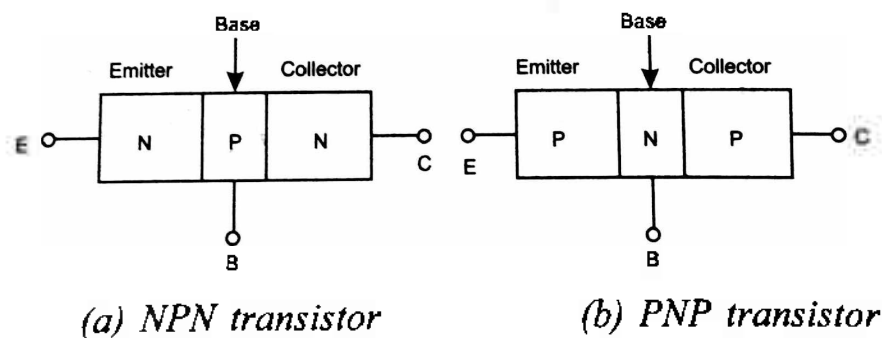


Fig. 4.44

The transistor (*NPN* or *PNP*) has three regions of doped semiconductors, namely Emitter, Base and Collector.

1. **Emitter.** The region of a transistor that supplies charge carriers (in case of *NPN* transistor electrons or in case of *PNP* transistor holes) is called the emitter. The emitter is always forward biased with respect to base so that it can supply large number of majority carriers. Fig. 4.45 shows the biasing condition of a transistor.
2. **Base.** The middle section of transistor is called base, it is very thin and lightly doped region. The base-emitter junction is forward biased, allowing low resistance for the emitter circuit. The base-collector junction is reverse biased and offers high resistance in the collector circuit.
3. **Collector.** The other side that collects the charges is called the collector. The collector is always reverse biased. The doping level of collector is between the heavy doping of emitter and light doping of base. The collector region is physically longer than the emitter region.

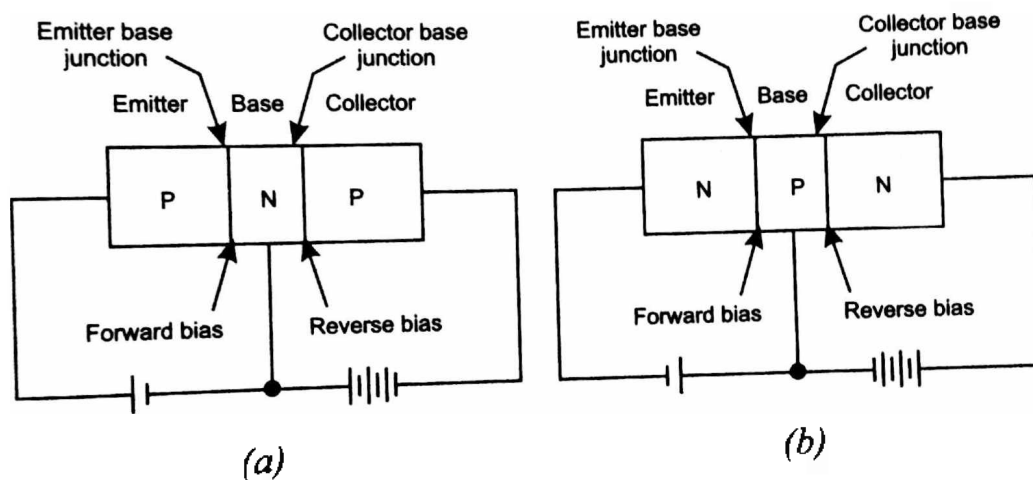
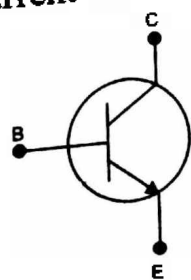


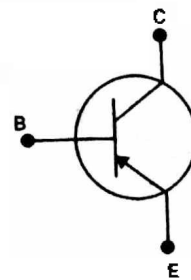
Fig. 4.45 Transistor biasing

4.24 TRANSISTOR SYMBOLS

Shows the symbols of transistors Fig. 4.46. Arrow is always placed on the emitter terminal and direction of arrow indicates the direction of conventional current flow. For example in *NPN* transistor, the arrow head points from the *P* region towards the *N*-region. Therefore, in an *NPN* transistor the conventional current will flow from base to emitter.



(a) *NPN* transistor



(b) *PNP* transistor

Fig. 4.46 Symbol of Transistors

4.25 UNBIASED TRANSISTOR

When no external supply is connected to a transistor, the transistor is in unbiased condition. Fig. 4.47 shows the depletion region formation of transistor when the transistor is unbiased. In this case, the diffusion of charge carriers across the junction produces two depletion regions. The width of depletion region is not same on both sides of junction because of different doping levels of the transistor. The depletion region penetrates more in lightly doped regions. Therefore penetration of depletion region is less in the heavily doped collector and emitter regions and more in the base region.

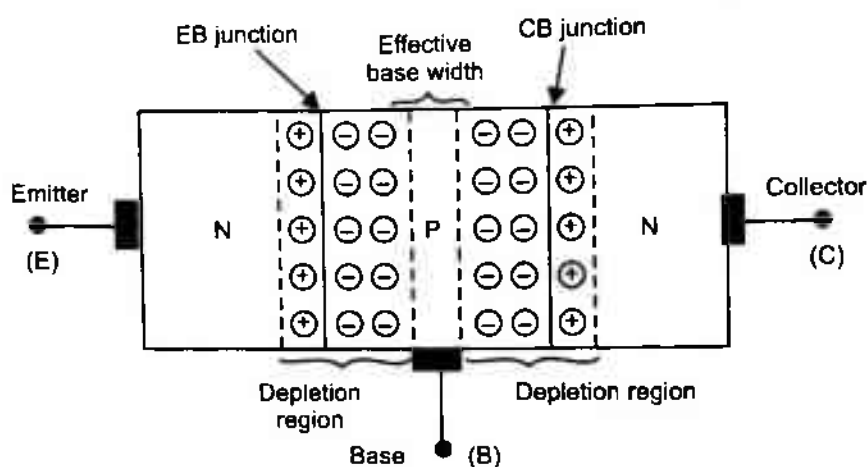


Fig. 4.47

4.26 TRANSISTOR ACTION

The method of applying external voltages to a transistor is called **transistor biasing**. The transistor operates in three different regions based on the biasing of transistor. Table 4.1 shows the different regions of operation with their corresponding biasing.

Table 4.1

Cases	Emitter-base junction	collector-base junction	Region of operation
1.	Forward-biased	Reverse-biased	Active
2.	Forward-biased	Forward biased	Saturation
3.	Reverse biased	Reverse biased	Cut-off

1. **Active region.** In this case the emitter base junction is forward biased and collector base junction is reverse biased. In this region the transistor is used for amplification. The collector current is dependent upon the base current.
2. **Saturation region.** In this case, both junctions are forward biased. In this region of operation, the collector current becomes independent of the base current. Now the transistor acts like a closed switch.

3. **Cut off region.** In this case, both junctions are reverse biased. In this region, the transistor has zero current, because the emitter does not emit charge carriers into base and no charge carriers are collected by the collector except a few thermally generated carriers. Now the transistor acts like an open switch.

4.27 OPERATION OF NPN TRANSISTOR

The NPN transistor circuit is shown in the Fig. 4.48. The emitter base junction is forward biased while collector base junction is reverse biased. The forward bias voltage V_{EB} is very small where as reverse biased voltage V_{CB} is considerably high.

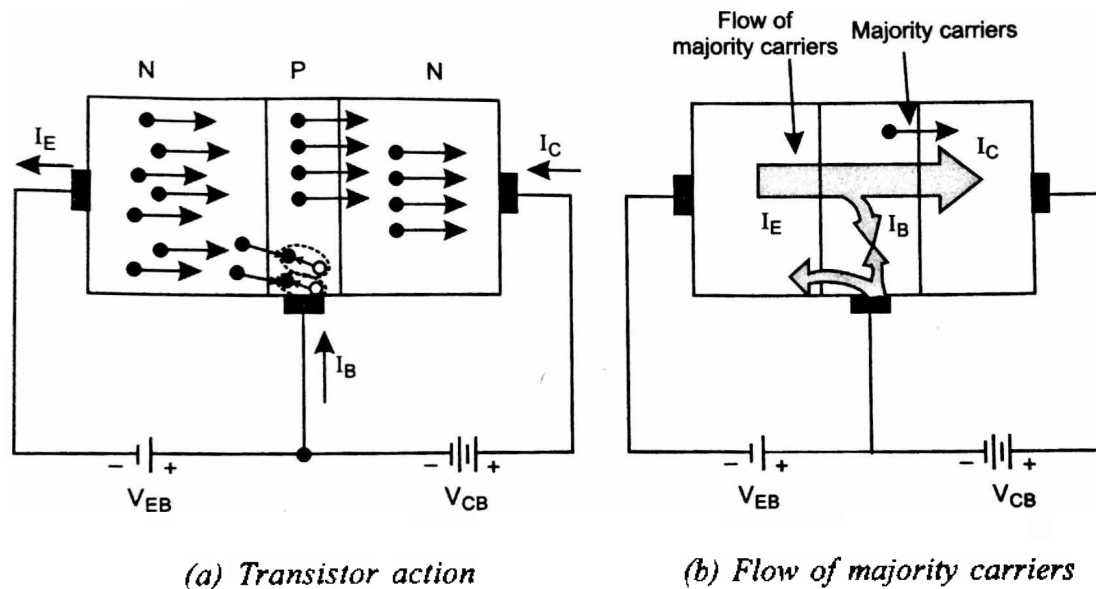


Fig. 4.48

Due to the forward bias on emitter base junction, a large number of electrons (majority carriers) in the emitter are injected towards the base. This constitutes the emitter current (I_E). When these electrons enter the base region, they combine with holes. Since the base is lightly doped and very thin, only a few electrons (less than 5%) combine with holes to constitute base current I_B . The remaining electrons (more than 95%) diffuse across the thin base region and reach the collector. These electrons then come under the influence of a positively biased N-region (collector) where electrons are collected by the collector. This constitutes collector current I_C .

However, the emitter current is the sum of collector current and base current

$$\text{Emitter current, } I_E = I_B + I_C$$

There is another component of collector current due to thermally generated minority carriers (holes in this case) which pass towards the base. This small current component is called **reverse saturation current**.

4.28 OPERATION OF PNP TRANSISTOR

A PNP transistor is shown in Fig. 4.49. The emitter base junction is forward biased while collector-base junction is reverse biased. The forward biased voltage V_{EB} is small whereas the reverse biased voltage V_{CB} is considerably high.

The forward bias of emitter base junction causes the holes in the P type emitter to flow towards the base. This constitutes the emitter current I_E . As the holes cross into N type base, they tend to combine with the electrons. As the base is lightly doped and very thin, therefore, only a few holes (less than 5%) combine with the electrons. The remained (more than 95%) cross into the collector region to constitute collector current I_C . In this way, almost the entire current flows to the collector circuit. It may be noted that current conduction within PNP transistor is by holes. However, the emitter current is the sum of collector current and base current.

$$\text{Emitter current } I_E = I_C + I_B \quad \dots(4.1)$$

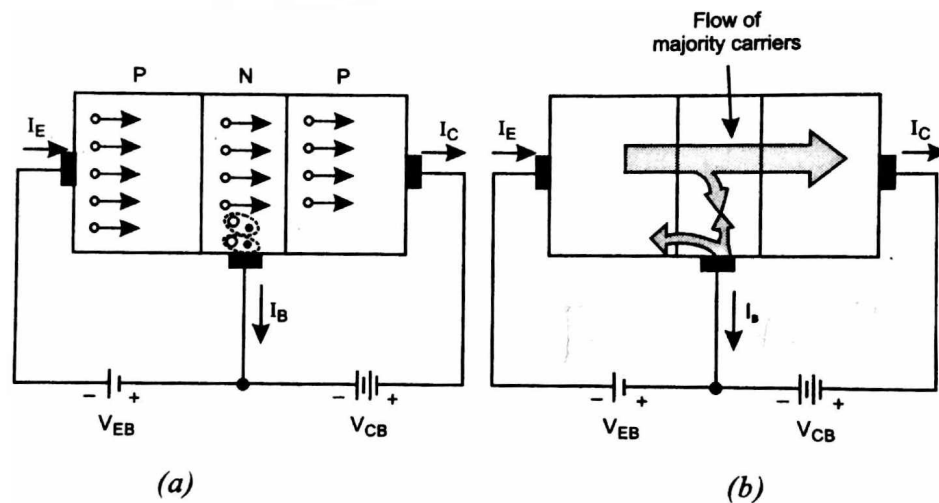


Fig. 4.49

4.29 TRANSISTOR CONFIGURATION

A transistor has three terminals, namely, emitter, base and collector. We require four terminals when a transistor is to be connected in a circuit, two for the input and two for the output. This is achieved by making one terminal of the transistor common to both input and output terminals. The input is fed to this common terminal and any one of the other two terminals. The output is obtained between the common terminal and the remaining terminal. Transistor is configured based on the common terminal.

1. common-base configuration
2. common-emitter configuration
3. common-collector configuration

4.33 V-I CHARACTERISTICS OF A TRANSISTOR

The behaviour of a transistor is observed with the help of V-I characteristics. There are two types of characteristics.

1. Input characteristics
2. Output characteristics

1. **Input characteristics.** Input characteristics of a transistor give the relationship between input current and input voltage for a constant output voltage.
2. **Output characteristics.** Output characteristics of a transistor give the relationship between output current and output voltage for constant input current.

4.33.1 V-I Characteristics in CB Configuration

The experimental setup for input and output characteristics is shown in the Fig. 4.58. In this circuit NPN transistor is used.

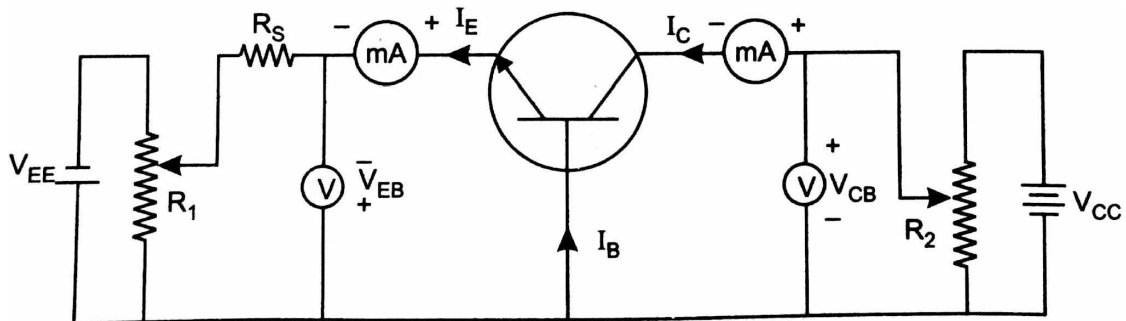


Fig. 4.58 Circuit arrangement for input and output characteristics

Input Characteristics

The input characteristic is a curve plotted between emitter current I_E and emitter base voltage V_{EB} at constant collector base voltage V_{CB} . The emitter current I_E is taken along y-axis and emitter base voltage V_{EB} along x-axis.

Keeping V_{CB} constant ($V_{CB} = 2 \text{ V}$), note down emitter current I_E for various values of V_{EB} . Then plot the readings to obtain the input characteristics as shown in Fig. 4.66. This gives input characteristic at $V_{CB} = 2 \text{ V}$. Follow the same procedure for different values of V_{CB} and plot the graph to obtain family of input characteristics. The input characteristics are also called as driving point characteristics.

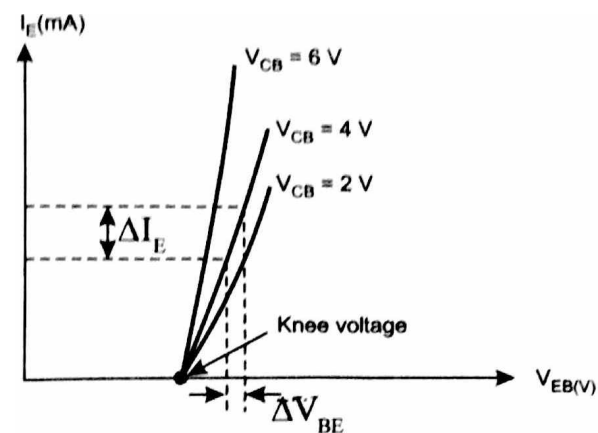


Fig. 4.59

Observations

1. Emitter current is very small (negligible), when the emitter base voltage V_{EB} is below to the knee voltage.
2. After knee voltage, the emitter current I_E increases rapidly with small increase in emitter base voltage.

Input resistance

It is the ratio of change in emitter-base voltage (ΔV_{EB}) to the resulting change in emitter current (ΔI_E) at constant collector-base voltage (V_{CB}).

$$\text{Input resistance } r_i = \frac{\Delta V_{EB}}{\Delta I_E} \text{ at constant } V_{CB}.$$

As small V_{EB} is enough to produce a large flow of emitter current I_E . Therefore the input resistance is very small.

Output Characteristics

It is the curve plotted between collector current I_C and collector base voltage V_{CB} at constant emitter current. Generally collector current I_C is taken along y-axis and collector-base voltage along x-axis.

Keeping emitter current I_E fixed at some value say 3 mA, note the collector current I_C for various values of collector base voltage V_{CB} . Then plot the readings to obtain the output characteristics as shown in Fig. 4.59.

Observations

1. The collector current I_C varies with V_{CB} only at very low voltage ($V_{CB} < 0$). The transistor is never operated in this region.

- Collector current I_C is almost constant when the voltage is increased beyond the cut in voltage. It means that now I_C is independent of V_{CB} and depends upon I_E .

Output Resistance

It is the ratio of change in collector base voltage (ΔV_{CE}) to change in collector current (ΔI_C) at constant emitter current I_E .

$$\text{Output resistance } r_o = \frac{\Delta V_{CE}}{\Delta I_C} \text{ at constant } I_E$$

The output resistance of CB circuit is very high of the order of several tens of kilo-ohms.

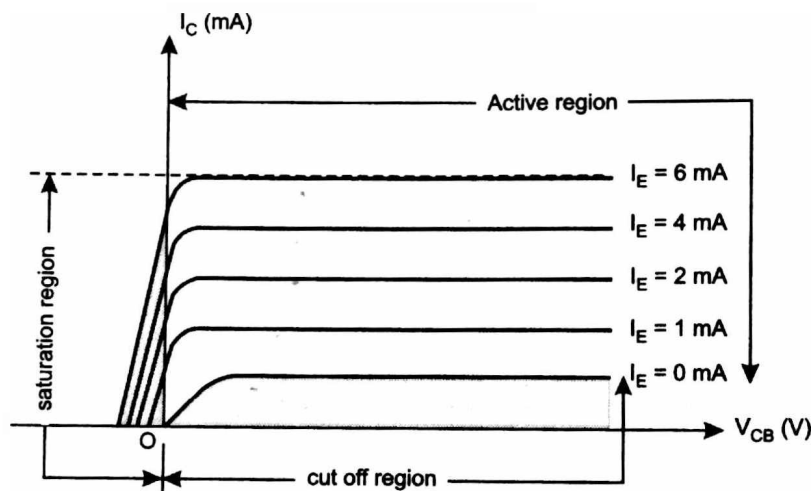


Fig. 4.60 Output characteristics of CB configuration

4.3.1.2 V-I Characteristics of CE Configuration

For any transistor configuration, the transistor has two characteristics namely input characteristics and output characteristics. Fig. 4.61 shows the circuit arrangement for input and output characteristics of CE configuration.

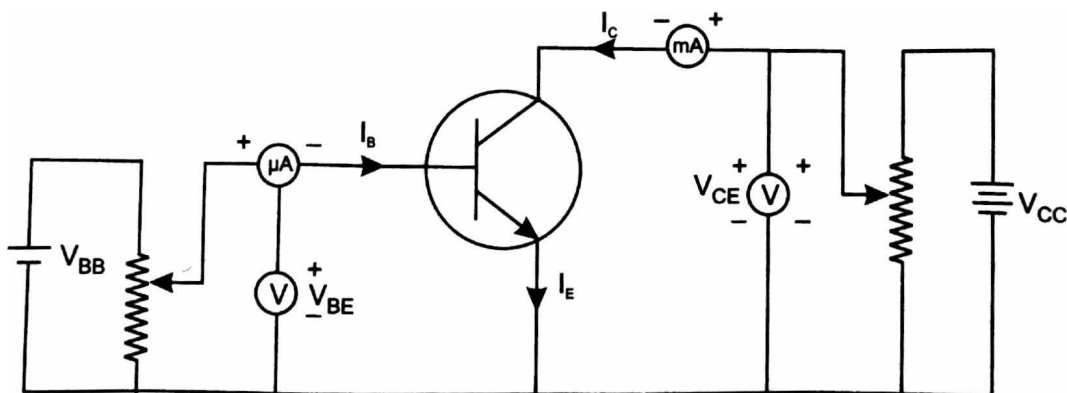


Fig. 4.61 Circuit arrangement for input and output characteristics of CE configuration

Input Characteristics

It is the curve between base current I_B and base emitter voltage V_{BE} at constant collector-emitter voltage V_{CE} . Keeping V_{CE} as constant, note down current I_B for various values of V_{BE} . Then plot the readings to obtain input characteristics shown in Fig. 4.62.

Observation

As compared to CB configuration, I_B increases less rapidly with increase of V_{BE} . Therefore input resistance of a CE configuration is higher than CB configuration.

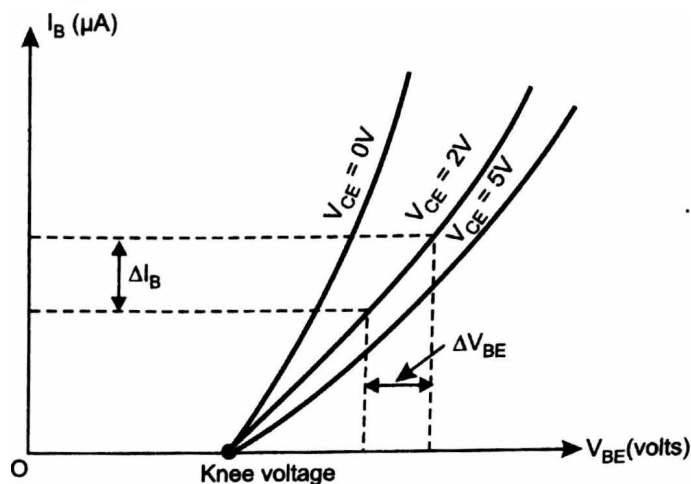


Fig. 4.62 Input characteristic of CE configuration

Input resistance

It is the ratio of change in base emitter voltage (ΔV_{BE}) to the change in base current (ΔI_B) at constant V_{CE} .

$$\text{Input resistance } r_i = \frac{\Delta V_{BE}}{\Delta I_B} \text{ at constant } V_{CE}.$$

The value of input resistance for a CE circuit is of the order of a few hundred ohms.

Output Characteristics

It is the curve plotted between collector current I_C and collector emitter voltage V_{CE} at constant base current I_B . Keeping base current I_B fixed at some value, note the collector current I_C for various values of V_{CE} . Plot the reading on a graph to obtain output characteristics as shown in Fig. 4.63.

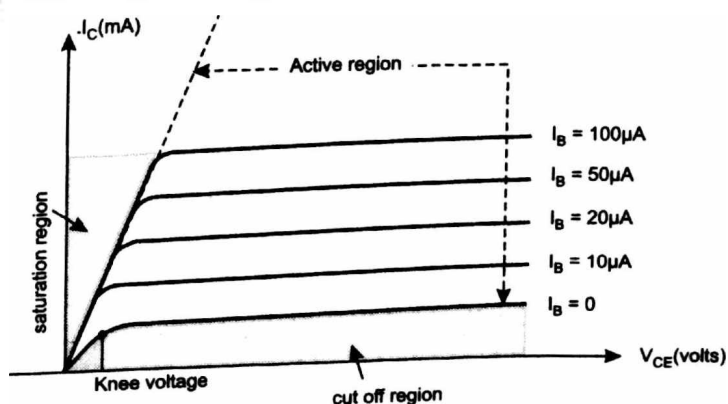


Fig. 4.63 Output characteristics of CE configuration.

Observations

1. The collector current I_C varies with V_{CE} between 0 and 1 V only. After this collector current is almost constant, independent of V_{CE} .
2. For any value of V_{CE} above knee voltage, the collector current I_C is approximately equal to βI_B .
3. The transistors are operated in the region above knee voltage.

Output resistance

It is the ratio of change in collector-emitter voltage (ΔV_{CE}) to change in collector current (ΔI_C) at constant I_B .

$$\text{Output resistance } r_o = \frac{\Delta V_{CE}}{\Delta I_C} \text{ at constant } I_B.$$

4.32 COMPARISON OF VARIOUS TRANSISTOR CONFIGURATIONS

S.No.	Characteristic	Common base	Common emitter	Common Collector
1.	Input resistance	Low (about 100 Ω)	Low (about 750 Ω)	Very high (about 750 k Ω)
2.	Output resistance	Very high (about 450 k Ω)	High (about 45 k Ω)	Low (about 50 Ω)
3.	Voltage gain	Low (about 150)	High (about 500)	Very low (Less than 1)
4.	Leakage current	Very small (5 μ A for Ge 1 μ A for Si)	Very large (500 μ A for Ge 20 μ A for Si)	Very large (500 μ A for Ge 20 μ A for Si)
5.	Application	For high frequency application	For audio frequency applications	For impedance matching

Conclusions

From the three configurations, the **common emitter circuit is the most efficient** (it is used about 90 to 95% of all transistor applications). The reason for use of this CE configurations is

- ♦ High current gain
- ♦ High voltage and power gain and
- ♦ Moderate output to input ratio

4.33 APPLICATION OF TRANSISTOR

1. Transistors are used as switches.
2. They are used in amplifiers to amplify the weak signals and
3. They are also used in oscillator circuits.

4.34 TRANSISTOR AS AN AMPLIFIER

Definition : Amplifier is an electron circuit constructed by using transistor to increase the magnitude of given weak signal.

Figure 4.64 shows the common emitter NPN transistor amplifier circuit. dc voltage V_{BB} is connected in the input circuit in addition to the signal voltage. The dc voltage is known as bias voltage and its magnitude is such that it always keeps the emitter base junction in forward bias. The dc supply V_{CC} keeps the collector base junction in reverse bias.

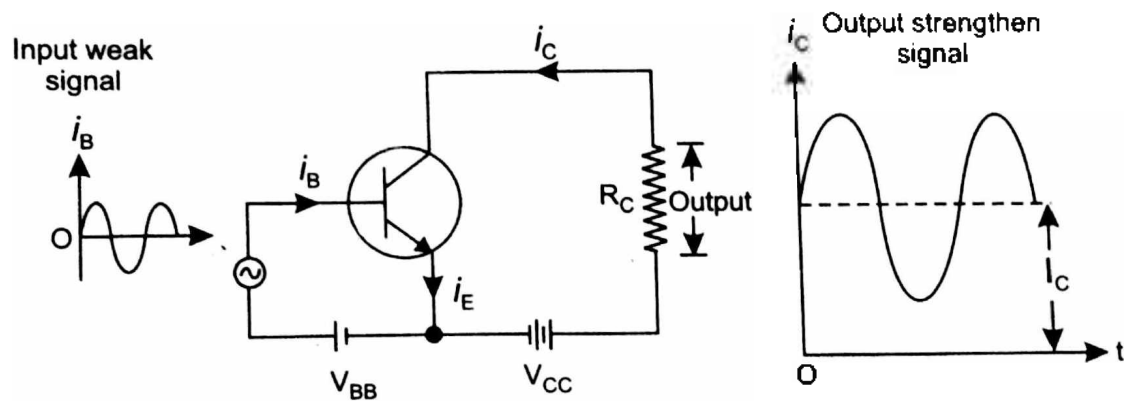


Fig. 4.64 CE amplifier

Operation of an Amplifier

During the positive half cycle of the input weak signal, the forward bias across the emitter base junction is increased. Therefore more electrons flow from the emitter to the collector through the base region. This causes an increase in collector current. This increased collector current produces higher voltage drop across the load resistor R_C , now we get the positive half cycle of strengthened output signal. During negative half cycle of the input signal, forward bias across emitter-base junction is decreased. Therefore collector current decreases. Due to this decreased collector current, the output voltage is also decreased. The amplified output is obtained across the load resistor R_C .

The positive going pulse of input source increases the emitter voltage. As base voltage is constant, the forward bias of emitter base junction reduces. This reduces I_B , thus reducing I_C and hence a drop across R_C occurs. Since output voltage $V_o = V_{CC} - I_C R_C$ the reduction in I_C results in an increase in V_o . Therefore, we can say that positive going input produces positive going output and similarly negative going input produces negative going output and there is no phase shift between input and output in common base amplifier.

4.38 RC PHASE SHIFT OSCILLATOR [JAN-2018]

Shows a transistorized RC phase shift oscillator Fig. 4.68 which uses a common emitter single stage amplifier and a phase shifting network consisting of three identical RC sections. The amplifier used causes a phase shift of 180° then the feedback network should create a phase shift of 180° , to satisfy the Barkhausen Criterion. Hence in a phase shift oscillators, three sections of RC circuit are connected in cascade, each introducing a shift of 60° , thus introducing a total phase shift of 180° , due to feedback network. A shift of 180° is developed by transistor amplifier and additional 180° phase shift is produced by the RC network, hence the total phase shift is 360° .

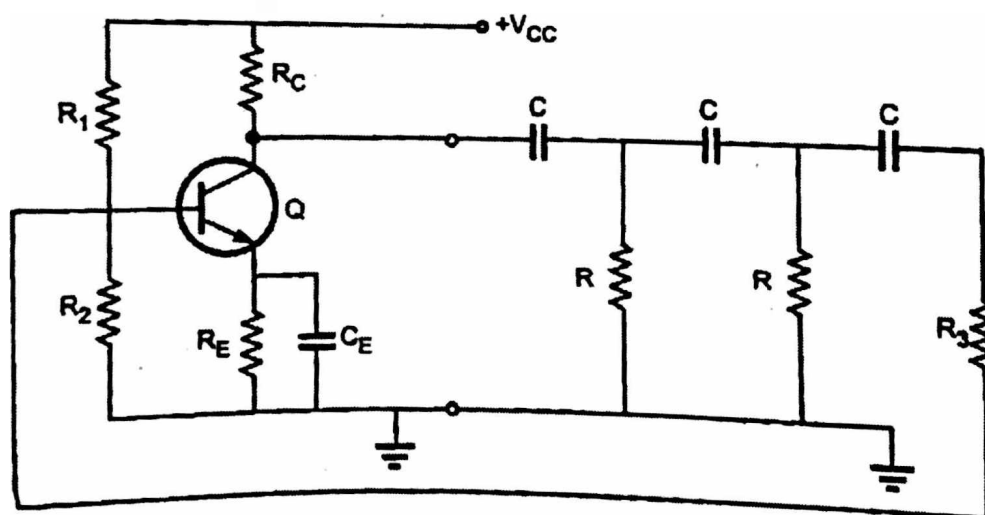


Fig. 4.68 RC Phase Shift Oscillator

Frequency of oscillation is
$$F = \frac{1}{2\pi RC\sqrt{6}}$$

Where,

$$R_1 = R_2 = R_3 = R$$

$$C_1 = C_2 = C_3 = C$$

4.39 HARTLEY OSCILLATOR

LC oscillator which uses two inductance and one capacitance in its feedback network is called Hartley Oscillator. Figure 4.69 shows the Hartley Oscillator using transistor.

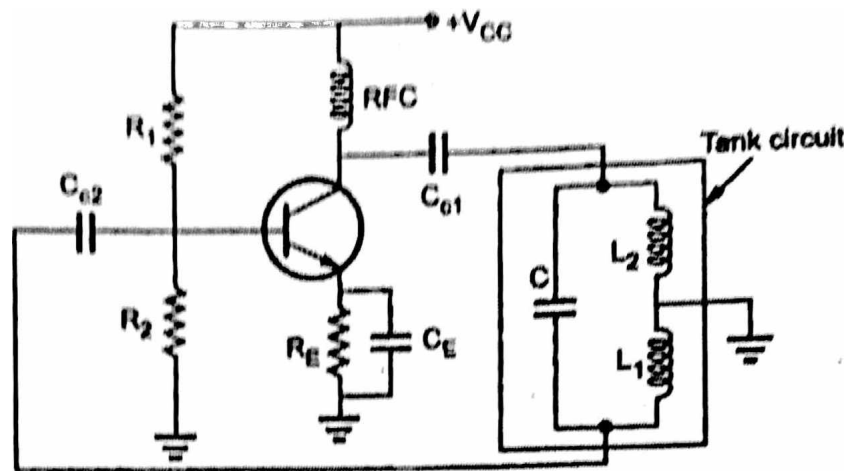


Fig. 4.69 Hartley Oscillator

The resistance R_1 and R_2 are the biasing resistances. The RFC is the radio frequency choke. Its reactance value is very high for high frequency, hence it can be treated as open circuit. While for d.c. conditions, the reactance is zero hence cause no problem for d.c. capacitors. Hence due to RFC, the isolation between a.c. and d.c. operation is achieved. R_E is also a biasing circuit resistance and C_E is the emitter bypass capacitor. C_{c1} and C_{c2} are the coupling capacitor.

The common emitter amplifier provides a phase shift of 180° . As emitter is grounded, the base and the collector voltages are out of phase by 180° . As the centre of L_1 and L_2 is grounded, when upper end becomes positive, the lower becomes negative and vice versa. So the LC feedback network gives an additional phase shift of 180° , necessary to satisfy oscillation conditions.

Frequency of oscillation

$$F = \frac{1}{2\pi\sqrt{L_{eq}C}}$$

$$\text{Where } L_{eq} = L_1 + L_2$$

4.40 FIELD EFFECT TRANSISTOR

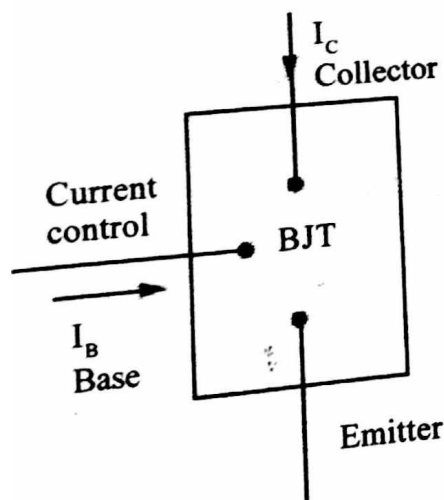
The field effect transistor (FET) is a three terminal device used for a variety of applications. They are very much similar to a bipolar junction transistor (BJT). The bipolar transistor has two disadvantages.

1. It has a low input impedance because of forward biased emitter junction.
2. It has considerable noise level.

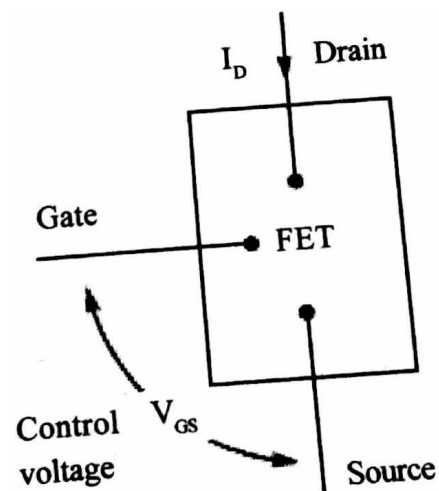
The field effect transistor (FET) by virtue of its construction and biasing has large input impedance which may be more than 100 megaohms. *The FET is much less noisy than bipolar transistor.* FET is also called as junction field effect transistor (JFET).

The major difference between the two types of transistors is that the *BJT is a current controlled device*, while the *JFET is a voltage controlled device*. In other words, the collector current I_C is a direct function of the base current I_B in the BJT. For FET the drain current I_D will be a function of the gate to source voltage V_{GS} applied to the input circuit is shown in Fig. 4.70.

- Terminals of FET : Drain (D), source (S) and gate (G)
- Voltage control device : The current I_D (drain current) depends on the control voltage V_{GS} applied between gate and source.
- Unipolar Device : FET is a unipolar device, that means the current flowing through it is only due to one type of carriers or majority carriers (holes or electrons).



(a) BJT



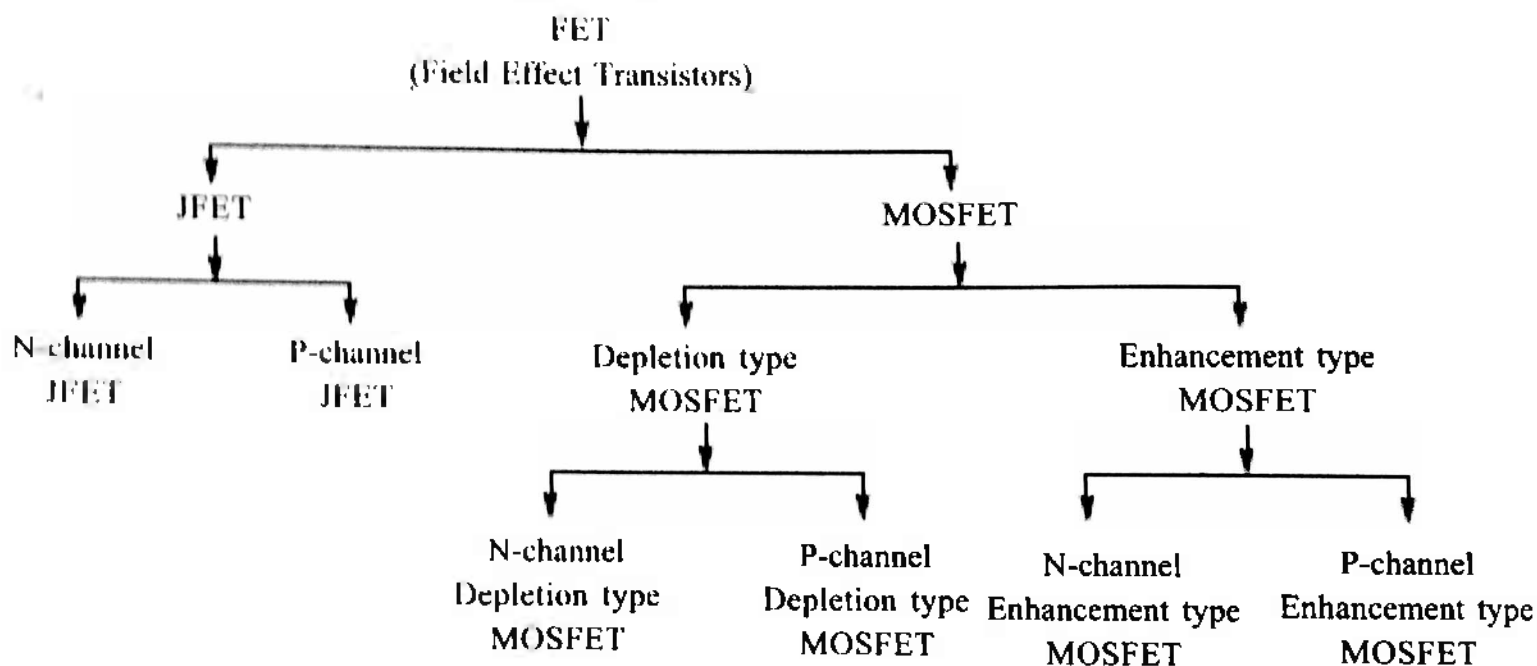
(b) FET

Fig 4.70.

<i>In BJT</i>		<i>In FET</i>
Emitter	$\Leftrightarrow$	Source
Base	$\Leftrightarrow$	Gate
Collector	$\Leftrightarrow$	Drain

4.41 CLASSIFICATION OF FETs

The field effect transistors are classified as follows



4.42 COMPARISON BETWEEN JFET AND BJT

S.No.	Field Effect Transistor (FET)	Bipolar Junction Transistor (BJT)
1.	It is a unipolar device (<i>i.e.</i> current in the device is carried either by electrons or holes)	It is a bipolar device (<i>i.e.</i> current in the device is carried by both electrons and holes).
2.	It is a voltage controlled device	It is a current controlled device.
3.	Input resistance is very high and is of the order of mega ohms	Input resistance is very low as compared to FET and is of the order of few kilo-ohms.
4.	It has negative temperature coefficient at high current levels	It has a positive temperature coefficient at high current level.
5.	It is less noisy than BJT and vacuum tube	More noisy than a FET
6.	Simpler to fabricate as an IC and occupies a less space on IC chip	Difficult to fabricate as an integrated circuit and occupies more space on IC chip as compared to that of FET.

S.No.	Field Effect Transistor (FET)	Bipolar Junction Transistor (BJT)
7.	Cost is more	Cost is low
8.	Transfer characteristics is non linear	Transfer characteristic is linear
9.	Good thermal stability because of absence of minority	Temperature dependent, thermal runaway may cause poor stability
10.	Small gain bandwidth product	Greater than FET

4.43 BASIC CONSTRUCTION OF JFETs [April - 2017]

A JFET consists of a P -type or N -type silicon bar containing two PN junctions on the sides as shown in the Fig. 4.71. The bar forms the conducting channel for the charge carriers. If the bar is of N -type, it is called N -channel JFET as shown in Fig. 4.71(a) and if the bar is of P -type, it is called a P -channel JFET as shown in Fig. 4.71(b). The two PN junctions forming diodes are connected internally and a common terminal called gate is taken out. Other terminals are source and drain taken out from the bar shown. JFET has three terminals gate (G), source (S) drain (D).

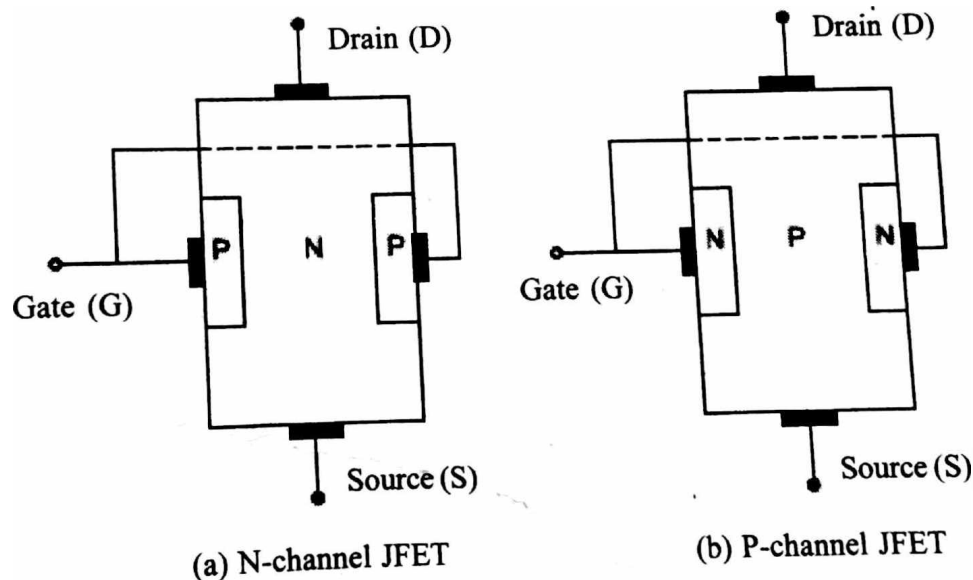


Fig. 4.71

4.43.1 Biasing of JFET

Figure 4.72 (a) shows N -channel JFET polarities whereas Fig. 4.72(b) shows the P -channel JFET polarities. Note that in each case, the voltage between the gate and source is such that the gate is always reverse biased.

This is the normal way of JFET connection. Therefore V_{GS} should be negative. With increase in the reverse voltage applied to the gate to source PN junction, the output current which is the drain current I_D will decrease. The drain and source terminals are interchangeable i.e., either end can be used as source and the other end as drain.

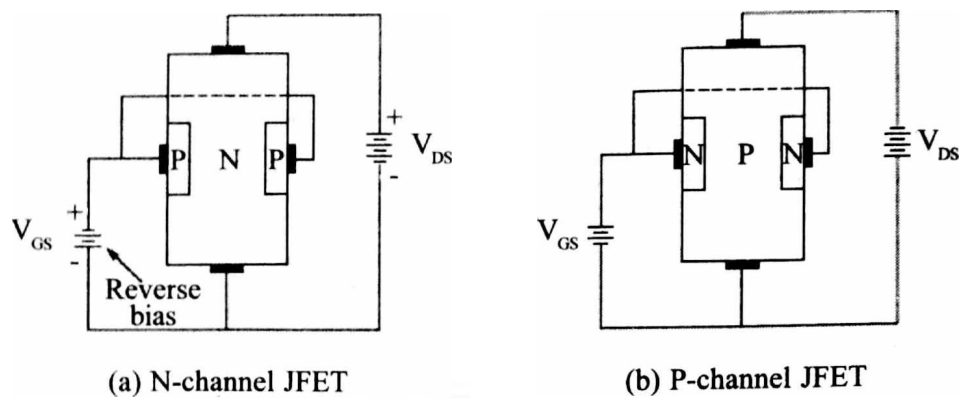


Fig. 4.72 Biasing of JFET

4.43.2 Symbols of JFET

Figure 4.73 shows the schematic symbol of JFET. The vertical line in the symbol may be thought of as a channel. Source (*S*) and drain (*D*) connected to this line. If the channel is *N*-type, arrow on the gate points towards the channels as shown in Fig. 5.4(a). However for *P*-type channel the arrow on the gate points from the channel to gate as shown in Fig. 5.4(b).

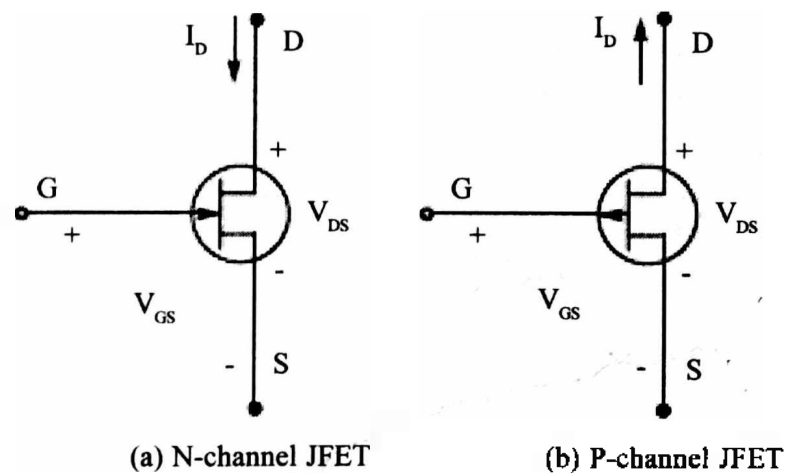


Fig. 4.73 Symbol of JFET

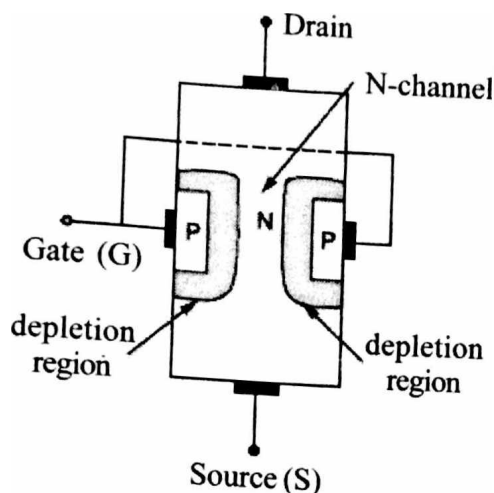
4.44 WORKING PRINCIPLE OF JFET

The working principle of JFET is explained by the following three conditions.

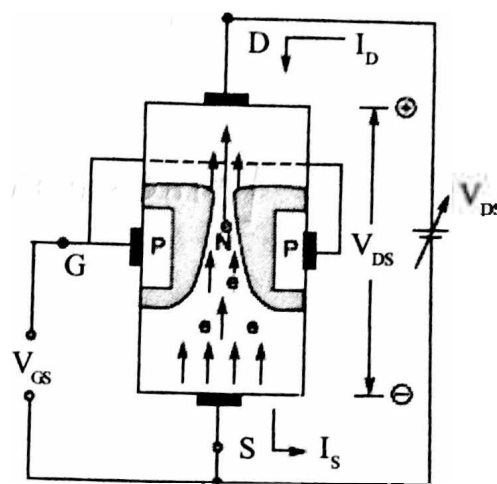
1. No bias condition
2. $V_{GS} = 0$, $V_{DS} > 0$
3. $V_{GS} < 0$, $V_{DS} > 0$

1. No bias condition

In the absence of any applied potential the JFET has two PN junctions under no-bias conditions. The result is a depletion region at each junction as shown in Fig. 4.74(a) that resembles the same region of a diode under no bias condition. Recall also that a depletion region is that region which is void of free carriers and therefore unable to support conduction through the region.



(a) No bias condition



(b) $V_{GS} = 0$ and $V_{DS} > 0$

Fig. 4.74

2. $V_{GS} = 0$ V and $V_{DS} > 0$ V

When a voltage V_{DS} is applied between drain and source terminals and voltage on the gate is zero as shown in Fig. 4.74(b). The two pn junctions at the sides of the bar establish depletion layers. The electrons will flow from source to drain through a channel between the depletion layers. The size of these layers determines the width of the channel and hence the current conduction through the bar. The instant voltage V_{DS} is applied, the electrons will be drawn by the drain terminal, establishing the conventional current I_D with defined direction of Fig. 4.74(b).

3. $V_{GS} < 0$ and $V_{DS} > 0$ V

When a reverse voltage V_{GS} is applied between the gate and source (Fig. 4.75), the width of the depletion layers is increased. This reduces the width of conducting channel there by increasing the resistance of N -type bar. Consequently, the current from source to drain is decreased.

On the other hand, if the reverse voltage on the gate is decreased, the width of the depletion layers also decreases. This increases the width of the conducting channel and hence source to drain current.

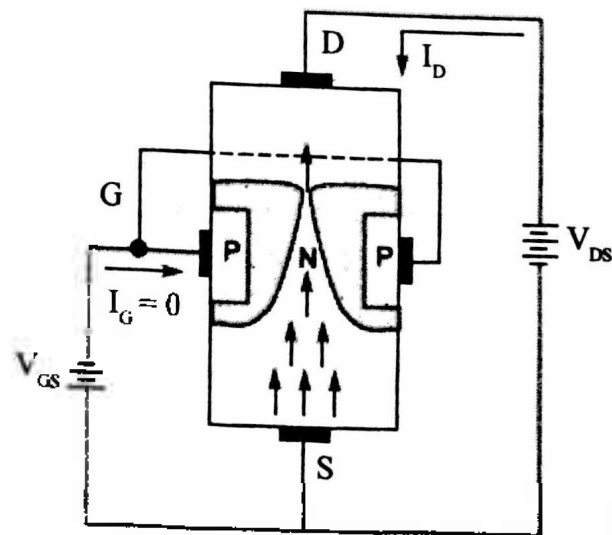


Fig. 4.75 Circuit operation $V_{GS} < 0$ and $V_{DS} > 0$

4.45 METAL OXIDE SEMICONDUCTOR FIELD EFFECT TRANSISTOR

Metal Oxide Semiconductor Field Effect Transistor (MOSFET) is an important semiconductor device and is widely used in many circuit applications. The main advantage of MOSFET over JFET is that the input impedance of a MOSFET is much more than that of a JFET because of very small gate leakage current. Since the gate of MOSFET is insulated from the channel, it is also termed as Insulated Gate Field Effect Transistor (IGFET). MOSFETs are further classified into depletion type and enhancement type based on their basic mode of operation.

4.46 THE DEPLETION-MODE MOSFET (D-MOSFET) [UPTU 2005-06]

4.46.1 Construction of DMOSFET

The basic structure of an n -channel depletion type MOSFET is shown in Fig. 4.76. A slab of p -type material is formed from a silicon base as the substrate. In some cases the substrate is internally

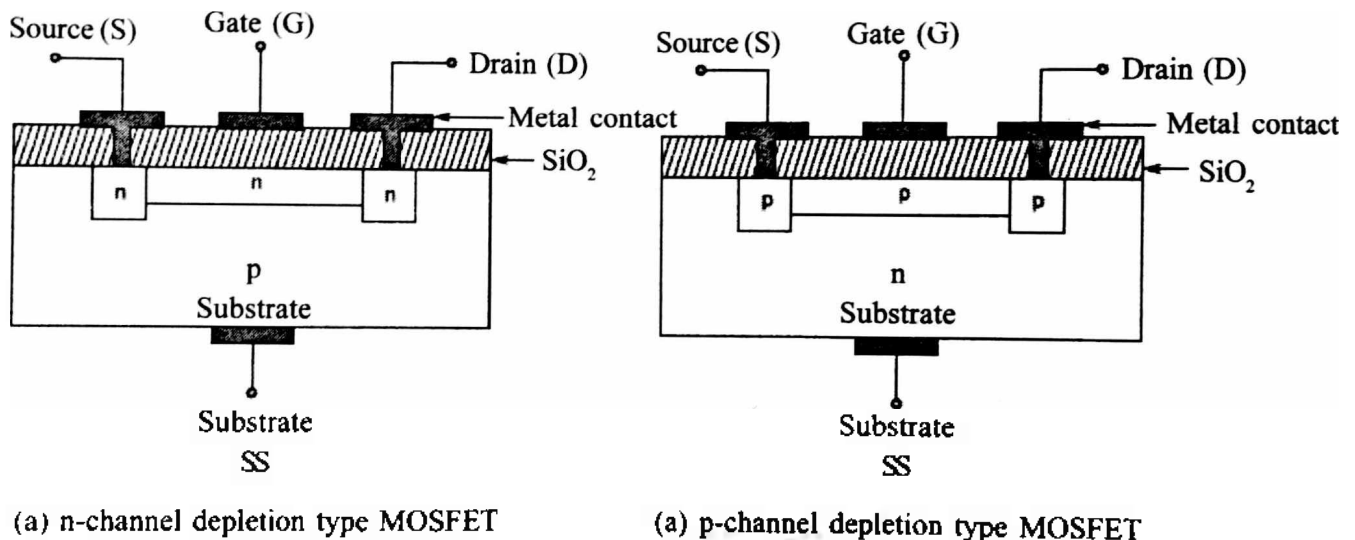


Fig. 4.76

connected to the source terminal. However many discrete devices provide an additional terminal labelled SS, resulting in a four terminal device. The source and drain terminals are connected through an n channel as shown in Fig. 4.76. The gate is also connected to a metal contact surface but remains insulated from the n -channel by a very thin silicon dioxide (SiO_2) layer. *There is no electrical connection between the gate terminal and the channel of MOSFET.* Due to the presence of the SiO_2 layer between gate terminal and n -type channel, the input impedance of MOSFET is very high.

For n -channel depletion type MOSFET there will be an n -type silicon region (channel) connecting the source and drain at the top of the p -type substrate as shown in Fig. 4.76(a). Similarly for p -channel depletion type MOSFET there will be a p -type region channel connecting the source and the drain at the top of the n -type substrate as shown in Fig. 4.76(b).

4.46.2 Operation of n -channel Depletion MOSFET The direct connection from gate terminal to source terminal sets the gate to source voltage to zero and a voltage V is applied across the drain to source terminals as shown in Fig. 6.2. This results in free electrons from the channel are attracted towards positive potential of drain terminal. This establishes current flow through the channel to be denoted as I_{DSS} at $V_{GS} = 0$ V. The depletion MOSFET can be operated in two different modes as given below

1. Depletion mode : The device operates in this mode, when the gate voltage is negative
2. Enhancement mode : The device operates in this mode, when the gate voltage is positive.

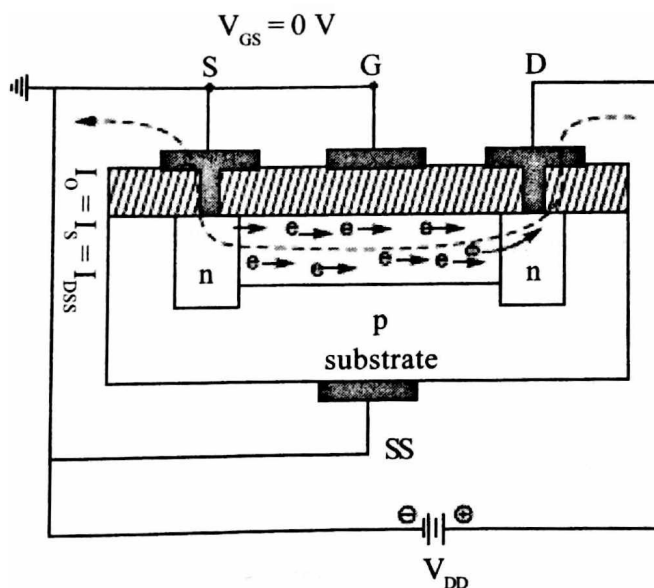


Fig. 4.77 N -channel depletion MOSFET operation when V

1. Depletion mode of operation Fig. 4.76(a) shows a MOSFET with a negative gate to source voltage. The negative potential at the gate will tend to pressure or repel electrons towards the p -type substrate as shown in Fig. 4.77(a level of recombination between repelled electrons and attracted holes will occur that will reduce the number of free electrons in the n -channel for conduction. The resulting drain current is reduced by increasing the negative gate voltage.

In other words we can say that, due to recombinations, n -channel is depleted of some of its electrons, thus decreasing the channel conductivity. The greater the regative voltage applied at the gate, the greater the depletion of n -channel electrons. The level of drain current will reduce with increasing negative bias for V_{GS}

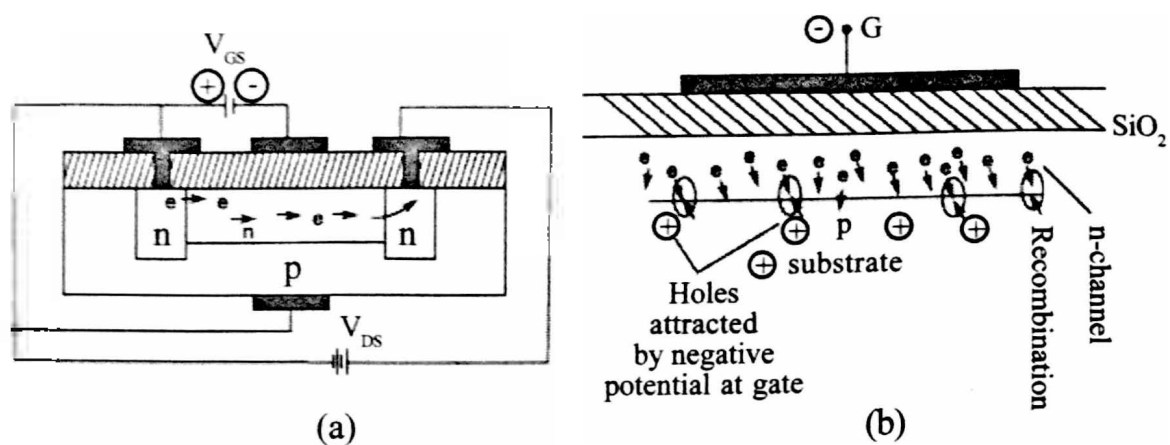


Fig. 4.78 Depletion mode of operation

2. Enhancement mode of operation

Figure 4.79 shows the depletion MOSFET with a positive gate to source voltage. For positive values of V the positive gate will draw additional electrons from the p -type substrate due to the reverse leakage and establish new carriers through the collisions resulting between accelerating particles. As the gate to source voltage continues to increase, the drain current will increase at a rapid rate. Because of this, as gate to source voltage increases in positive direction, the drain current also increases. The application of a positive gate to source voltage has "enhanced" the level of free electrons in the channel as compared to that encountered with $V_{GS} = 0$ V. For this reason, this mode of operation is called as enhanced mode of operation.

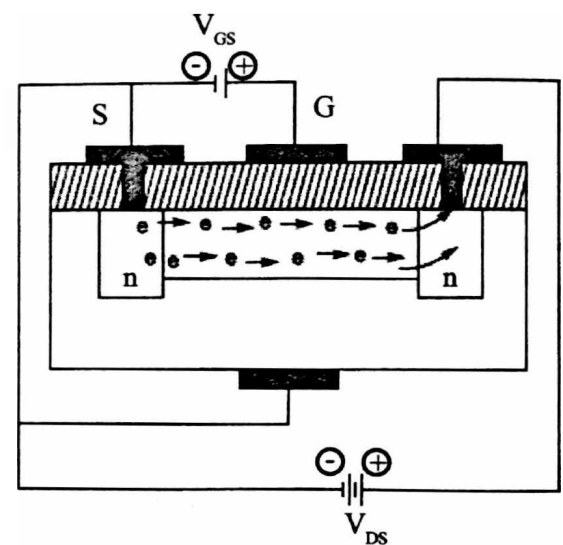


Fig. 4.79 Enhancement mode of operation

4.47. THE ENHANCEMENT MODE MOSFET (EMOSFET)

The other type of MOSFET is enhancement MOSFET can also be classified into n -channel and p -channel.

4.47.1 Construction of n -channel EMOSFET (NMOS)

The basic construction of enhancement MOSFET (EMOSFET) is shown in Fig. 4.79. A p -type material is used as substrate of this device. No channel is connected between source and drain. This is the primary difference between DMOSFET and EMOSFET. The SiO_2 layer is present to isolate the gate metallic platform from the region between the drain and source, but now it is simply separated from a section of p -type material. The summary of an enhancement type

MOSFET is quite similar to that of the depletion type MOSFET, except for the absence of a channel between the drain and source terminals.

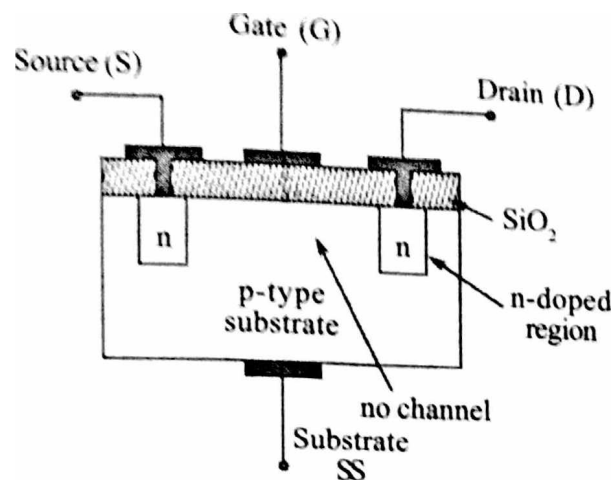


Fig. 4.80 Construction of EMOSFET

4.47.2 Operation of *n*-channel Enhancement MOSFET

The operation of enhancement type MOSFET can be explained with two different operating conditions 1. $V = 0$ 2. $V_{GS} > 0$

1. Operation with $V_{GS} = 0$ V If the gate source voltage V_{GS} is set at 0 V and a positive voltage V_{DS} is applied between the drain and source of the device as shown in Fig. 6.10. The absence of a channel will result in current of effectively zero amperes. This is totally different from depletion type MOSFET and JFET where $I = I_{DSS}$. When gate to source voltage is zero, the V_{DS} supply tries to force free electrons from source to drain. But the presence of *p* region (absence of channel) does not permit the electrons to pass through it. Thus there is no drain current for $V = 0$. Due to this fact the enhancement type MOSFET is also called normally OFF MOSFET.

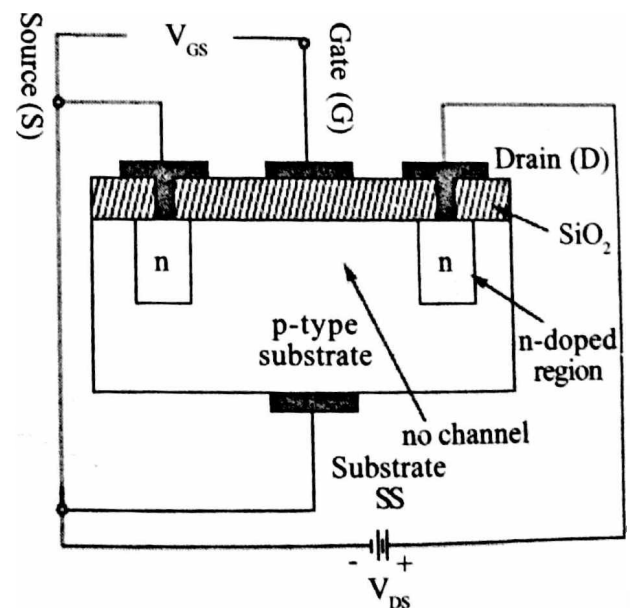


Fig. 4.81 Operation of EMOSFET when $V_{GS} = 0$

2. Operation with $V > 0$

Set the gate source voltage V_{GS} greater than 0 V and V_{GS} is also connected as shown in Fig 6.11. The positive voltage at the gate will pressure the holes in the substrate along the edge of the SiO_2 DS layer to leave the area and form a depletion region. However, the electrons in the P-type substrate (the minority carriers of the channel) will be attracted to the positive gate and accumulate in the region near the surface of the SiO_2 layer. The SiO_2 layer and its insulating material will prevent the negative carriers from being absorbed at the gate terminal. As V_2 increases, the concentration of electrons near the SiO_2 surface increases until eventually induced n-type region can support a current flow between drain and source. that results in the significant increase in drain current is called **threshold voltage** and is represented by the symbol V_{th} . The level of V_{GS} (th). As V is increased beyond the threshold level, the density of free electrons in the induced channel will increase, resulting in an increased level to drain current.

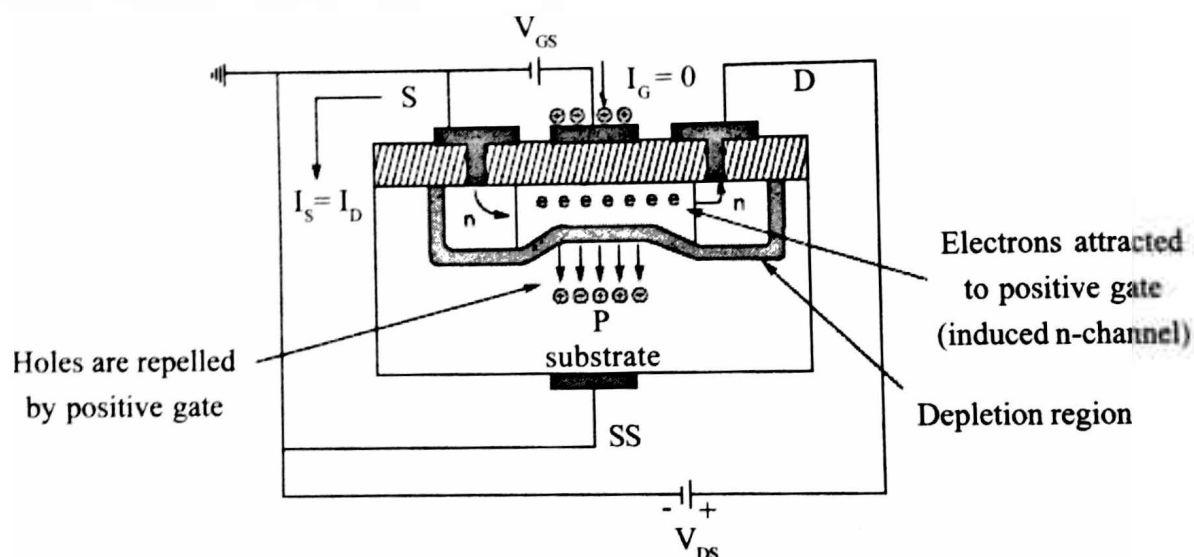


Fig. 4.82 The operation of EMOSFET when $V_{GS} > 0$

Effect of increasing V_{DS}

Figure 4.83 shows the effect of increasing V_{DS} with constant V_{GS} . The gate source voltage V is kept constant and drain to source voltage V_{DS} is increased gradually. Due to this, the gate terminal becomes less positive as compared to drain. So less number of electrons are attracted towards the gate terminal and the "induced channel" becomes narrow as shown in Fig. 6.82. Eventually the channel will be reduced to the point of pinch off and saturation will be established. In other words, any further increase in V_{DS} at the fixed value of V will not affect the saturation level of I until breakdown conditions are encountered. D

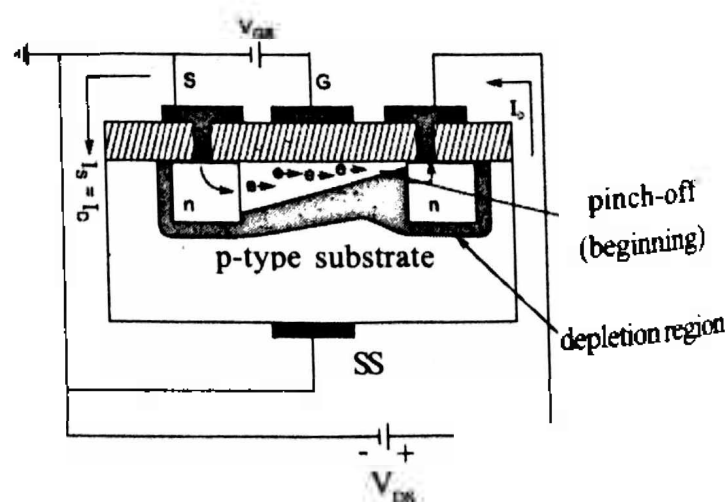


Fig. 4.83 Effect of increasing V_{DS}

REVIEW QUESTIONS

1. Draw a RC phase shift oscillator circuit diagram and explain its operation. Derive the expression for the frequency of oscillations. Design one such to produce 1000 Hz.
2. Write a difference between intrinsic and extrinsic semiconductor.
3. Draw the VI characteristics of PN diode and explain.
4. Draw the VI characteristics of Zener diode and explain.
5. Draw the VI characteristics of JFET and explain.
6. Draw the VI characteristics of UJT and explain.
7. Draw the RC coupled amplifier and explain.
8. What is FET ? Write its types? Explain Enhancement MOSFET?
9. Define Oscillator, draw and explain Hartley Oscillator.
10. Write short notes on Colpitts oscillator.
11. Draw & Explain the VI characteristics of MOSFET?
12. Draw a Hartley oscillator circuit diagram and explain its operation. Derive the expression for the frequency of oscillations. Design one such to produce 1000 kHz.

5.1 INTRODUCTION

Generally the electronics is divided into two basic areas : analog and digital electronics. The digital electronics is a newer and greater performance as compared to analog electronics. The main difference between analog and digital signals can be stated simply as Analog signals are continuously variable between defined values, and digital signals can vary only by distinct or discrete steps.

5.2 ANALOG SYSTEM AND DIGITAL SYSTEM

A system which are capable of processing continuous range of values or analog signal is called as analog system or analog electronics.

Ex. Analog meters, physical balance etc.

System which process discrete values or digital signal is called as digital systems or digital electronics.

Ex. Digital calculator, Microprocessor, Digital Computer etc.

Advantages of Digital systems

1. Digital systems are easy to design.
2. Information storage is easy.
3. High accuracy and precision can be achieved.
4. It is less affected by noise.
5. The circuit can be easily fabricated in integrated circuits.

5.3 LOGIC GATES [JAN - 2018, 16]

A logic gate is a circuit that performs a logic function on a number of binary input signals. The signals at the input and output terminals of gate either at HIGH level voltage or at a LOW level voltage. A gate will produce one output level whenever any other combination

is present at the inputs. The different types of basic gates available are AND, OR and NOT gate. The logic functions are defined by a truth table. This table lists all the possible combinations of the input variables and output variable applied to a gate. Each variable can only be at either 0 (logic 0 level) or 1 (logic 1 level).

The total number of possible combinations of binary inputs to a gate is determined by

$$N = 2^n$$

Where

N is the number of possible combinations

n is the number of bits.

For two variables : $N = 2^2 = 4$ combinations

For three variables : $N = 2^3 = 8$ combinations

5.3.1 AND Gate

The AND gate is composed of two or more inputs and a single output, performs logical multiplication. The standard symbol for the AND gate is shown in Fig.5.1 and its truth table is shown in Table 5.1. The logical operation of the AND gate is such that the output is HIGH (1) when all the inputs are HIGH, otherwise it is LOW (0). The AND Gate expression represent the function $Y = A.B$.

Table 5.1 Truth table for AND gate

Input		Output
A	B	$Y=A.B$
0	0	0
0	1	0
1	0	0
1	1	1

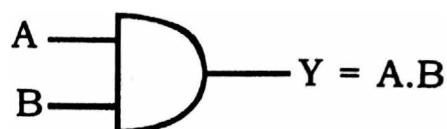


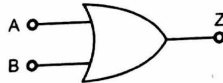
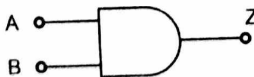
Fig.5.1 Standard logic symbol for AND gate

5.3.2 OR Gate

OR gate, which is composed of two or more inputs and a single output, performs logical addition. The standard symbol for the OR gate is shown in Fig. 5.2 and its truth table is shown in Table 5.2. The logical operation of the OR gate is such that the output is HIGH (1) when any of the inputs are HIGH, otherwise it is LOW (0). OR Gate expression represent the function $Y=A+B$.

39.12 SUMMARY OF LOGIC GATES

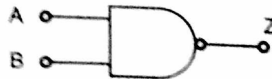
39.12 SUMMARY OF LOGIC GATES

Gate	Symbol	Function	Truth Table																		
OR		$Z = A \text{ OR } B$ $= A + B$	<table> <tr> <th colspan="2">Inputs</th><th>Output</th></tr> <tr> <th>A</th><th>B</th><th>Z</th></tr> <tr> <td>0</td><td>0</td><td>0</td></tr> <tr> <td>0</td><td>1</td><td>1</td></tr> <tr> <td>1</td><td>0</td><td>1</td></tr> <tr> <td>1</td><td>1</td><td>1</td></tr> </table>	Inputs		Output	A	B	Z	0	0	0	0	1	1	1	0	1	1	1	1
Inputs		Output																			
A	B	Z																			
0	0	0																			
0	1	1																			
1	0	1																			
1	1	1																			
AND		$Z = A \text{ AND } B$ $= A \cdot B$	<table> <tr> <th colspan="2">Inputs</th><th>Output</th></tr> <tr> <th>A</th><th>B</th><th>Z</th></tr> <tr> <td>0</td><td>0</td><td>0</td></tr> <tr> <td>0</td><td>1</td><td>0</td></tr> <tr> <td>1</td><td>0</td><td>0</td></tr> <tr> <td>1</td><td>1</td><td>1</td></tr> </table>	Inputs		Output	A	B	Z	0	0	0	0	1	0	1	0	0	1	1	1
Inputs		Output																			
A	B	Z																			
0	0	0																			
0	1	0																			
1	0	0																			
1	1	1																			
NOT		$Z = \text{NOT } A$ $= \overline{A}$	<table> <tr> <th>Inputs</th><th>Output</th></tr> <tr> <th>A</th><th>Z</th></tr> <tr> <td>0</td><td>1</td></tr> <tr> <td>1</td><td>0</td></tr> </table>	Inputs	Output	A	Z	0	1	1	0										
Inputs	Output																				
A	Z																				
0	1																				
1	0																				
NOR		$Z = A \text{ NOT OR } B$ $= \overline{A \text{ OR } B}$ $= \overline{A + B}$	<table> <tr> <th colspan="2">Inputs</th><th>Output</th></tr> <tr> <th>A</th><th>B</th><th>Z</th></tr> <tr> <td>0</td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1</td><td>0</td></tr> <tr> <td>1</td><td>0</td><td>0</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </table>	Inputs		Output	A	B	Z	0	0	1	0	1	0	1	0	0	1	1	0
Inputs		Output																			
A	B	Z																			
0	0	1																			
0	1	0																			
1	0	0																			
1	1	0																			

(Contd...)

Logic Gates and Boolean Algebra

897

Gate	Symbol	Function	Truth Table																		
NAND		$Z = A \text{ NOT AND } B$ $= A \text{ NAND } B$ $= \overline{A \cdot B}$	<table border="1"> <thead> <tr> <th colspan="2">Inputs</th> <th>Output</th> </tr> <tr> <th>A</th> <th>B</th> <th>Z</th> </tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table>	Inputs		Output	A	B	Z	0	0	1	0	1	1	1	0	1	1	1	0
Inputs		Output																			
A	B	Z																			
0	0	1																			
0	1	1																			
1	0	1																			
1	1	0																			
EX-OR		$Z = A \text{ EX-OR } B$ $= A \oplus B$ $= \overline{A}B + A\overline{B}$	<table border="1"> <thead> <tr> <th colspan="2">Inputs</th> <th>Output</th> </tr> <tr> <th>A</th> <th>B</th> <th>Z</th> </tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table>	Inputs		Output	A	B	Z	0	0	0	0	1	1	1	0	1	1	1	0
Inputs		Output																			
A	B	Z																			
0	0	0																			
0	1	1																			
1	0	1																			
1	1	0																			
EX-NOR		$Z = \overline{A \text{ EX-OR } B}$ $= A \text{ EX-NOR } B$ $= \overline{A \oplus B}$ $= AB + \overline{A} \overline{B}$	<table border="1"> <thead> <tr> <th colspan="2">Inputs</th> <th>Output</th> </tr> <tr> <th>A</th> <th>B</th> <th>Z</th> </tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>1</td><td>1</td></tr> </tbody> </table>	Inputs		Output	A	B	Z	0	0	1	0	1	0	1	0	0	1	1	1
Inputs		Output																			
A	B	Z																			
0	0	1																			
0	1	0																			
1	0	0																			
1	1	1																			

5.11 LAWS OF BOOLEAN ALGEBRA

Commutative Laws

The commutative law of addition for two variables is algebraically expressed as

$$A + B = B + A$$

The commutative law of multiplication for two variables is expressed as

$$AB = BA$$

Associative Laws

The associative law of addition of three variables is expressed as

$$A + (B + C) = (A + B) + C$$

The associative law of multiplication of three variables is expressed as

$$A(BC) = (AB)C$$

Distributive Law

The distributive law of three variables is expressed as

$$A(B + C) = AB + AC$$

This law states that OR ing several variables and AND ing the result is equivalent of AND ing the single variable with each of the variables in the grouping, then ORing is the result.

5.12 RULES OF BOOLEAN ALGEBRA

By combining the laws of Boolean algebra, we form several useful rules that may be used in manipulating and simplifying Boolean algebra expressions. Table 5.15, note that in each case, A, B, or C can either represent a single variable or a combination of variables.

Table 5.15 Rules of Boolean Algebra

Rule Number	Boolean Expression
1	$A + 0 = A$
2	$A + 1 = 1$
3	$A \cdot 0 = 0$
4	$A \cdot 1 = A$
5	$A + A = A$
6	$A + \overline{A} = 1$
7	$A \cdot A = A$
8	$A \cdot \overline{A} = 0$
9	$\overline{\overline{A}} = A$
10	$A + AB = A$
11	$A + B = B + A$
12	$(A + B)(A + C) = A + BC$

Among these rules, in particular rules 1 to 9, are easily verified using truth tables. Let us examine two methods by which we can prove the relationships of rules 10-12. First we use the laws and rules of Boolean algebra. Second we employ the use of truth tables.

Example 5

Simplify the boolean function

$$A + AB$$

Solution:

Method 1:

$$\begin{aligned} A + A.B &= A(1+B) \\ &= A \cdot 1 \\ &= A \end{aligned}$$

Method 2:

A	B	AB	A + AB
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1

Example 6

Simplify the Boolean Function

$$(A + B)(A + C)$$

Method 1:

$$\begin{aligned} (A + B)(A + C) &= AA + AC + BA + BC \\ &= A + AC + BA + BC \\ &= A(1 + C) + BA + BC \\ &= A \cdot 1 + BA + BC \\ &= A + BA + BC \\ &= A(1 + B) + BC \\ &= A \cdot 1 + BC \\ &= A + BC \end{aligned}$$

Method 2:

Table illustrating $(A + B)(A + C) = A + BC$

A	B	C	A + B	A + C	(A + B)(A + C)	BC	A + BC
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	1	1	1	1
1	0	0	1	1	1	0	1
1	0	1	1	1	1	0	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

5.13 DE MORGAN'S THEOREM

For N variables, DeMorgan's theorems are expressed in the following formulas:

Theorem 1

The complement of the product is equivalent to the sum of the complements

$$\overline{ABC \dots N} = \overline{A} + \overline{B} + \overline{C} + \dots + \overline{N}$$

Theorem 2

The complement of the sum is equivalent to the product of the complements

Example 7

Applying Demorgan's theorem to the expression $\overline{(A + B + C)D}$

$$\overline{(A + B + C)D} = \overline{A + B + C} + \overline{D}$$

$$= \overline{A} \overline{B} \overline{C} + \overline{D}$$

Example 8

Applying Demorgan's theorem to the expression $\overline{ABC + DEF}$

$$\overline{ABC + DEF}$$

Solution:

$$\begin{aligned}\overline{ABC + DEF} &= (\overline{ABC})(\overline{DEF}) \\ &= (\overline{A} + \overline{B} + \overline{C})(\overline{D} + \overline{E} + \overline{F})\end{aligned}$$

5.14 LOGIC CIRCUITS

Recall that the functions associated with logic gates to describe their operation. Accordingly, the interconnected assembly of these gates may be used to create any logic function. The resulting circuit formed by the grouping of these gates is referred to as a logic circuit. The truth tables for logic circuits are constructed by applying each of the possible $N=2^n$ input combinations to the circuit inputs and determining the corresponding output. Figure 5.7 illustrates the logic circuit diagram for the function $X = AB + B$ and the corresponding truth table is listed in Table 5.16.

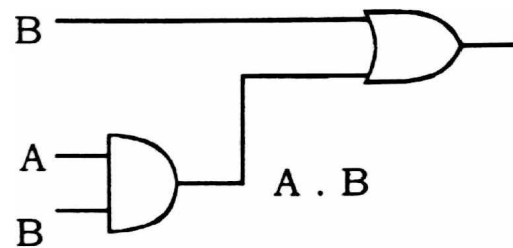


Fig. 5.7 Logic circuit for Function $X = AB + B$

Table 5.16 Truth table for Function $X = AB + B$

Input		Output
A	B	$X = AB + B$
0	0	0
0	1	1
1	0	0
1	1	1

Example 9

Simplify the expression $Y = AB + AB \cdot (A+C)$

Solution :

$$\begin{aligned}
 Y &= AB + AB \cdot (A + C) \\
 &= AB + AB \cdot (A + C) \\
 &= AB + ABA + ABC \\
 Y &= AB(1+A) + ABC \quad (1 + A = 1) \\
 &= AB(1) + ABC \\
 &= AB + ABC \\
 &= AB(1+C) \\
 Y &= AB
 \end{aligned}$$

Example 10

Simplify the given logic circuit shown in fig. Ex. 8(a) and implement the simplified logic circuit using logic gates.

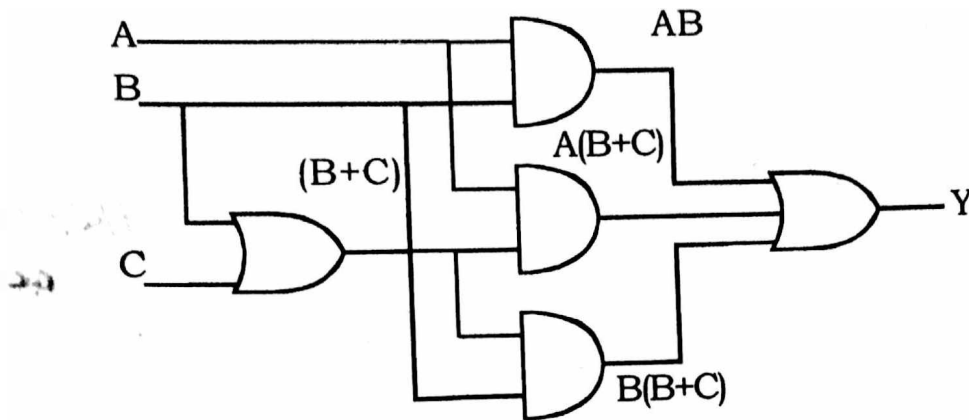


Fig. Ex. 8(a)

Solution :

From the circuit, the expression for output of logic circuit

$$\begin{aligned} Y &= AB + A(B + C) + B(B + C) \\ &= AB + AB + AC + BB + BC && (A + A = A) \\ &= AB + AC + B + BC && (B.B = B) \\ &= AB + AC + B(1 + C) && ((1 + C) = 1) \\ &= AB + AC + B \\ &= B(1 + A) + AC \\ Y &= B + AC \end{aligned}$$

Simplified logic circuit

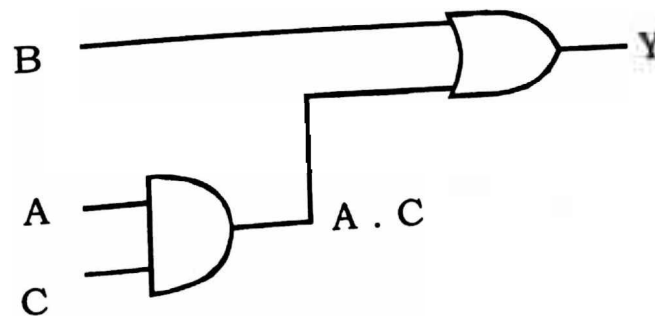


Fig. Ex. 8(b)

5.16 ADDERS [JAN - 2018]

Adder is a combinational logic circuit, which performs the arithmetic addition of binary digits. Adder produce two outputs namely sum and carry. Adder circuits can be classified into two types depending on the number of bits in the input variable.

1. Half adder and
2. Full adder

5.16.1 Half Adder

Half adder is a logic circuit, which is used to perform two bit binary addition. Figure 5.11(a) shows the block diagram of half adder and the Table 5.17 shows the truth table of half adder.

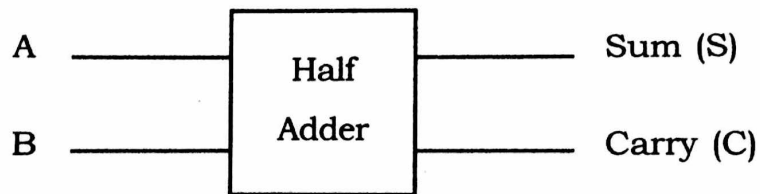


Fig. 5.11 (a) Block diagram of Half Adder

Table 5.17 Truth table of Half adder

Inputs		Outputs	
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Observe from Table 5.17 that we may derive logical expressions for the Sum (S) and carry output (C_o) bits. Namely, $C_o = AB$, and $S = A \oplus B$. From these two expressions, the implementation of the half adder function is conducted as illustrated in Fig.5.10(b).

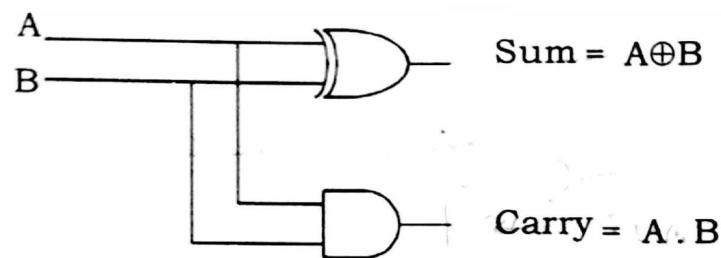


Fig. 5.11(b) Logic Circuit for Half Adder

5.16.2 Full Adder

Full adder is a combinational circuit which performs the arithmetic addition of three input bits. The logical symbol for the full-adder is shown in Fig.5.12(a) and the associated truth table is Table 5.18 .

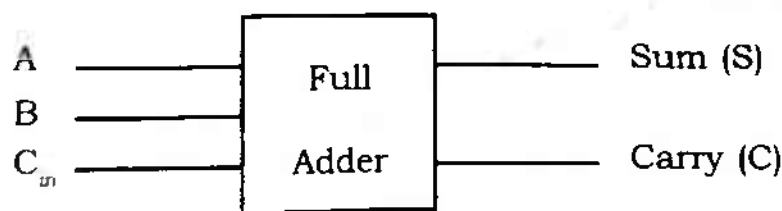


Fig. 5.12(a) Block diagram of Full Adder

Recall that for the half-adder the sum of A and B is the XOR of those two variables,. Similarly, for the three variables A, B and C_{in} the sum becomes . This is illustrated in Fig.5.11(b).

Table 5.18 Truth table of Full Adder

Inputs			Outputs	
A	B	C_{in}	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

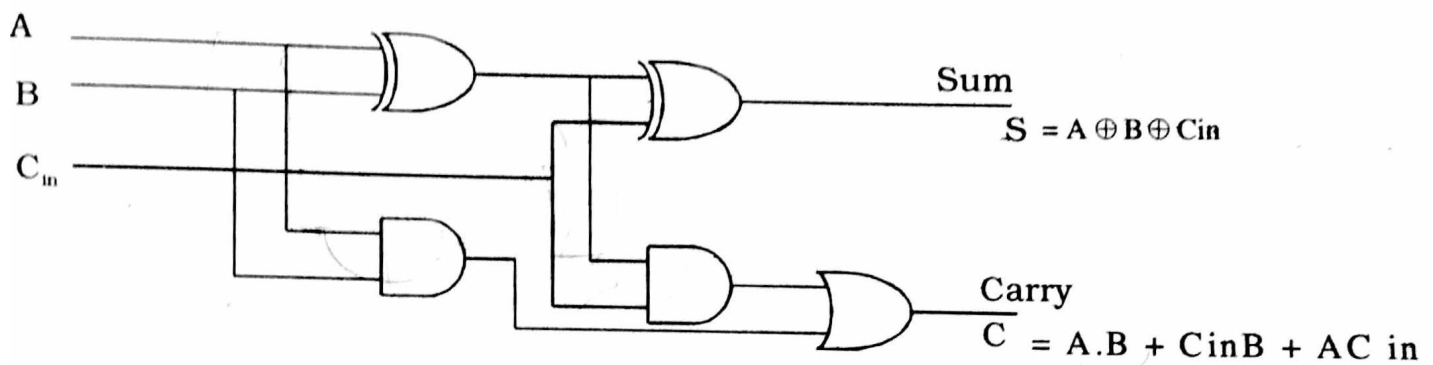


Fig. 5.12(b) Logic Circuit for the full Adder

5.17 SUBTRACTORS [APRIL - 2017]

Subtractor is a combinational logic circuit, it performs the subtraction operation. The subtractor will produce a difference and borrow. Subtractor is classified based on the number of bits performed.

1. Half Subtractor and
2. Full Subtractor

5.17.1 Half Subtractor

A half subtractor is a combinational logic circuit that performs subtraction involving two bits and produce their outputs as, difference and borrow. The half subtractor needs two inputs : minuend and subtrahend bits and two output : Borrow and difference Fig. 5.13(a) shows the block diagram of half subtractor and Table 5.19 shows the truth table of half subtractor.

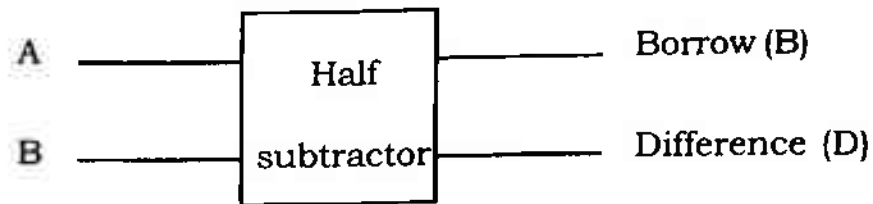


Fig. 5.13(a) Block Diagram for Half Subtractor

Table 5.19 Truth table of Half subtractor

Inputs		Output	
A	B	Difference(D)	Borrow(B)
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Logic Diagram

The logic diagram is constructed for the following expression by using logic gates, it is shown in Fig. 5.13(b)

$$\text{Difference} = A \oplus B$$

$$\text{Borrow} = \bar{A}B$$

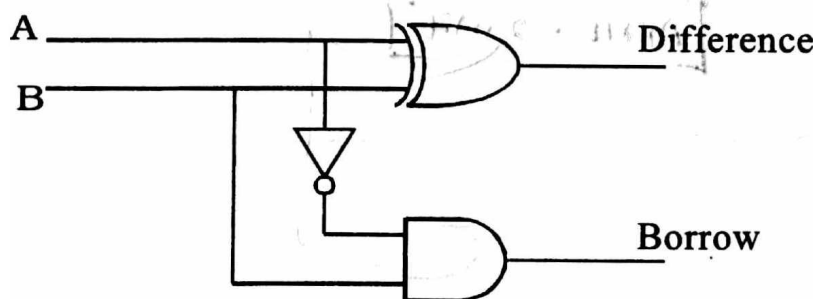


Fig. 5.13(b) Logic diagram of Half Subtractor

5.17.2 Full Subtractor

A full subtractor is a combinational logic circuit that performs subtraction involving three bits, namely minuend bit, subtrahend bit and the borrow from the previous stage. Figure 5.13(a) shows the block diagram of full subtractor and table 5.20 shows the truth table for full subtractor.

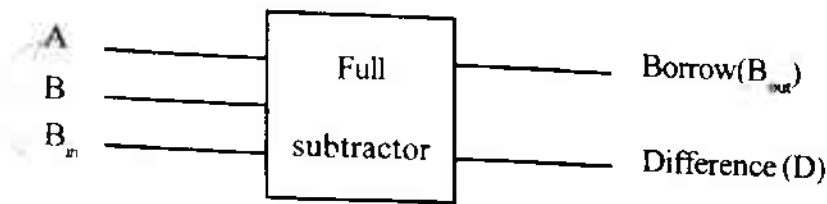


Fig. 5.14 (a) Block diagram of Full Subtractor

Table 5.20 Truth table for Full subtractor

Inputs			Outputs	
A	B	B _{in}	D	B _{out}
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

The full subtractor can be implemented with two half subtractor and one OR gate, as shown in fig. 5.14(b). The difference output from the second half subtractor is the exclusive OR of B_{in} and the output of the first half subtractor, which is same as difference output of full subtractor.

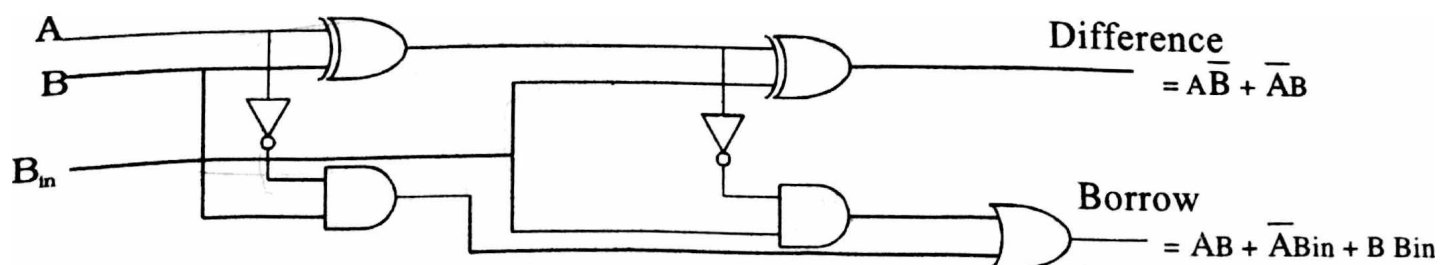


Fig.5.14(b) Logic Diagram for full Subtractor

5.27 FLIP FLOPS

Flip flops are synchronous bistable storage devices capable of storing one bit. In this case, synchronous means that the output state only changes at a specified point on a triggering input called the clock (C). That is, the output changes are synchronised with the clock signal.

5.27.1 RS Latch

The simplest type of flip flop is the Reset Set (RS) flip flop. It can be constructed using two NAND gates or two NOR gates. Figure 5.15 shows the basic RS flip flop using two NOR gates. The two NOR gates are cross coupled so that the output of NOR gate 1 is connected to one of the inputs of NOR gate 2 and vice versa.

The latch has two outputs Q and $\bar{Q}$, and two inputs, Set and Reset (S and R). The basic operation of flip flop as explained in the four cases as follows

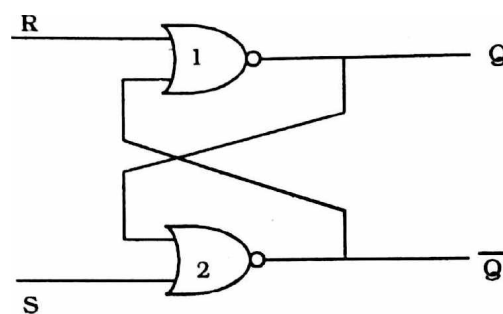


Fig. 5.15 Basic latch using NOR gates

Table 5.25 Truth Table for RS Flip flop using NOR Gate

R	S	Q	$\bar{Q}$	State
0	0	NC	NC	No change
0	1	1	0	Set
1	0	0	1	Reset
1	1	1/0	0/1	*

* Indeterminate condition

Case 1 $R = 0$ and $S = 1$

In this case, the previous outputs are assumed as $Q = 1$. So the S input of NOR gate 2 is at logic 1, hence its output, $\bar{Q}$ is at logic 0. Both inputs to NOR gate 1 are now at logic 0. So that its output, Q is at logic 1.

Case 2 $R = 1$ and $S = 0$

In this case, R input of NOR gate 1 is at logic 0, hence its output Q is at logic 1 when the previous outputs are considered as $Q = 0$ and $\bar{Q} = 1$. Both inputs of NOR gate 2 is logic 0. So that its output $\bar{Q}$ is at logic 1.

Case 3 $R = 0$ and $S = 0$

Let us assume that initially $Q = 0$ and $\bar{Q} = 1$. With $\bar{Q} = 0$ both inputs of NOR gate 1 are at logic 0. So its output Q is at logic 1. With $Q = 1$, one input of NOR gate 2 is at logic 1, hence its output $\bar{Q}$ is at logic 0. This shows that when S and R both are low, output state does not change.

Case 4 $R = 1$ and $S = 1$

When R and S both are at logic 1, they force the outputs of both NOR gates to the low state ($Q = 0$ and $\bar{Q} = 0$). So we call this an indeterminate or prohibited state. This condition also violates the basic definition of a flip flop requires $\bar{Q}$ to be the complement of Q. Thus in normal operation this condition must be avoided by making sure that 1's are not applied to the both inputs simultaneously. The truth table of basic flip flop using NOR gates is shown in Truth table 5.16 to simplify the operation. To make synchronisation of flip flop, clock is also applied to the input of flip flop. This type of flip flop is called clocked flip flop. Figure. 5.24 shows the clocked RS flip flop.

5.27.2 Latch using NAND Gates

The basic flip flop may be constructed using two NAND gates as shown in Fig. 5.25 and the truth table of this flip flop as shown in Table 5.17. In this case $\bar{S} = \bar{R} = 1$ makes the logic levels at the two gate outputs are the same (logic level 1) and hence is not used. Since NAND gates are employed, the inputs exercise control at logic level 0 to maintain the present state. That is, the flip flop changes state when either input is put to logic 0 or logic 1. Since this situation is inconsistent with the preference discussed earlier, we have labeled the control inputs $\bar{S}$ and $\bar{R}$ rather than S and R.

To understand how this circuit functions, recall that a low on any input to a NAND gate forces its output high. Thus a low on the $\bar{S}$ input will be set the flip flop ($Q = 1$ and $\bar{Q} = 0$). A low input of R will be reset it ($Q = 0$ and $\bar{Q} = 1$). If both $\bar{R}$ and $\bar{S}$ are high, the flip flop will remain in its previous state. When both $\bar{R}$ and $\bar{S}$ inputs are low, the output is unpredictable

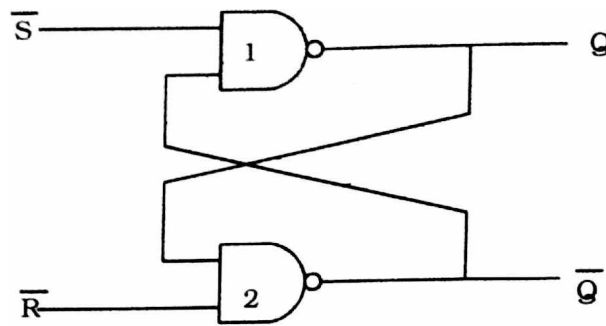


Fig. 5.17 RS Latch using NAND Gate

5.27.3 Clocked RS Flip Flop

It is not always desirable to allow a latch to change states at random times. In synchronous operation one pair of gates are connected as an RS flip flop. A second pair; called the steering gates, can be enabled by clock signal. This clock signal allows one or the other of these gates to pass a SET or RESET signal to the flip flop.

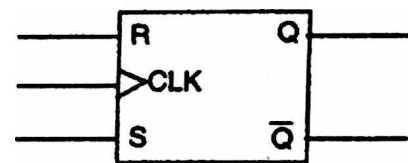
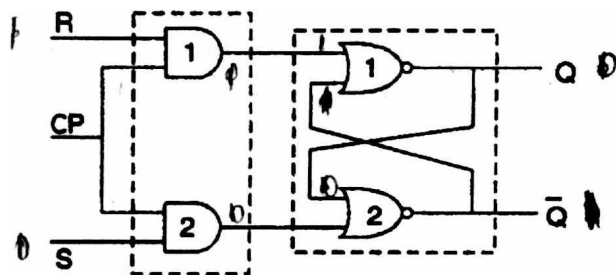


Fig. 5.18 (a) Clocked RS flip flop Fig.5.18 (b) Logic symbol of Clocked RS flip flop

The S and R inputs are free to change at random, but the flip flop output will change only when the clock is high. If the clock is LOW, the flip flop is disabled even if all the inputs appear at the input terminal. Fig. 5.18 (a) shows the clocked RS flip flop and Table shows the truth table of clocked rS flip flop. Fig. 5.18 (b) shows the logic symbol of clocked RS flip flop.

Table 5.26 Truth table of clocked RS flip flop

Clock CLK	Flip flop inputs		Outputs		States
	R	S	Q	Q	
0	X	X	Q	Q	No change
1	0	0	NC	NC	No change in the previous output
1	0	1	1	0	Set
1	1	0	0	1	Reset
1	1	1	0	0	Indeterminate state

Operation

When clock is absent or $CLK = 0$, the circuit will not work as RS flip flop. In the presence of clock i.e., $CLK = 1$ the operation of RS flip flop is explained for the following four cases.

Case I : If $R = 0$ and $S = 0$ and assume that $Q = 1$

The output of both AND gates are '0' which are given as one of the inputs to the NOR gate. If $Q = 1$ and $\bar{Q} = 0$, the output of NOR gate 2 is equal to 0, i.e. same as the previous states or values. This state is called 'no change' state.

Case II : If $R = 1$, $S = 0$

The output of AND gate 1 is equal to 1 AND gate 2 is '0' with these inputs, the NOR gate 1 output(Q) = 0 and NOR gate 2 output($\bar{Q}$) = 1. This state is called "Reset state".

Case III : If $R = 0$, $S = 1$

The output of AND gate 1 is '0' and AND gate 2 is 1. With these inputs, the NOR gate 1 output(Q) = 1 and NOR gate 2 output($\bar{Q}$) = 0. This is called "set state".

Case IV : If $R = 1$, $S = 1$

The output of both AND gates is 1 with these inputs, irrespective of the other input, the output of both NOR gates is '0', i.e., $Q = 0$ and $\bar{Q} = 0$, which is invalid output. This state is called 'indeterminate state'.

5.27.4 D Flip Flop (Delay Flip Flop)

Looking at the truth table of SR flip flop we can realize that when both inputs are same the output either does not change or it is unpredictable (Inputs - 00, no change and inputs - 11 unpredictable). In many practical applications, these input conditions are not required. These input conditions can be avoided by making them complement to each other. This modified clocked SR flip flop is known as D flip flop. As shown in Fig. 5.19, D input goes directly to the S input, and its complement is applied to the R input, through gate 5. Therefore, only two input conditions exist, either $S = 0$ and $R = 1$ or $S = 1$ and $R = 0$. When $D = 1$, $S = 1$ and $R = 0$ and when $D = 0$, $S = 0$ and $R = 1$. Therefore, during the occurrence of clock pulse if $D = 1$, the Q output is set and if $D = 0$, the output is reset. The truth table of clocked D flip flop is shown in Table 5.27.

Table 5.27 Truth table of Clocked D flip flop

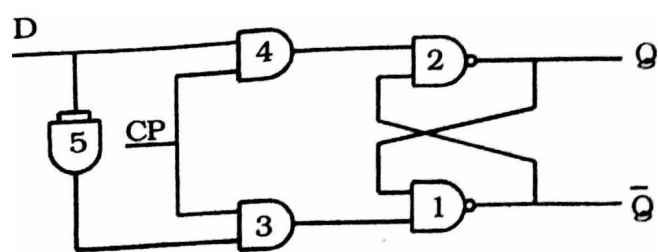


Fig. 5.19 D Flip Flop

CP	D	Q_{n+1}
0	x	Q_n
1	0	0
1	1	1

The output does not exist when the clock pulse is absent. Q_n is the binary state (present state) of the flip flop before the occurrence of a clock pulse. It is known as present state. Q_{n+1} is the state of the flip flop after the occurrence of a clock pulse which is known as next state.

5.27.5 JK Flip Flop

A JK flip flop is a refinement of the RS Flip flop. In JK flip flop the unpredictable state in the RS flip flop is defined inputs J and K behave like inputs S and R to set and reset the flip flop respectively as shown in Fig. 5.20.

Case 1

When J and K both are low, both AND gates are disabled. Therefore, clock pulses have no effect and, Q and $\bar{Q}$ retain their last values. The truth table of JK FF as shown in Table 5.28.

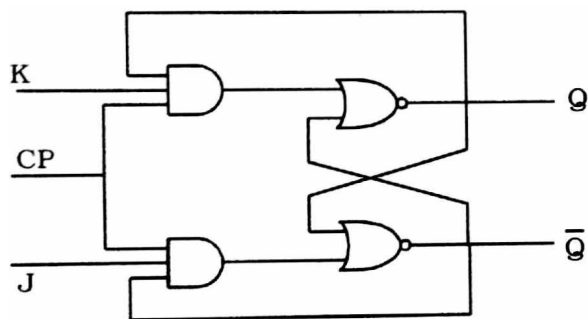


Table 5.28 Truth table of Clocked JK flip flop

CP	J	K	Q_{n+1}
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	toggle

Fig. 5.20 Logic circuit for Clocked JK Flip Flop

Case 2

When J is low and K is high. The lower AND gate is disabled. So there is no way to set the flip flop. The only possibility is reset. When K is high the upper gate passes a Reset trigger as shown as the next possible clock pulse arrives. This forces Q to become low. Therefore $J = 0$ and $K = 1$ means that the next positive clock pulse resets the flip flop unless Q is already reset.

Case 3

When J is high and K is low, the upper gate is disabled, so there is no way to reset the flip flop. The only possibility is to set the flip flop if it is not previously set with $Q = 0$, Q is high and hence the lower gate passes a SET trigger on the next positive clock pulse. This drives Q into the high state. Therefore, $J = 1$ and $K = 0$ means that the next positive clock pulse set the flip flop unless Q is already set.

Case 4

When J and K both are high (Recall that this is an indeterminate condition with an RS flip flop) it's possible to set or reset the flip flop. If Q is high, the upper gate passes a RESET trigger on the next positive clock edge. On the otherhand, when Q is low, the lower gate passes a SET trigger on the next positive clock edge. Either way, Q changes to the complement of the last state. Therefore, $J = K = 1$ means output of the flip flop will toggle on the next positive clock edge.

5.28 COUNTER

A counter is used to count the number of clock pulses given. A counter is constructed by chain of flip flops. These flip flops go through the sequence of states, when pulses are applied to its clock inputs. This is the main difference between a register and a counter. (The counter is used to count the number of clock pulse though the sequence may be ascending, descending and non sequence).

5.28.1 Classification Counters

There are two types of counters

1. Synchronous counter

In synchronous counter, the common clock input is connected to the flip flops and all the flip flops are clocked simultaneously.

2. Asynchronous counter

In asynchronous counter (called ripple counter), the first flip flop is clocked by the external clock pulse and each successive flip flops are clocked from the previous flip flop outputs (may Q or $\bar{Q}$). Therefore the asynchronous counter is not clocked simultaneously.

5.28.2 Synchronous Vs Asynchronous Counter

Table 5.29 shows the comparison between synchronous and asynchronous counters

Synchronous counter	Asynchronous counter
1. In this type of counters there is no connection of first flip flop output to clock input of the next flip flop.	1. In this counter flip flops are connected in such a way that the first flip flop output is used as a clock for the next flip flop
2. All the flip flops are clocked simultaneously	2. All the flip flops are not clocked simultaneously
3. Designing of logic circuits becomes complex for more number of states	3. Design is very simple even for more number of states
4. High speed	4. Low speed

5.29 RIPPLE COUNTER (ASYNCHRONOUS COUNTER) [JAN-2016, 17]

If starting from the initial state, the counter returns to the initial state after completed sequence of the input wave form, the counter is called as modulo (modulus n) counter or a base r counter. The ripple counter as shown in Fig.5.21 with timing diagram. Three JK flip flops are connected for the toggling mode ($J = K = 1$ or connect with $+5V$) the timing diagram shown in Fig. 5.22. As per the definition the first flip flop is clocked by the external clock pulse and remaining flip flops are clocked by using the previous flip flop outputs.

We assume that the flip flops toggle on the negative going transition of the wave form at the clock input. Each flip flop changes the state when the preceding flip flop or input wave form makes changes from logic 1 to logic 0. Considering the initial state $Q_A Q_B Q_C = 000$.

The count sequence of the counter can be read simply (000, 001, 010, 011, 100, 101, 110, 111)

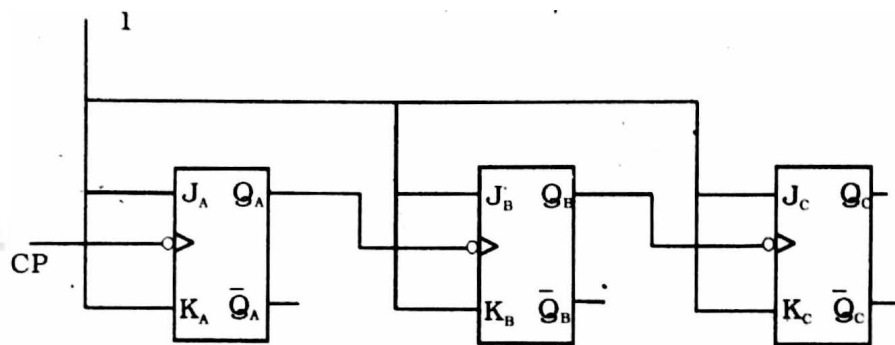


Fig. 5.21 3 - bit Ripple Counter

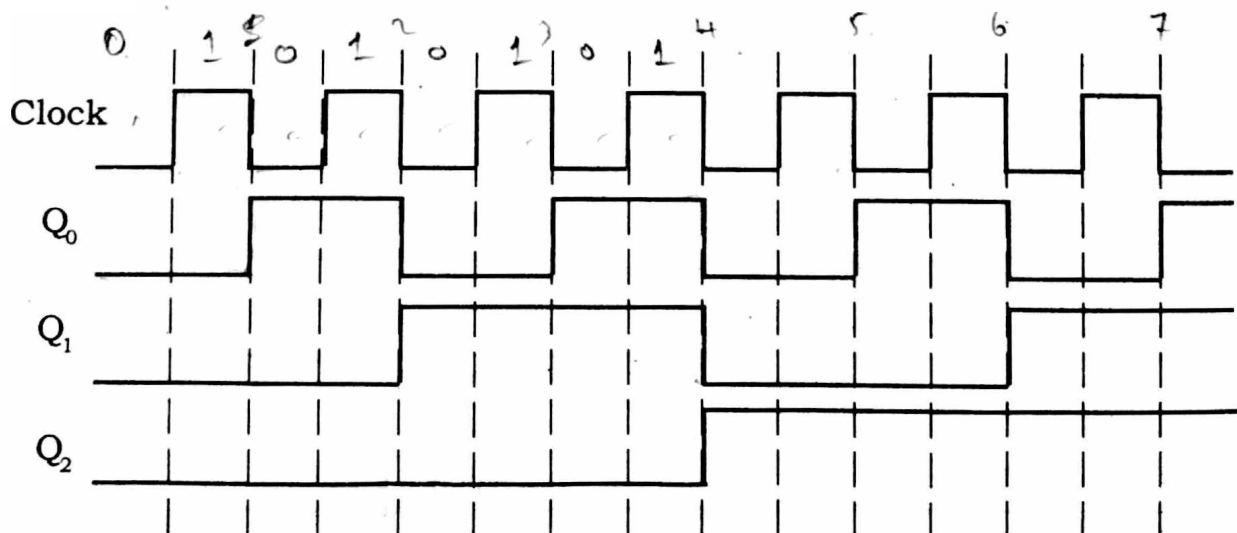


Fig. 5.22 Timing diagram for 3 bit Ripple Counter

5.30 ASYNCHRONOUS BCD COUNTER [APRIL-2017]

Counters can be designed to have a number of states in their sequence that is less than 2^n . The resulting sequence is called a *truncated sequence*. One common modulus for counters with truncated sequences is ten. Such counters are referred to as *decade* counters. Furthermore, a counter which counts a sequence of zero (0000) through nine (1001) is referred to as a *BCD decade* counter because its ten state sequence is the BCD code.

To obtain a truncated sequence it is necessary to force the counter to cycle before going through all of its normal states. For example, the BCD decade counter must recycle back to the state 0000 after state 1001. Note that a decade counter will require 4 flip-flops.

One method that can be used to achieve recycling after the count of nine (1001) is to decode ten (1010) with a NAND gate and connect the output of the NAND gate to the clear inputs of all the flip-flops as illustrated in Figure 5.23.

Notice that only Q_2 and Q_4 are connected to the NAND gate inputs. This arrangement is an example of *partial decoding*, in which two unique states ($Q_2=1$ and $Q_4=1$) are enough to decode the count of 1010. This is because none of the other states (zero through nine) have Q_2 and Q_4 high at the same time. Consequently, when the counter goes into count ten (1010) the decoding gate output goes LOW and asynchronously Resets all of the flip-flops.

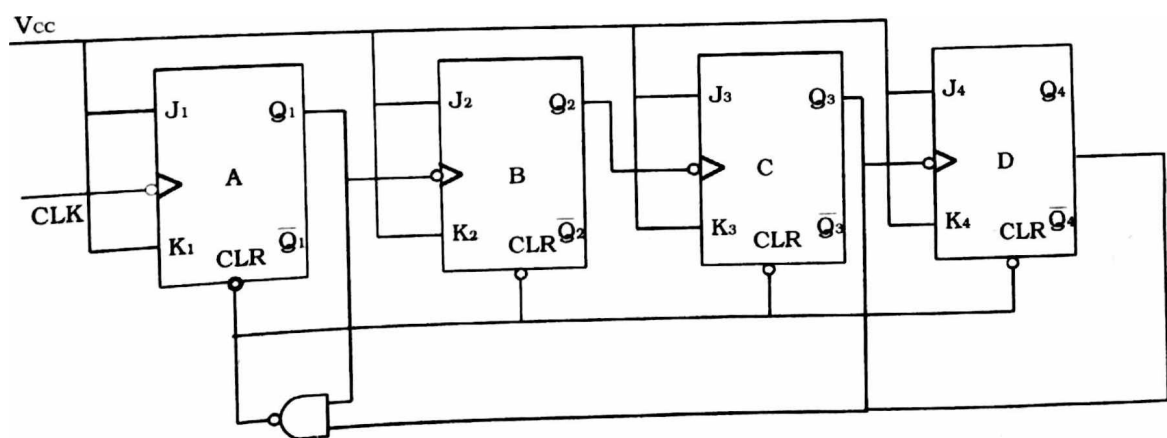
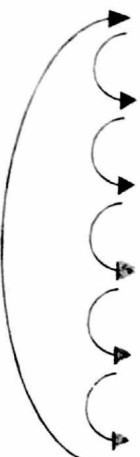
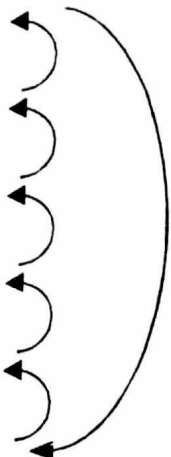


Fig.5.23 An asynchronous BCD counter

5.31 UP/DOWN SYNCHRONOUS COUNTER

An up/down counter is one that is capable of progressing in ascending order or descending order through a certain sequence. An up/down counter is also called bidirectional counter. Usually up/down operation of the counter is controlled by UP/DOWN signal. When this signal is HIGH, counter goes through UP sequence, i.e., 0,1,2,3,...,n. When UP/DOWN signal is LOW counter follows reverse sequence i.e. n, n-1, n-2,,1,0. For 3-bit counters these sequences are : 0,1,2,3,4,5,6,7 for up sequence and 7,6,5,4,3,2,1,0 for down sequence.

Table 5.30 Three bit UP/DOWN counter Table

CP	UP	Q _C	Q _B	Q _A	DOWN
0		0	0	0	
1		0	0	1	
2		0	1	0	
3		0	1	1	
4		1	0	0	
5		1	0	1	
6		1	1	0	
7		1	1	1	

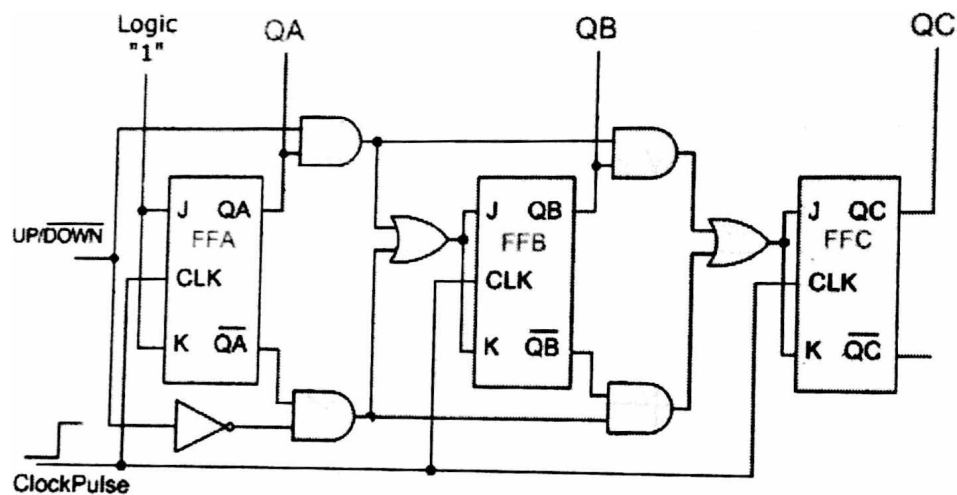


Fig.5.24 Logic diagram for synchronous up/down counter

The arrows in the Table 5.30 indicate the state-to-state movement of the counter for both its UP and its DOWN modes of operation. Figure 5.24 shows the three bit up/down counter.

5.32 SHIFT REGISTERS [APRIL - 2015, 17]

A register that is capable of shifting data one bit at a time is called a shift register. The logical configuration of a serial shift register consists of a chain of flip-flops connected in cascade, with the output of one flip-flop being connected to the input of its neighbour. The operation of the shift register is synchronous; thus each flip-flop is connected to a common clock. Using D flip-flops forms the simplest type of shift-registers.

The basic data movements possible within a four-bit shift register is shown in Fig. 5.25.

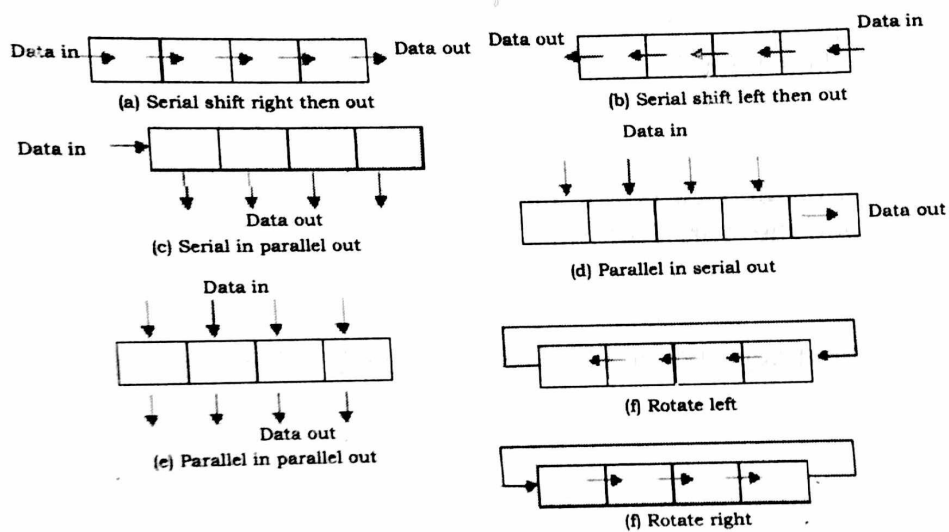
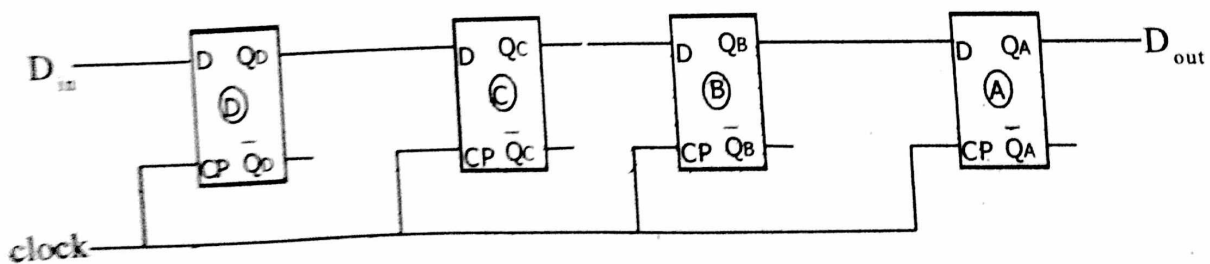


Fig. 5.25 Data movement diagrams

Figure 5.26 shows the circuit diagram for a four-bit serial in-serial out shift register implemented using D flip-flops. Assuming that the register is initially clear.



Chapter

6

Communication and Computer Systems

6.1 INTRODUCTION

Communication is the transfer of information from one place to another place. Distribution of data, message or information from one location to another location with high reliability and security is the role of communication system.

6.2 ELEMENTS OF COMMUNICATION SYSTEM (COMMUNICATION MODEL)

Communication System consists of the following components which acts together to accomplish information transfer or exchange. Fig.6.1 shows the block diagram of the communication system.

1. Input Transducer

2. Transmitter

3. Channel

4. Receiver and

5. Output Transducer

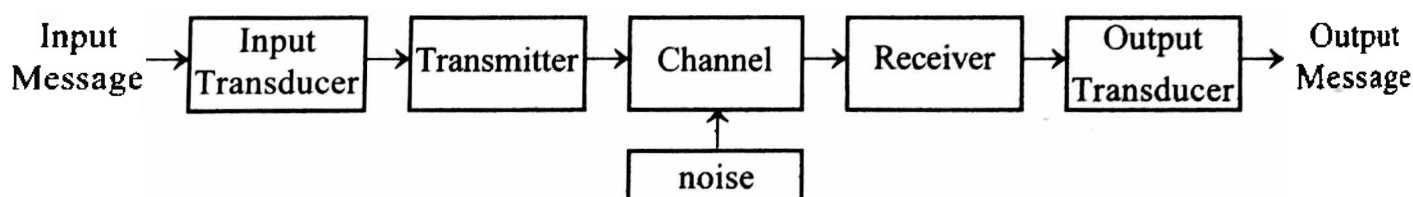


Fig.6.1 Block diagram of communication system

Input Transducer

The input message produced by a source must be converted by a transducer to a suitable form such that it is accepted by the transmitter.

In electrical communication, speech signals are converted to voltage variation by a microphone.

Transmitter

The transmitter processes the input signal to produce a transmitted signal suited to the characteristics of the transmission channel.

Signal processing for transmission always involves modulation. In addition to modulation, other functions performed by the transmitter are amplification, filtering and coupling the modulated signal to the channel.

Channel

The channel can have different forms: the atmosphere or free space, coaxial cable, fiber optic, waveguide, etc.

The signal undergoes some amount of degradation in the channel due to noise interference and distortion resulting from band limitation and nonlinearities.

Types of channel :

1. Wired - eg. Twisted pair, coaxial cable, optical fiber
2. Wireless - eg. Microwave, satellite, mobile

Receiver

The receiver function is to extract the desired signal from the received signal at the channel output and to convert it to a form suitable for the output transducer.

Other functions of the receiver are amplification (the received signal may be extremely weak), demodulation and filtering.

Output Transducer

The function of the output transducer is to convert the electric signal at its input into the form desired by the system user.

For eg. Loudspeaker, acts as a output transducer to convert received electrical signal to voice signal.

6.3 ANALOG AND DIGITAL COMMUNICATION SYSTEM

There are many kinds of information sources, which can be divided into two distinct message categories, analog and digital.

An **analog** message is a physical quantity that varies with time, usually in a smooth and continuous fashion.

Since the information resides in a time-varying waveform, an analog communication system should deliver this waveform with a specified degree of **reliability** or **fidelity**.

A **digital** message is an ordered sequence of symbols selected from a finite set of discrete elements.

Since the information resides in discrete symbols, a digital communication system should deliver these symbols with a specified degree of **accuracy** in a specified amount of time.

6.3.1 Comparison between Analog and Digital Communications

Table 6.1 Comparison of digital and analog communication systems

Digital Communication System	Analog Communication System
Advantages : 1. Inexpensive digital circuits 2. Privacy preserved (data encryption) 3. Can merge different data (voice, video and data) and transmit over a common digital transmission system 4. Error correction is possible by Coding	Disadvantages: 1. Expensive analog components: L & C. 2. No privacy 3. Cannot merge data from different sources 4. No error correction capability
Disadvantages: 1. Larger bandwidth synchronization problem is relatively difficult	Advantages : 1. Smaller bandwidth synchronization problem is relatively easier

6.4 MODULATION

Modulation is the process by which the characteristics of carrier wave is varied in accordance with the instantaneous value of message signal.

Figure 6.2 shows the block diagram of modulation process.

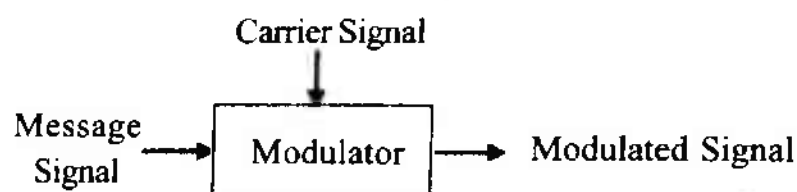


Fig. 6.2 Block diagram of modulation processes

The characteristics of high frequency carrier, which has been modified to carry the lower frequency information, this process is called modulation.

Once this information is received, the lower frequency information must be recovered from the high frequency carrier. This process is called demodulation.

6.4.1 Need for modulation

1. Ease of radiation of the signals are translated to high frequencies, it becomes relatively easier to design amplifier circuit as well as antenna system at these increased frequencies.
2. To overcome hardware limitations.
3. Transmitting such lower frequencies require antennas with miles in wavelength.
4. Modulation to reduce noise which result in the optimization of Signal to Noise ratio (S/N)
5. To minimize the effects of interference

6. For multiplexing and frequency assignment
7. For efficient radio transmission
8. To reduce antenna height

6.4.2 Classification of Modulation

Classification of modulation by the nature of modulating signal

1. **Analog modulation** - the modulating signal is an analog signal
Examples: AM, FM, Phase Modulation (PM), Pulse Amplitude Modulation
2. **Digital modulation** - the modulating signal is a digital signal.
Examples: Amplitude Shift Keying (ASK), FSK, Phase Shift Keying (PSK)

Classification of modulation by the nature of carrier

1. **Continuous wave (CW) modulation** - Carrier is a sinusoidal wave
Examples: AM, FM, ASK, FSK, PSK
2. **Pulse modulation** - Carrier is a train of pulses
Examples : Pulse amplitude modulation (PAM), Pulse width modulation (PWM), Pulse position modulation (PPM)

Advantage of continuous wave (cw) modulation

1. It produces frequency translation ie. converted from one frequency to another frequency range. Thus, the CW modulation process will have all the benefits of frequency translation.
2. The effect of noise and interference can be reduced by using large transmission bandwidth.

6.5 ANALOG MODULATION

Modulation is the process by which the characteristics of carrier wave is varied in accordance with the instantaneous values of message signal

Let a sinusoidal carrier wave in analog modulation is given by the carrier signal below

$$S_c(t) = A_c(t) \cos(2\pi f_c t + \phi)$$

where

$A_c(t)$ = carrier amplitude

f_c = carrier frequency

ϕ = carrier phase angle

The above parameters may be varied for the purpose of transmitting information giving respectively the modulation methods; namely

1. Amplitude Modulation (AM)

It is the process of by which the amplitude of carrier wave is varied in accordance with the instantaneous value of message signal is known as amplitude modulation.

2. Frequency Modulation (FM)

It is the process of by which the frequency of carrier wave is varied in accordance with the instantaneous value of message signal is known as frequency modulation.

3. Phase Modulation (PM)

It is the process of by which the phase of carrier wave is varied in accordance with the instantaneous value of message signal is known as phase modulation.

6.5.1 TYPES OF COMMUNICATION

Based on the nature of communication channel, the process of communication may be broadly classified into two types.

1. Wire-line Communication system
2. Wireless communication system

6.6 WIRED COMMUNICATION

Definition : Reliable end to end connection is possible by means of wires. eg. Twisted pair, co-axial cable, optical fiber.

In a wire, these waves are induced and guided by an electrical current passing along with electrical conductor, but that is not the only way of propagating electromagnetic (EM) waves. By using a very strong electrical signal as a transmitting source and electromagnetic wave can be made to spread far and wide through the air. The radio waves are produced by transmitters, which consist of a radio wave source connected to some form of antenna. Figure 6.3 shows the wired communication system.

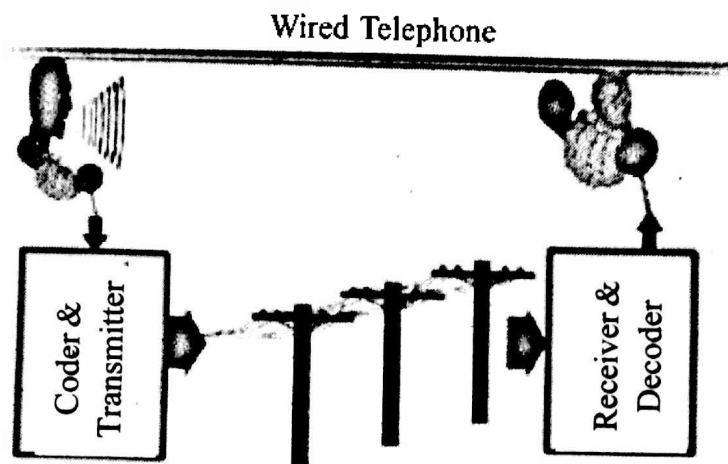


Fig. 6.3 Wired communication system

6.7 WIRELESS COMMUNICATION SYSTEM

The transmission channel is the main issue of communication system.

Conventionally, it is the set of hard wired cables that connect all the lines of the wire line phone.

In Wireless systems, the cables are replaced by free space, but only at the cost of requiring the erection of antennas that allow the line of sight communication.

Long distance communication is possible with wireless communication. Eg., Radio communication, TV reception.

Figure 6.4 shows the wireless communication system and the Fig 6.5 shows the block diagram of wireless communication system.

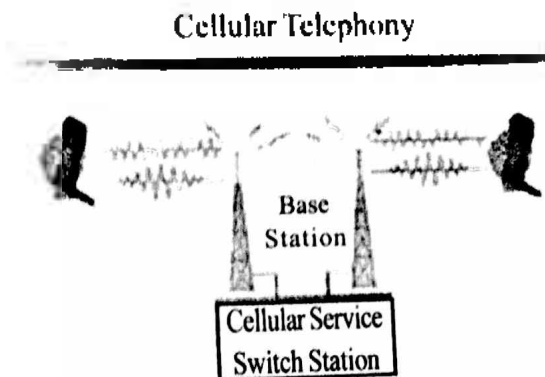


Fig.6.4 wireless communication system

An antenna is a transducer which converts the electrical energy to electromagnetic energy to ease transmission message through free space. At the receiver we make use of a receiving antenna and convert the electromagnetic signal back to electrical signal.

Without antenna wireless communication is not possible. The radiation properties of antenna are proportional directly to the wavelength of the signal. The wavelength of very low frequency signals are in order of kms and so to the size of the antenna to transmit them.

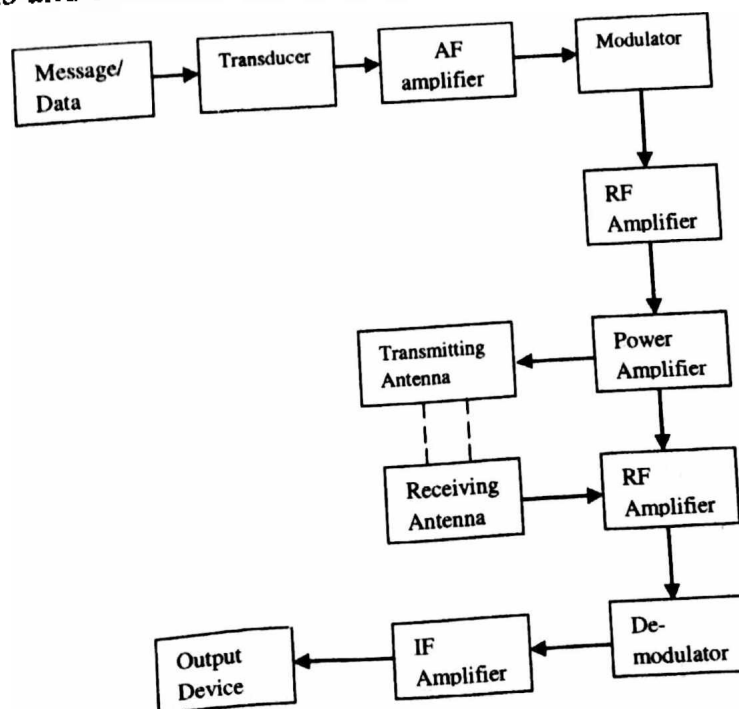


Fig 6.5 Block diagram for wireless communication system

6.7.1 Advantage of wireless communication

The majority of radio frequency (RF) technology has permitted the use of electromagnetic radiation links as the major trunk channel for long distance communication.

The use of microwave links has major advantages over cabling systems.

1. Freedom from land acquisition rights

The acquisition of rights to lay cabling, repair cabling, and permanent access to repeater stations is a major cost in the provision of cable communications.

The use of radio links, that require only the acquisition of the transmitter/receiver station, removes this requirement.

It also simplifies the maintenance and repair of the link.

2. Ease of Communication over difficult terrain

Some terrains make cable laying extremely difficult and expensive, even if the land acquisition cost is negligible.

6.7.2 Disadvantages of wireless communication

1. Bandwidth allocation is extremely limited

The competition for RF bandwidth from various competing users leads to very strict allocations of bandwidth.

Unlike cabling systems, that can increase bandwidth by laying more cables, the radio frequency (RF) bandwidth allocation is finite and limited.

2. Atmospheric effects

The use of free-space communication results in susceptibility to weather effects, particularly rain.

3. Transmission path needs to be clear

Microwave communication requires line-of-sight, point-to-point communication. The frequency of repeater stations is determined by the terrain.

Care must be taken in the system design to ensure freedom from obstacles.

In addition, links must be kept free of future constructions that could obstruct the link.

4. Interference and efficiency of propagation

The microwave system is open to RF interference.

The various propagation models used to understand the effect of terrain and weather, are used to aid the setup properties of the antennas (height, gain, power etc...)

5. Restrictive Costs

The cost of design, implementation and maintenance of microwave links is high.

6.7.3 Electromagnetic Spectrum for various Communication services

Table 6.2 shows the various frequency spectrum used for different communication services

Table. 6.2 Electromagnetic Spectrum for various Communication services

Classification	Frequency Range	Wavelength	Uses
Very low Frequencies (VLF)	10-30 kHz	30-10 km	Long Distance point- point communications
Low Frequencies(LF)	30-300 kHz	10-1 km	Long Distance point-point communications, Marine, Navigation, Power Line carrier communication and Broadcast
Medium Frequencies(MF)	300-3000 kHz	1000-100 meters	Power Line carrier communication, Broadcast, Marine communication, navigation and harbour telephone
High Frequencies(HF)	3-30 MHz	100-10 meters	Moderate and long range communication of all types, broadcast.
Very High Frequencies(VHF)	30-300 MHz	10-1 meters	Short distance communications, TV and FM broadcast, Data communication, Mobile and Navigation systems.
Ultra High Frequencies(UHF)	300-3000 MHz	1-0.1 meters	Short distance communications, TV and broadcast, Radar, Mobile and Navigation and microwave relay systems.
Super High Frequencies(SHF)	3-30 GHz	10-1 cm	Radar, Navigation and microwave relay systems.
Extremely High Frequencies(EHF)	30-300 GHz	1-0.1 cm	Radar, Satellite, Mobile, Navigation and microwave relay systems.

6.8 MICROWAVE COMMUNICATION [APRIL - 2016]

Electromagnetic waves on the frequency range of 1 GHz to 30 GHz are referred to as microwaves. Fig 6.6 shows the block diagram of microwave communication. Several signal channels are modulated into micro-wave carrier frequencies and transmitted to the repeater stations spaced 50 to 70 km apart. As microwaves travel only on line-of-sight paths, the transmitter and receiver should be visible to each other. Hence, it is necessary to provide repeater stations in between the terminal stations at about 50km intervals.

Repeater antennas are placed on roads and hilltops so that there will not be any obstruction in the line of sight between the repeater antennas. This system is called '**Line of sight microwave link**'. As microwave communication offers a large transmission bandwidth, many thousands of telephone channels along with a few TV channels can be transmitted over the same route using the same facilities. Normally carrier frequencies in the 3 to 12 GHz range are used for microwave communication. The transmitter output powers can be low because highly directional high gain antennas are used.

The general block diagram of line of sight transmission system is shown in Fig. 6.6

- The base signal is first conditioned using conditional circuits.
- The conditioned signal is amplified using amplifiers to increase the signal strength.
- The signal is fed into modulator to produce a frequency modulated wave.
- This signal is mixed with local oscillator signal.
- The output from the mixer is intermediate frequency (IF) wave and is fed into the transmitter where the IF wave is transferred into microwave range using microwave carrier.
- Then it is transmitted through the antenna after amplification.
- The receiver separates the IF signals and demodulates to recover the original data or message.

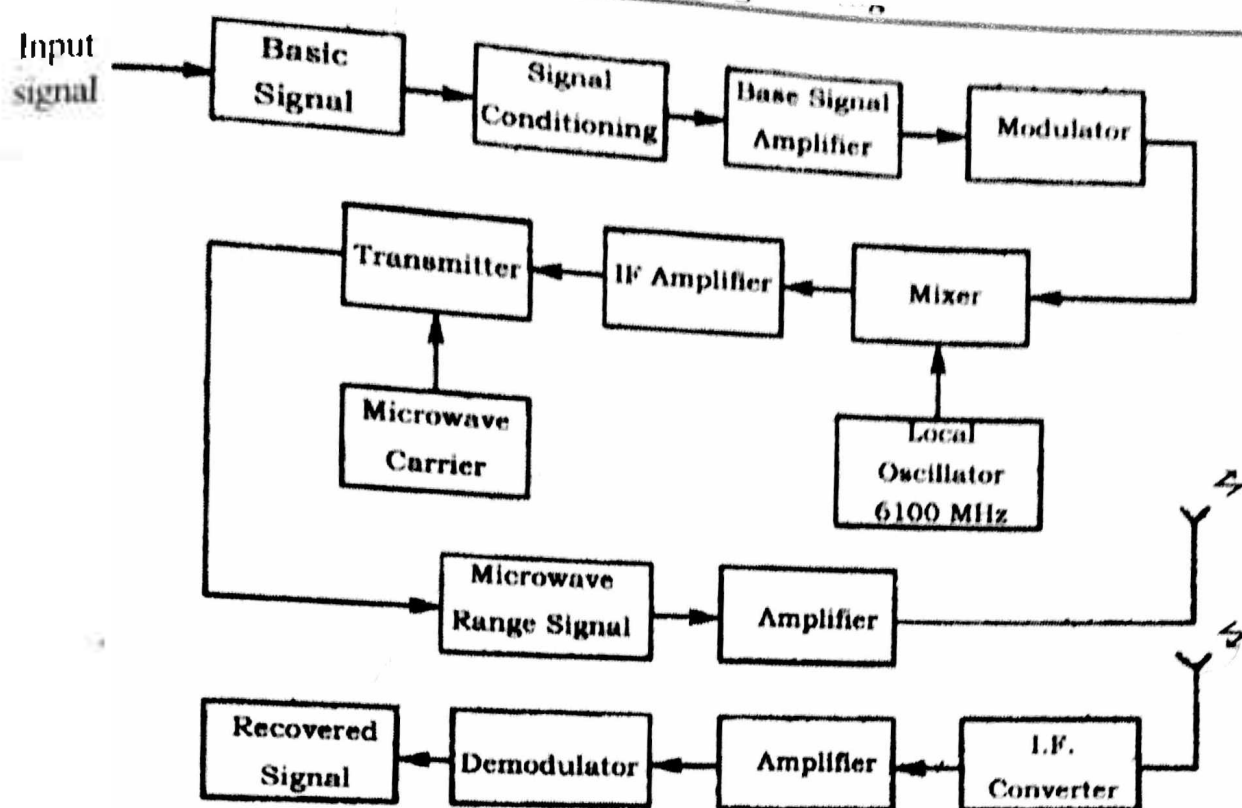


Fig. 6.6 Block diagram of a Microwave communication system

- Presently, microwave communications are widely used for telephone networks, in broadcast and television systems and in several other communication applications by services, railways, etc.,

6.9 SATELLITE COMMUNICATION [3A~2018]

The satellite is essentially a microwave station placed in orbital space. The broadcast signals are beamed up to the satellite from an earth station through the directional dish antenna that is synchronized to the position of the satellite, a device called transponder is used in the satellite to receive the weak signals and are amplified.

The amplified signal further retransmit the back to another earth station in a different location on the earth. The earth stations typically transmit their signals to satellites on carrier frequencies ranging from 5.92 to 6.43 GHz band called the **uplink frequency**.

The satellite transponders convert these signals to a range from 3.7 to 4.2 GHz band. These frequencies are referred to as **downlink frequencies**. In this orbit, the satellite is made to travel at a velocity equal to the rotation of earth in order to maintain a fixed position relative to a point above equator. The general diagram is shown in Fig 6.7.

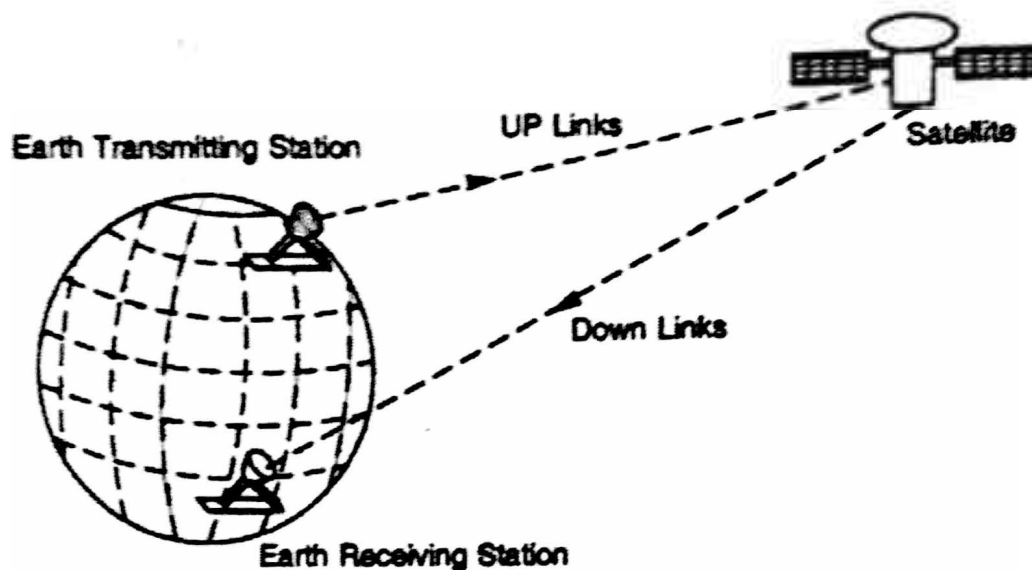


Fig. 6.7 General diagram of Satellite systems

A passive satellite simply reflects a signal back to earth and there are no gain devices on-board to amplify the signal. On the other hand, an active satellite receives, amplifies and retransmits the signal back towards earth.

6.9.1 Orbital Patterns

Once launched, a satellite remains in orbit because the centrifugal force caused by its rotation around the earth is counter-balanced by the earth's gravitational pull.

The closer to earth the satellite rotates, the greater the gravitational pull, and greater the velocity required to keep it from being pulled to earth.

6.9.2 Low Altitude Satellites

Close to earth (150 to 500km in height) travel at approx 28,500km/Hr. At this speed, it takes approx 90 minutes to rotate around the entire earth. Consequently, the time that the satellite is in the Line of sight (LOS) of a particular earth station is 15 minutes or less per orbit.

6.9.3 Medium Altitude Satellites

9,500 to 19,000 km in height have a rotation period of 5 to 12 Hrs and remain in LOS of a particular earth station for 2 to 4 Hrs per orbit.

6.9.4 High Altitude Satellites

High altitude, geosynchronous satellite is a satellite which is placed at a height of 35,786 km from the earth's surface and has an orbital velocity equal to that of the earth's.

A geosynchronous satellite that lies on the earth's equatorial plane is called as the geostationary satellite.

A geo-stationary satellite remains in a fixed position with respect to a given earth station and has 24 hrs availability time. Three geostationary satellites spaced 120° apart can cover the whole world. Based on the area of the coverage, the orbits are classified as

1. **Polar orbit** : When the satellite rotates in an orbit that takes it over the north and south poles, it is called polar orbit.
2. **Equatorial Orbit** : When the satellite rotates in an orbit above the equator, it is called an equatorial orbit
3. **Inclined orbit** : Any other orbital path is called an inclined orbit.

6.9.5 Satellite System

A satellite system, consists of three basic sections,

1. The uplink (transmitting earth station)
2. The satellite transponder
3. The downlink (receiving earth station)

Typical frequencies for telecommunication services in a satellite system are 6/4 GHz and 14/12 GHz,

Where 6 and 14 GHz represent uplink frequencies and 4 and 12 GHz represent downlink frequencies.

Uplink Model

A typical earth station transmitter consists of the following as shown in Fig 6.8

- ♦ **IF modulator**
It converts the input base band signals to either an FM, or a PSK modulated intermediate frequency.
- ♦ **An IF-to-RF microwave up converter**
It translates the IF to appropriate RF carrier frequency.
- ♦ **A high power amplifier (HPA)**
It provides adequate output power to propagate the signal to the satellite transponder.

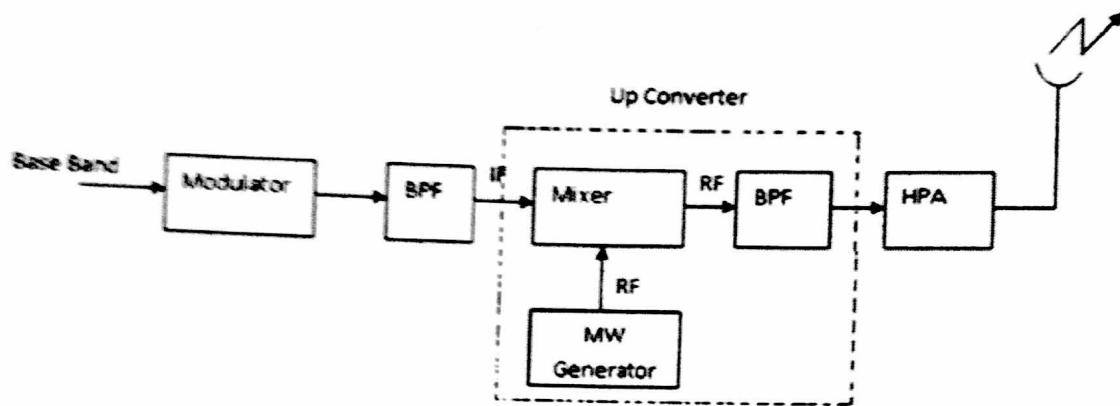


Fig. 6.8 Block diagram of Earth Station Transmitter

Satellite Transponder: (The transponder is an RF-to RF repeater)

A Satellite Transponder consists of

- ♦ a Band Pass Filter(BPF)
- ♦ An input low noise amplifier(LNA)
- ♦ A frequency translator
- ♦ and a high power amplifier

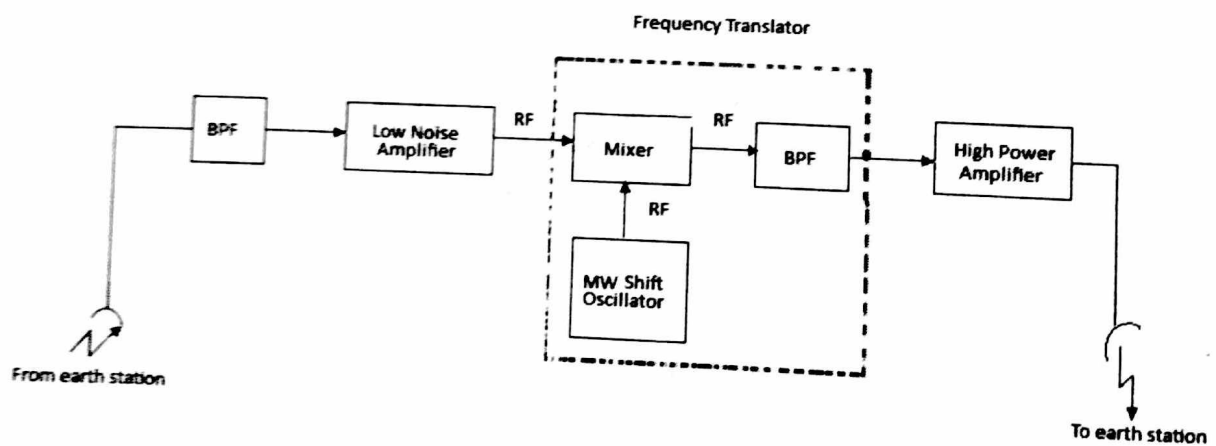


Fig.6.9 shows the block diagram of the satellite transponder

downlink Model

A typical earth station receiver (as shown in Fig. 6.10) consists of

- ♦ an input BPF
it restricts the input noise power to the LNA
- ♦ LNA and an RF-to-IF down converter
it is a mixer-BPF-combination which converts the received RF signal to an IF frequency.

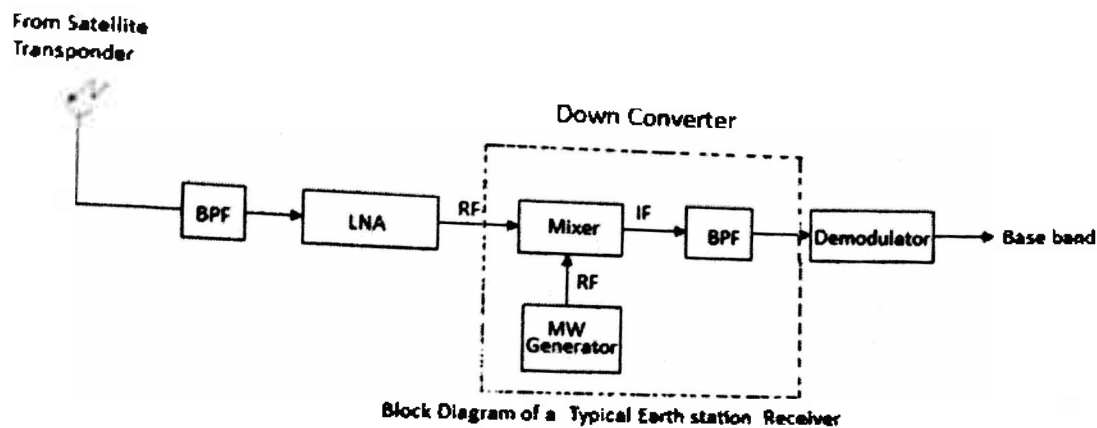


Fig. 6.10 Block Diagram of a typical earth station receiver

Communication Satellites

INTELSAT (USA based International Telecommunication Satellite Organization which provide services to 119 member countries)

INSAT (Indian Satellites which provide services to Indian region)

Advantages

It provides Point to multipoint communication

Offers telecommunication links which includes telephone, TV, telegraphy, telex, FAX, video conferencing, video text, digital transmission services, etc.,

Global Positioning Systems (GPS)

Disadvantages:

There is a large time-delay of 250 millisecs between the transmission and reception of a signal.

The malfunctions in the satellite are highly difficult to correct.

The initial cost involved is quite large.

6.10 OPTICAL TRANSMISSION [JAN-2015/17]

The principle behind new communication systems are

- (i) To improve transmission fidelity
- (ii) To increase the data rate
- (iii) To increase the transmission distance between relay stations.

Optical frequencies lie in the range 10^{14} to 10^{15} Hz. A fiber can carry approximately 10 million TV channels.

In optical fiber communication shown in Fig 6.11, electromagnetic waves in the optical frequency region are used as the carrier.

The message to be transmitted is converted into a suitable electrical form by the electrical transmit section.

This electrical signal is allowed to modulate the light output from the optical source, which may be either a Light Emitting Diode (LED) or an Injection Laser Diode (ILD).

The modulated light is launched into the optical fiber which is the communication channel linking the transmitter with the receiver.

At the receiving end, the input optical signal is converted into suitable electrical variations by the optical detector, which may be either a PIN photodiode or Avalanche photodiode.

These electrical variations are converted to the original message form in the electrical receive section and given to the destination.

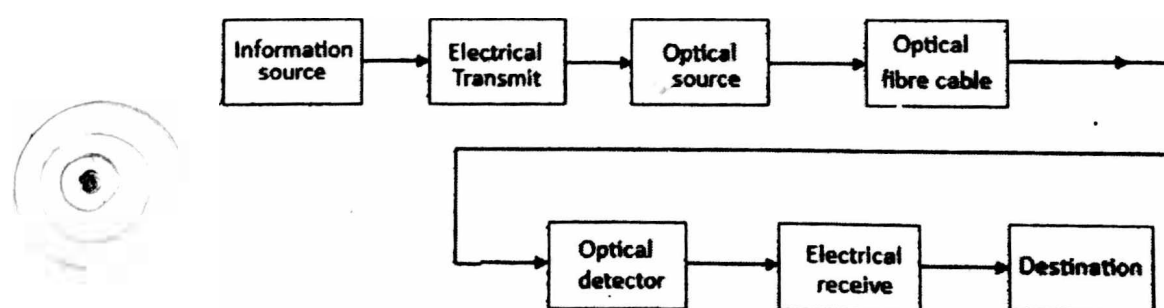


Fig. 6.11 Schematic block Diagram of Optical Fibre Communication System

6.10.1 Optical Fibre

An optical fibre is a piece of very thin (hair-thin), highly pure glass, with an outside cladding of glass that is similar, but because of a slightly different chemical composition, has a different refractive index as shown in Fig 6.12.

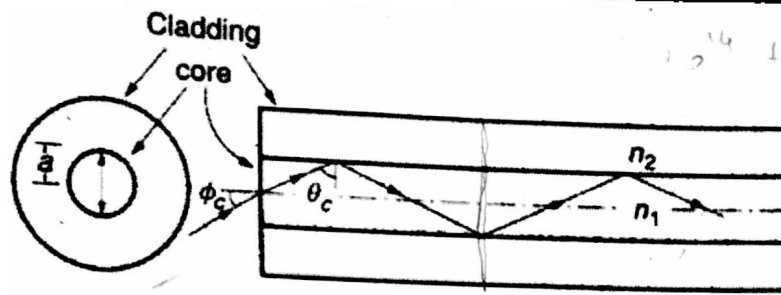


Fig. 6.12 Optical Fiber

The simplest optical fiber consists of a central cylindrical core of constant refractive index n_1 and a concentric cladding surrounding the core of slightly lower refractive index n_2 .

An optic fiber cable is quite similar in appearance to the coaxial cable system. This type of fibre is called step index fiber, whose core diameter is in the range of 2 to 200 μm , as the refractive index makes a step change at the core-cladding interface.

In Graded index fiber, the refractive index gradually reduces from the centre to the outside of the fiber cross-section. These were initially easier to manufacture.

Total internal reflection

If a light ray travelling in the core of higher refractive index is incident at the core-cladding interface with an angle of incidence, with respect to the normal, greater than the critical angle, it will be reflected back into the originating dielectric medium, i.e. core, with high efficiency (around 99%). This phenomenon is known as total internal reflection.

6.10.2 Advantages of Optical fiber Communication

- ♦ Enormous potential bandwidth
- ♦ Small size and weight
- ♦ Free from Electromagnetic Interference
- ♦ Provides high degree of signal security and cross talk between parallel fibers is avoided.
- ♦ Highly reliable and easy to maintain.
- ♦ Flexible, compact and extremely rugged.
- ♦ Low cost line communication.
- ♦ High tolerance to temperature

6.10.3 Disadvantages of Optical fiber Communication

- ♦ Connections and taps are more difficult to make than copper wire
- ♦ Fiber is not as flexible as a copper wire.

6.10.4 Applications of Optical fiber Communication

- ♦ Computers
- ♦ LAN
- ♦ Industrial Electronics
- ♦ Telecommunications

6.11 SWITCHING TECHNIQUES [JAN - 2018]

6.11.1 Circuit-Switching

Communication using circuit switching implies that there is a dedicated communication path between the two stations. The path is connected through a sequence of links between network nodes. On each physical link, a logical channel is dedicated to the connection. Circuit switching is the commonly used technique in telephony, where the caller sends a special message with the address of the called (i.e. by dialing a number) to state its destination. It involves the following three distinct steps, as shown in the Fig. 6.13

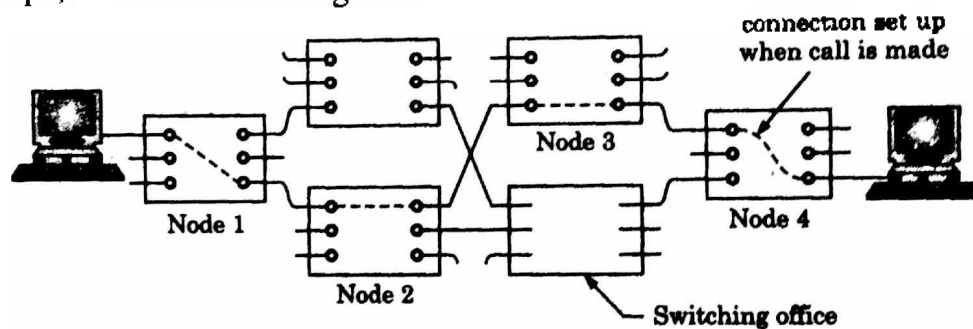


Fig. 6.13 Circuit switching

Circuit Establishment : To establish an end-to-end connection before any transfer of data. Some segments of the circuit may be a dedicated link, while some other segments may be shared.

Data transfer

- ♦ Transfer data is from the source to the destination.
- ♦ The data may be analog or digital, depending on the nature of the network
- ♦ The connection is generally full-duplex

Circuit disconnect

- ♦ Terminate connection at the end of data transfer
- ♦ Signals must be propagated to de-allocate the dedicated resources

Advantages of circuit switching

- ♦ Fixed bandwidth, guaranteed capacity
- ♦ Low variance end to end delay

Disadvantages of circuit switching

- ♦ Connection setup and tear down introduces extra overhead
- ♦ Circuit switching is inflexibility in dealing with computer oriented data
- ♦ A circuit uses a fixed amount of bandwidth, regardless of whether it is used or not
- ♦ The circuit switching is that the network is only capable of supporting a limited number of simultaneous circuits. When this limit is reached, the network blocks further attempts for connection until some of the existing circuits are reused.

6.11.2 Packet Switching Networks

Packet switching is often used in computer networks where individual users need the channel at frequent intervals. In Packet switching, messages are broken into short blocks and interleaved with other messages. Thus, user queue for the channel and share it with one another efficiently. Data is sent in individual packets. Each packet is forwarded from switch to switch, eventually reaching its destination. Each switching node has a small amount of buffer space to temporarily hold packets. If the outgoing line is busy, the packet stay in queue until the line becomes available. Packet switching handles bursty traffic cell.

*No physical link b/w
source and destination
Data is virtual*

Two variations of packet switching exists

1. Virtual circuit and
2. datagram

6.11.3 Virtual Circuit Packet Switching

The virtual circuit method (also known as connection-oriented) is closer to circuit switching. Here a complete route is worked out prior to sending data packets. The route is established by sending a connection request packet along the route to the intended destination. This packet informs the intermediate nodes about the connection and the established route so that they will know how to route subsequent packets. The result is a circuit somewhat similar to those in circuit switching, except that it uses packets as its basic unit of communication. Hence it is called a virtual circuit.

Each packet carries a virtual circuit identifier or header which enables a node to determine to which virtual circuit it belongs and hence how it should be handled. (The virtual circuit identifier is essential because multiple virtual circuits may pass through the same node at the same time.)

Because the route is fixed for the duration of the call, the nodes spend no effort in determining how to route packets.

Figure 6.14 illustrates the virtual circuit method using the switch. When the two hosts initiate a connection, the network layer establishes a virtual circuit (denotes by shaded switches) which is maintained for the duration of the connection. When the hosts disconnect, the network layer releases the circuit. The packets in transit are displayed as dark boxes within the buffers. These packets travel only along the designated virtual circuit.

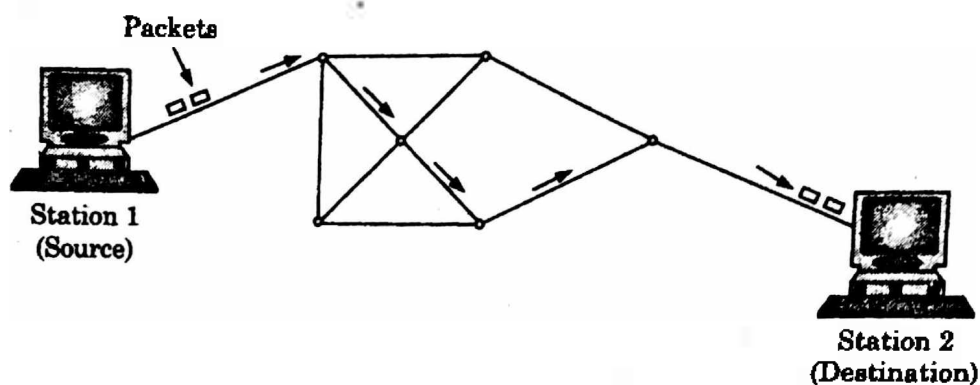


Fig. 6.14 Virtual circuit packet switching

6.11.4 Datagram Packet Switching

The datagram method (also known as connectionless) does not rely on a preestablished route, instead each packet is treated independently. Therefore, it is possible for different packets to travel along different routes in the network to reach the same

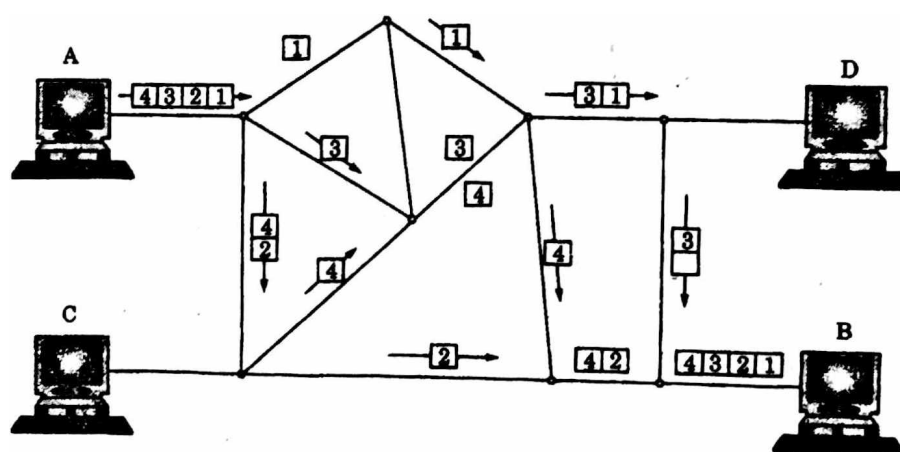


Fig. 6.15 Datagram packet switching

final destination. As a result, packets may arrive out of order, or even never arrive (due to node failure). It is up to the network user to deal with lost packets, and to rearrange packets to their original order. Because of the absence of a pre-established circuit, each packet must carry enough information in its header to enable the nodes to route it correctly. Figure illustrates the datagram method, station A send packet to station B. Note how the packets exercise different routes.

Advantages of packet switching

- ◆ Uses resources more efficiently
- ◆ Very little setup or tear down time
- ◆ It is more flexible

Disadvantages of packet switching

- ◆ No guarantee in delay
- ◆ Algorithms are more complicated
- ◆ Difficult to bill customers
- ◆ If a node crashes momentarily, all of its queued packets are lost

Comparison between Virtual and Datagram

Sl.No.	Virtual Circuit (VC)	Datagram
1.	Circuit setup is required	Circuit setup is not required
2.	Each packet contains a short VC number as address	Each packet contains the full source and destination address
3.	Route chosen when VC setup and all packets follow this rout	Each packet is routed independently
4.	In case of router failure all VC that passed through the router are terminated	Only crashed packets lost
5.	Congestion control is easy using buffers	Difficult to control congestfon

6.12 OVERVIEW OF ISDN

[APRIL - 2015, 17, JAN - 2018, 16]

ISDN - Integrated Services Digital Network – “A network that provides end to end digital connectivity, used for supporting a wide range of services which includes

- ♦ Telephony (voice and music)
- ♦ Data (telemetry, E-mail and alarm)
- ♦ Text (telex, teletex and videotext)
- ♦ Image (Facsimile, TV conferencing, video phone)

It also provides supplementary services like

- ♦ Direct Dialing In
- ♦ Call Wait and
- ♦ Call Hold etc.,

The evolution of ISDN is based on two important technological developments

- (i) Digital transmission
- (ii) Digital switching

6.12.1 Architecture of ISDN

It consists of

- 1) Common physical interface
- 2) ISDN central office
- 3) Digital Subscriber Loop
- 4) ISDN Channels and
- 5) Integrated digital network

Fig 6.13 illustrates the overall concept of an ISDN

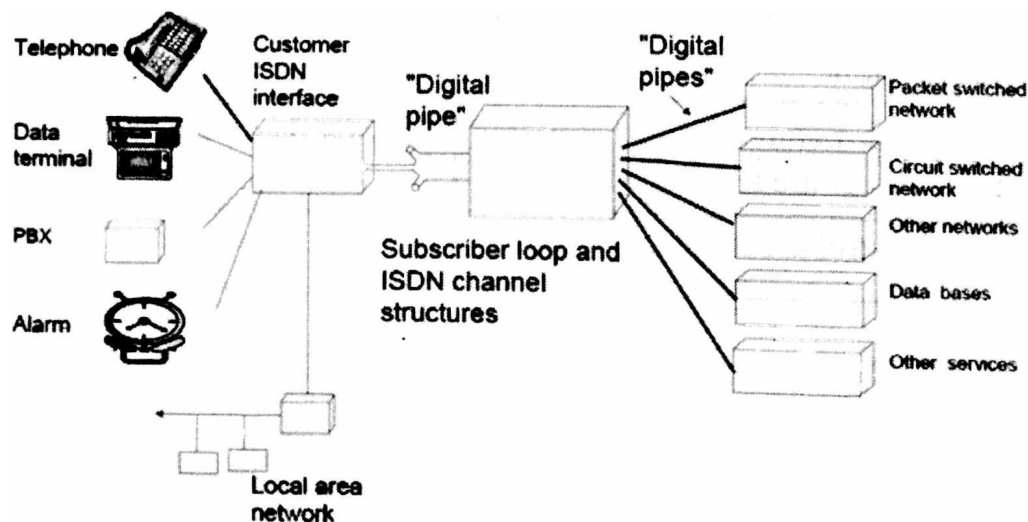


Fig.6.16 Conceptual view of ISDN

Common physical interface

In an ISDN, all devices such as Digital telephone, Alarm, Computer terminal, Videotext, facsimile, PBX and even a LAN can be connected to the transmission line using the common physical interface.

It essentially provides DTE-DCE connection, where DTE is Data Terminal Equipment and DCE is Data Communication Equipment.

ISDN central office

It connects the numerous ISDN subscriber loop signals to the ISDN.

It provides subscribers the access to circuit switched networks, packet switched networks, databases and other services.

Digital Subscriber Loop (DSL) and ISDN Channels

The DSL is the connection between common physical interface and the ISDN central office.

In ISDN, this is one or two twisted pairs of copper cable or a fiber optic link that provides full duplex digital transmission.

It has two different ISDN channel structures

1. Basic channel structure
2. Primary channel structure

Both the channels are constructed from the following channels

1. B-channel (64kbps)
2. D-channel (16 or 64 kbps)
3. H-channel (384 (or) 1536 and (or) 1920 kbps)

6.12.2 Advantages of an ISDN line over a standard telephone line

Some of the advantages of an ISDN line over a standard telephone line are:

- ◆ Digital service – should have less errors
- ◆ Connection is fast – direct connection with no dialing
- ◆ Higher bandwidth – less time downloading material
- ◆ Able to use ISDN for more than one task e.g. surf and telephone
- ◆ Permits use of newer technologies e.g. video phone
- ◆ Streaming video or audio without disruption

- ♦ Video conferencing - multiple recipients
- ♦ Multiple users
- ♦ Because it's a digital service, ISDN offers excellent line quality compared to a rather noisy analogue line.
- ♦ ISDN connections are made in typically less than a second - unlike modems and analogue lines which require 30 to 60 seconds before any data can be transmitted.
- ♦ Voice, data and video can all be sent over a single ISDN line

6.12.3 Limitations of ISDN

- ♦ ISDN is more expensive to install than a standard telephone.
- ♦ ISDN is more expensive to rent per month than a standard telephone line.
- ♦ An engineer has to fit the system in your home.
- ♦ Not all exchanges have ISDN enabled.
- ♦ If all you require is a telephone service, then the standard analogue service is fine.

It costs money to convert your telephone line back should you decide to revert to an analogue service.

6.12.4 Applications using ISDN Smart Networks

Based on these kinds of networks, we can achieve different services for different customers for minimum investment:

- ♦ Wide Area Network (WAN)
- ♦ Private network interconnection
- ♦ Telecommunication (work at home)
- ♦ Videotext
- ♦ Call waiting - automatic call in case of no answer
- ♦ Caller's identity by different ring tone
- ♦ Personal Mailbox
- ♦ Electronic directory and PC to PC screen sharing

6.13 NETWORK MODELS

[SAN - 2015, 17, 18]

Computer networks are created by different entities. Standards are needed so that these heterogeneous networks (different networks) can communicate with one another. The two best-known standards are the OSI model and the Internet model (discussed later). The OSI (Open Systems Interconnection) model defines a seven-layer network; the Internet model defines a five-layer network.

Computer network can be categorized as

1. Local-Area Networks (LAN)
2. Metropolitan Area Network (MAN) and
3. Wide-Area Networks (WAN)

6.13.1 Local Area Network

A local area network (LAN) is usually privately owned and links the devices in a single office, building, or campus as shown in the Fig. 6.17. Depending on the needs of an organization and the type of technology used, a LAN can be as simple as two PCs and a printer in someone's home office; or it can extend throughout a company and include audio and video peripherals. LAN size is limited to a few kilometers.

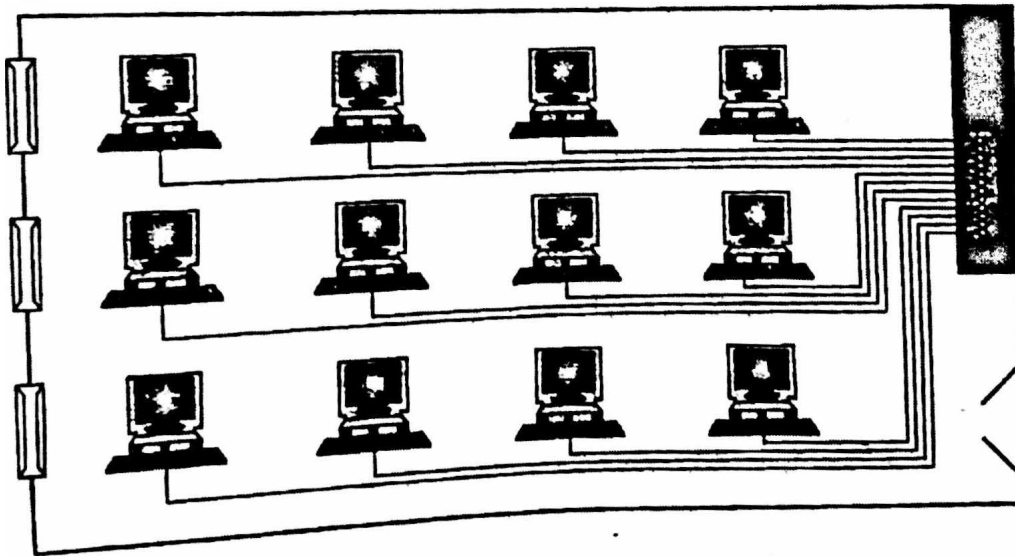


Fig. 6.17 Local Area Network

LANs are designed to allow resources to be shared between personal computers or workstations. The resources to be shared can include hardware (e.g., a printer), software (e.g., an application program), or data. One of the computers may be given a large capacity disk drive and may become a server to clients. Software can be stored on this central server and used as needed by the whole group.

Major Characteristics of LAN

- ♦ Every computer has the potential to communicate with any other computers of the network.
- ♦ High degree of interconnection between computers.
- ♦ Easy physical connection of computers in a network.
- ♦ Inexpensive medium of data transmission.
- ♦ High data transmission rate.

Advantages

- ♦ The reliability of network is high because the failure of one computer in the network does not effect the functioning for other computers.
- ♦ Addition of new computer to network is easy.
- ♦ High rate of data transmission is possible.
- ♦ Peripheral devices like magnetic disk and printer can be shared by other computers.

Disadvantages

If the communication line fails, the entire network system breaks down.

6.13.2 Metropolitan Area Network

MAN is designed to extend over the entire city. It may be a single network as a cable TV network. Or it may be means of connecting a number of LANs into a larger network so that resources may be shared as shown in the Fig. For example, a company can use a MAN to connect the LANs in all its offices in a city. MAN is wholly owned and operated by a private company or may be a service provided by a public company. Metropolitan area networks can span upto 50 km, devices used are modem and wire/cable.

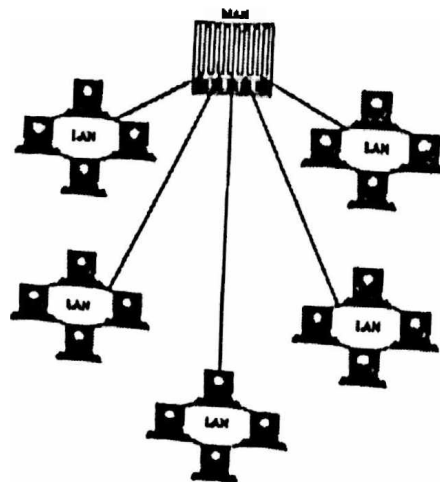


Fig. 6.18 Metropolitan Area Networks (MAN)

Major Characteristics of MAN

- ♦ MAN's cover much greater distance than LAN's do.
- ♦ Most MAN's can recover very quickly from a link or switch/router failure.
- ♦ Some MAN topologies are based on a ring. Data is passed around in a ring fashion and also the network routes and switches are interconnected in a ring fashion.
- ♦ Have the ability of a users to dynamically allocate more Bandwidth on demand.

6.13.3 Wide Area Network

A wide area network (WAN) provides long-distance transmission of data, images, audio, and video information over large geographic area that may comprise a country, a continent, or even the whole world. A WAN can be as complex as the backbones that connect the Internet or as simple as a dial-up line that connects a home computer to the Internet as shown in the Fig.

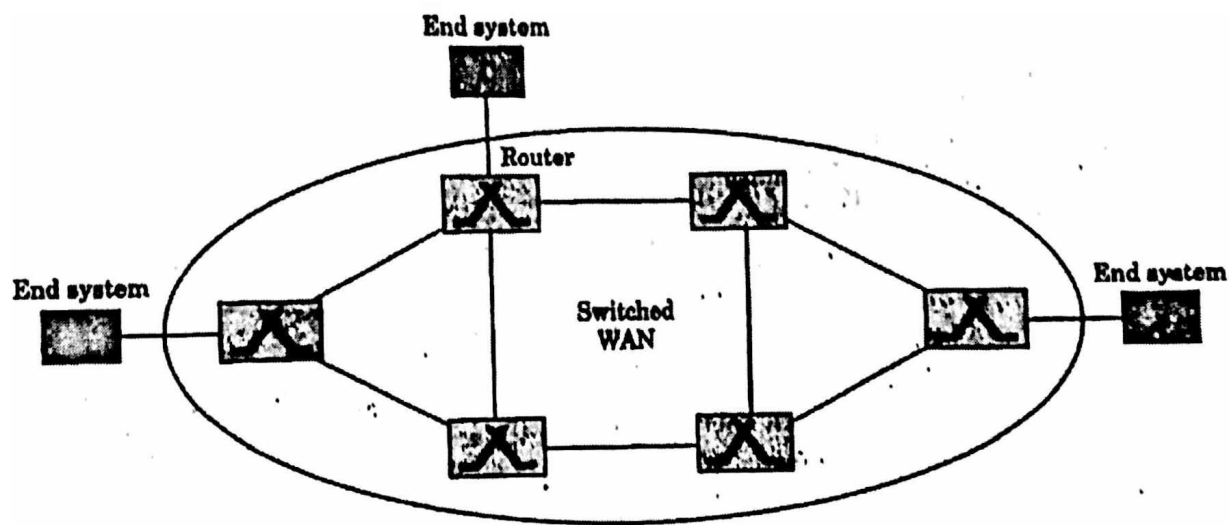


Fig. 6.18 WANs a switched WAN and a point-to-point WAN

The switched WAN connects the end systems, which usually comprise a router (internet-working connecting device) that connects to another LAN or WAN. The point-to-point WAN is normally a line leased from a telephone or cable TV provider that connects a home computer or a small LAN to an Internet service provider (ISP). This type of WAN is often used to provide Internet access.

Characteristics of WAN

Following are the major characteristics of WAN

1. Communication Facility : For a big company spanning over different parts of the country the employees can save long distance phone calls and it overcomes the time lag in overseas communications. Computer conferencing is another use of WAN where users communicate with each other through their computer system.

2. Remote Data Entry : Remote data entry is possible in WAN. It means that sitting at any location you can enter data, update data and query other information of any computer attached to the WAN but located in other cities. For example, suppose you are sitting at Chennai and want to see some data of a computer located at Delhi, you can do it through WAN.

3. Centralised Information : In modern computerised environment you will find that big organisations go for centralised data storage. This means if the organisation is spread over many cities, they keep their important business data in a single place. As the data are generated at different sites, WAN permits collection of this data from different sites and saves at a single site.

Difference between LAN and WAN

Sl.No.	LAN	WAN
1.	It is restricted to limited geographical area of few kilometers	WAN covers great distance and operate nationwide or even worldwide
2.	In this configuration, the computer terminals and peripheral devices are connected with wires and coaxial cables	In WAN there is not physical connection. Communication is done through telephone lines and satellite links
3.	Cost of data transmission in LAN is less because the transmission medium is owned by a single organisation	In case of WAN the cost of data transmission is very high because the transmission medium used are hired, either telephone lines or satellite links
4.	The speed of data transmission is higher than WAN. The transmission speed in LAN varies from 0.1 to 100 megabits per second	The speed of data transmission is less. In case of WAN the speed ranges from 1800 to 9600 bits per second (bps).
5.	Few data transmission errors occur in LAN, It is because in LAN the distance covered is negligible	Higher data transmission errors occur compared to LAN

6.14 MOBILE COMMUNICATION SYSTEM

The Global System for Mobile Communication (GSM) network is a cellular telecommunication network with a versatile architecture complying with the ETSI (European Telecommunication Standard Institute) GSM 900/GSM1800 standard. Siemens implementation is the digital cellular mobile communication system D900/1800/1900 that uses the very latest technology to meet every requirement of the standard."

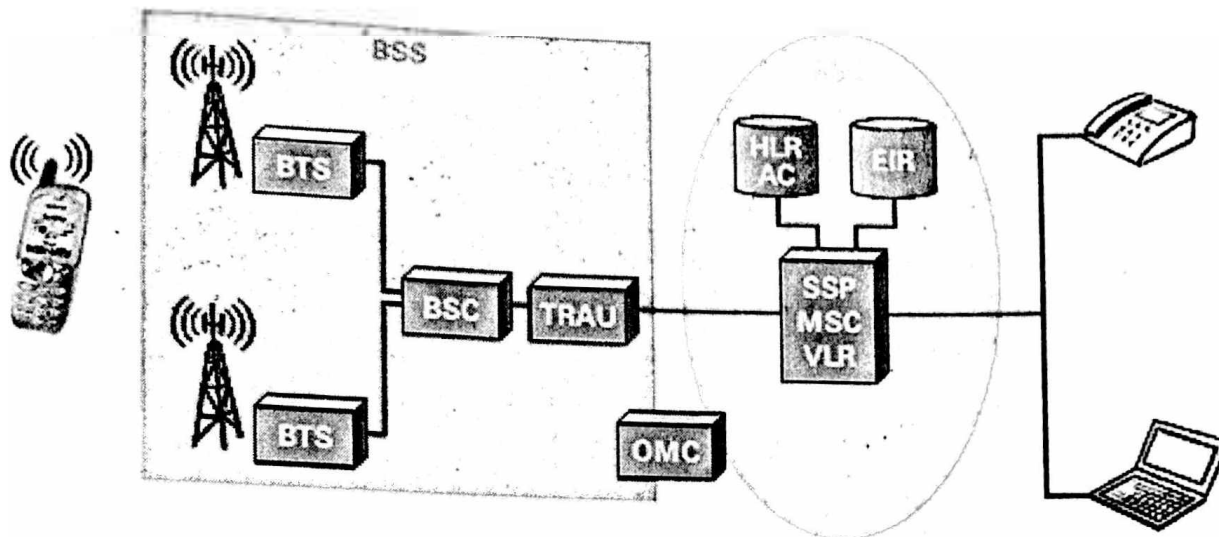


Fig. 6.19 Mobile Communication System

“The Base Station Subsystem consists of the Base Transceiver Station (BTS), the Base Station Controller (BSC), the Transcoding and Rate Adaptation Unit (TRAU) and the Operation and Maintenance Center (OMC-B).”

TRAU: Transcoding and Rate Adaptation Unit. The TRAU adapts the transmission speed of the digital voice and data traffic on the radio path to the standard speed of 64 kbit/s into the network. Complete redundancy of all boards ensures highest reliability. Furthermore, no hardware upgrades are necessary when features such as new speech codices are introduced.”

The TRAU, also known as a Trans Coding Unit or TCU, is a compressor and converter. It first compresses traffic coming from the mobiles through the base station controllers. That’s quite an achievement because voice and data have already been compressed by the voice coders in the handset. Anyway, it crunches that data down even further. It then puts the traffic into a format the Mobile Switch can understand. This is the transcoding part of its name, where code in one format is converted to another. The TRAU is not required but apparently it saves quite a bit of money to install one. Here’s how Nortel Networks sells their unit:

Reduce transmission resources and realize up to 75% transmission cost savings with the TCU.

The Trans Coding Unit (TCU), inserted between the BSC and MSC, enables speech compression and data rate adaptation within the radio cellular network. The TCU is designed to reduce transmission costs by minimizing transmission resources between the BSC and MSC. This is achieved by reducing the number of PCM links going to the BSC, since four traffic channels (data or speech) can be handled by one PCM time slot. Additionally, the modular architecture of the TCU supports all three GSM vocoders (Full Rate, Enhanced Full Rate, and Half Rate) in the same cabinet, providing you with a complete range of deployment options.

The voice coders or vocoders are built into the handsets a cellular carrier distributes. The carrier specifies which rate they want traffic compressed, either a great deal or just a little. The cellular system is designed this way, with handset vocoders working in league with the equipment of the base station subsystem. To read more about PCM [click here](#).

“SSS: Switching sub-system.”

“SSP: Service Switching Point. The SSP comprises the switch related Intelligent Network functions (IN). Together with the IN SCP operator services like Prepaid, VPN and Personal Number Service are implemented.”

REVIEW QUESTIONS

1. Explain the communication model in detail.
2. What is need for modulation.
3. Compare Analog and digital communication system.
4. Compare LAN and MAN
5. Explain mobile communication system.
6. What is meant by wireless communication system and explain microwave communication system.

PART - A

TWO MARKS

QUESTIONS AND ANSWERS



CHAPTER - 4 : ELECTRONIC CIRCUITS

1. **What is semiconductor ?**

The materials, whose electrical properties lie between those of conductors and insulators are known as semiconductor.

2. **What are the classification of energy band.**

Energy band is classified into three types.

(i) Conduction band

(ii) Forbidden band

(iii) Valence band

3. **What is energy band**

The range of energies possessed by an electron in a solid is known as energy band.

4. **What are the classification of semiconductors ?**

The semiconductor are classified into two types.

(i) Intrinsic semiconductors

(ii) Extrinsic semiconductors

5. **What is doping ?**

The process of adding impurity atoms to the intrinsic semiconductor is called doping.

6. **What are the classification of extrinsic semiconductor ?**

Extrinsic semiconductors are classified into two types.

(i) N – type semiconductor

(ii) P – type semiconductor

6. **What is N – type semiconductor ?**

A small amount of pentavalent impurity is added to a pure semiconductor is known as N – type semiconductor.

7. **What is P – type semiconductor ?**

A small amount of trivalent impurities is added to a pure semiconductor is known as P – Type semiconductor.

8. **Define drift current.**

When an electric field is applied across the semiconductor material, the charge carriers attain a certain drift velocity v_d , which is equal to the product of the mobility of the charge carriers and the applied electric field intensity E . The holes move towards the negative terminal of the battery and the electrons move towards the positive terminal of the battery. This combined effect of the movement of the charge carries constitutes a current known as drift current.

9. Define diffusion current.

An electric current can flow in a semiconductor even in the absence of the applied voltage provided a concentration gradient exists in the material. A concentration gradient is said to exist if the number of either electrons or holes is greater in one region of the semiconductor as compared to the rest of the region. In a semiconductor material, the charge carriers have the tendency to move from the region of higher concentration to that of lower concentration of the same type of charges carriers. This movement of charge carriers takes place resulting in a current called diffusion current.

10. Define diffusion length.

The average distance that a charge carrier can diffuse during its lifetime is called diffusion length L .

$$L = \sqrt{D\tau} \quad \text{where, } D = \gamma, \quad I = \gamma$$

11. Define P – N Junction.

When a P – Type semiconductor is joined to a N – type semiconductor the contact surface is called PN junction or PN diode.

12. What is static resistance ?

The static resistance of a diode is the ratio of voltage to current. It is denoted by 'R'. At any point on the current – voltage characteristics of the diode, static resistance R equals the reciprocal of the slope of the line joining the operating point to the origin. The static resistance R varies widely as the operating point shifts and it does not constitute a useful parameter.

13. What is dynamic resistance ?

The dynamic resistance of a diode is defined as the reciprocal of the slope of the current – voltage characteristics. It is given by

$$r = \frac{\text{change in voltage}}{\text{Resulting change in current}} = r = \frac{\Delta V}{\Delta I}$$

14. What is reverse resistance ?

The reverse resistance offered by PN Junction diode under reverse bias condition. It is very large compared to the forward resistance. Reverse resistance is in the range of $M\Omega$

15. What are the breakdown in PN Junction diode ?

The diode breakdown may occur in two methods

- i) Zener breakdown
- ii) Avalanche Breakdown

16. What is Zener breakdown ?

The breakdown using heavily doped P and N regions direct rupture of covalent bonds takes place. Because of the strong electric field at the junction of PN diode, the new electron – hole pairs so created increase the reverse current in a reverse biased PN diode.

17. What are PN diode application ?

- (i) Rectifiers in d.c. power supplier.
- (ii) Clamping circuits used as d.c. restorer in TV receivers and voltage multiplier
- (iii) PN Diodes in demodulation (detector) circuits.
- (iv) Switch in digital logic circuits used in computers.
- (v) Clipping circuits used as wave shaping circuits used in TV receivers, radio, computers and radars.

18. What is zener diode ?

A zener diode is a special purpose diode that is operated in reverse-biased conditions. Its operation depends on the zener breakdown phenomenon

19. What are the break down mechanism in zener diodes ?

The two kinds of breakdown that occur in zener diode are

- (i) Avalanche breakdown
- ii) Zener breakdown

20. What are the applications zener diodes ?

- (i) voltage regulator
- ii) fixing reference voltages in electronics circuits such as power supplies and transistor biasing
- (iii) clippers in wave shaping circuits
- (iv) square wave generation.

21. What is transistor ?

Transistor consist of two junctions formed by sandwiching either P-type or N-type semiconductor between a pair of opposite types

22. What are the advantages of transistor over the vaccum tube ?

Transistor has the following advantage than the vaccum tube

- (i) smaller in size
- (ii) no filament and no need if power for heating filament
- (iii) low operating voltage
- (iv) higher efficiency

23. **What are the types of transistor ?**

1. Unipolar junction transistor (UJT)
2. Bipolar junction transistor (BJT)

24. **Describe the basic structure of the BJT.**

Transistor is a solid state and current controlled device. A transistor consists of layer of p-type material sandwiches between two layer of N-type material. This forms an NPN-transistor. Similarly, if an N-type material layer is sandwiched between two layers of p-type material, it forms a PNP transistor. There are three terminal for a transistor emitter, base and controller – base junction.

25. **What are the three terminals in BJT?**

1. Emitter (E)
2. Collector (C)
3. Base (B)

26. **What is biasing ?**

Applying external voltage to a transistor is called biasing. In order to operate transistor properly as an amplifier, it is necessary to correctly bias the two pn junctions with external voltages

27. **What are the types of transistor configuration**

There are three types of configurations for the operation of a transistor

- (i) Common base (CB) configuration
- (ii) Common emitter (CE) configuration
- (iii) Common collector (CC) configuration

28. **Define input characteristics for CB configurations.**

The curve drawn between emitter current (I_E) and Emitter – Base voltage (V_{EB}) for a given value of collector – Base (V_{CB}) is known as input characteristics.

29. **Write the current amplification factor for a CE transistor.**

The current amplification factor for a CE transistor is given by,

$$\beta = \frac{\text{change in collector current}}{\text{change in basecurrent at instant } V_{CE}}$$

$$\gamma = \frac{\Delta I_E}{\Delta I_B} \text{ at constant } V_{CE}$$

30. **Write the relationship between α and β**

$$\alpha = \frac{\beta}{1 + \beta} \quad \text{or} \quad \beta = \frac{\alpha}{1 - \alpha}$$

31. ***What are two basic types of a transducer ?***
 1. Active transducer
 2. Passive transducer
32. ***What is an active transducer ?***
 The active or self generating transducer converts energy directly from one state to another without the need for an external power sources or excitation. For example thermocouple.
33. ***What is a passive transducer ?***
 A passive transducer does not convert energy directly but derives the power required for energy conversion from an external source. For example, potentiometer
34. ***Give four examples for active transducers.***
 1. Thermocouple
 2. Moving coil generator
 3. Photovoltaic cells
 4. Piezo-electric pick up
35. ***Give four example for passive transducers.***
 1. Potentiometer
 2. Strain gauge
 3. Photoconductive cell
 4. Thermistor
36. ***What is an LVDT ?***
 Linear variable differential transformer is a magnetic displacement transducer The magnitude of voltage induced in secondary would be the indicative of the amount of displacement that has taken place.
37. ***What is Hall effect ?***
 When a thin piece of semiconductor is placed in a magnetic field and a current is passed through it such that an electric field is at right angles to the magnetic field, then a voltage will appear across the remaining traverse face of the material.
38. ***What is a strain gauge ?***
 It is an example of a resistance changing passive transducer used for measuring strain. When the gauge stretched or compressed the resistance of the wire in the strain gauge changes.

CHAPTER - 6 : COMMUNICATION & COMPUTER SYSTEMS

1. What is communication?

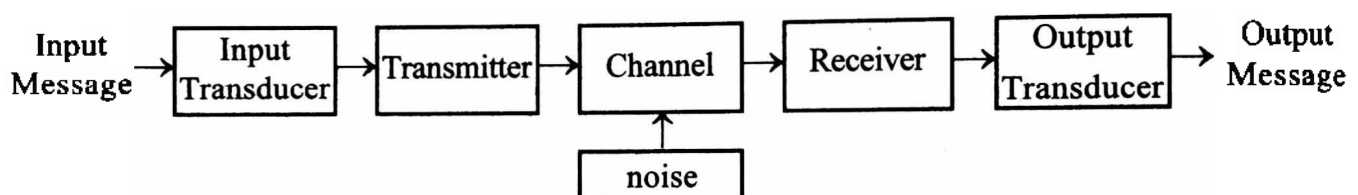
- **Communication** is the transfer of information from one place to another place.
- Distribution of data, message or information from one location to another location with high reliability and security is the role of communication system.

2. What are the elements of Communication system?

Communication System consists of the following components which acts together to accomplish information transfer or exchange.

1. Input Transducer
2. Transmitter
3. Channel
4. Receiver and
5. Output Transducer

3. Draw the basic block diagram of the communication system.



What are the basic types of communication?

There are two types

1. Analog communication system and

An **analog** message is a physical quantity that varies with time, usually in a smooth and continuous fashion. Since the information resides in a time-varying waveform, an analog communication system should deliver this waveform with a specified degree of **reliability** or **fidelity**.

2. Digital communication system

A **digital** message is an ordered sequence of symbols selected from a finite set of discrete elements. Since the information resides in discrete symbols, a digital communication system should deliver these symbols with a specified degree of **accuracy** in a specified amount of time.

5. What is the need for modulation?

1. Ease of radiation as the signals are translated to high frequencies, it becomes relatively easier to design amplifier circuit as well as antenna system at these increased frequencies.
2. To overcome hardware limitations.
3. Transmitting such lower frequencies require antennas with miles in wavelength.

Digital Communication System	Analog Communication System
Advantages : 1. Inexpensive digital circuits 2. Privacy preserved (data encryption) 3. Can merge different data (voice, video and data) and transmit over a common digital transmission system 4. Error correction is possible by Coding Disadvantages: 1. Larger bandwidth 2. Synchronization problem is relatively difficult	Advantages : 1. Smaller bandwidth 2. Synchronization problem is relatively easier Disadvantages: 1. Expensive analog components: L&C 2. No privacy 3. Cannot merge data from different Sources 4. No error correction capability

2. Frequency Modulation (FM),

It is the process of by which the frequency of carrier wave is varied in accordance with the instantaneous value of message signal is known as *frequency modulation*.

3. Phase Modulation (PM)

It is the process of by which the phase of carrier wave is varied in accordance with the instantaneous value of message signal is known as *phase modulation*.

9. What are the types of communication system based on nature?

Based on the nature of communication channel, the process of communication may be broadly classified into two types.

1. Wired Communication system
2. Wireless communication system

10. Write the advantage of wireless communication

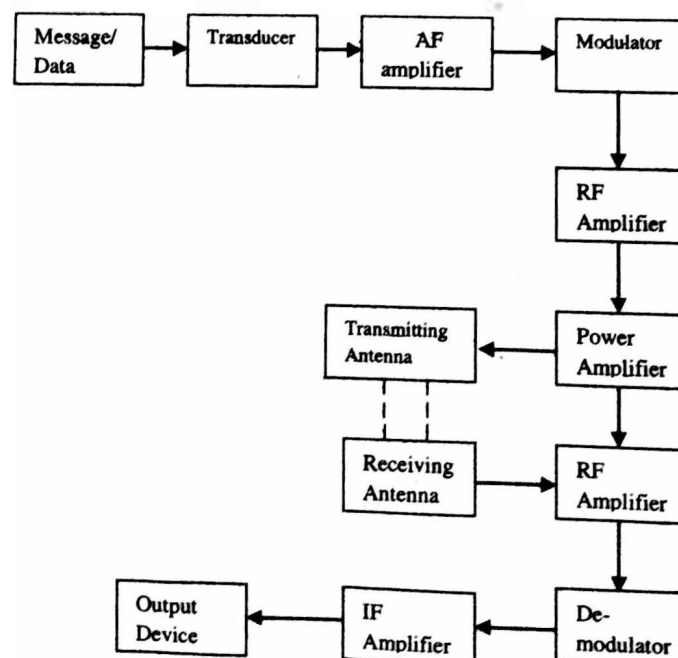
The use of microwave links has major advantages over cabling systems.

1. Freedom from land acquisition rights
2. Ease of Communication over difficult terrain

11. Write the disadvantages of wireless communication

1. Bandwidth allocation is extremely limited
2. Atmospheric effects
3. Transmission path needs to be clear
4. Interference and efficiency of propagation
5. Restrictive Costs
6. The cost of design, implementation and maintenance of microwave links is high.

12. Draw a Block Diagram of a Wireless Communication system



13. **Write a note on microwave communication**

Electromagnetic waves on the frequency range of 1 GHz to 30 GHz are referred to as microwaves. Several signal channels are modulated into micro-wave carrier frequencies and transmitted to the repeater stations spaced 50 to 70 km apart.

14. **What is Line-of-sight?**

Repeater antennas are placed on roads and hilltops so that there will not be any obstruction in the line of sight between the repeater antennas. This system is called '**Line of sight microwave link**'.

15. **Write the applications of microwave communications**

Presently, microwave communications are widely used for telephone networks, in broadcast and television systems and in several other communication applications by services, railways, etc.,

16. **What is uplink and downlink frequency?**

The earth stations typically transmit their signals to satellites on carrier frequencies ranging from 5.92 to 6.43 GHz band called the **uplink frequency**. The satellite transponders convert these signals to a range from 3.7 to 4.2 GHz band. These frequencies are referred to as **downlink frequencies**.

17. **Classify the orbits.**

Based on the area of the coverage, the orbits are classified as

1. Polar orbit

When the satellite rotates in an orbit that takes it over the north and south poles, it is called **polar orbit**.

2. Equatorial Orbit

When the satellite rotates in an orbit above the equator, it is called an **equatorial orbit**

3. Inclined orbit

Any other orbital path is called an inclined orbit.

18. **What are the main blocks of satellite system?**

A satellite system, consists of three basic sections,

- The uplink (transmitting earth station)
- The satellite transponder
- The downlink (receiving earth station)

Typical frequencies for telecommunication services in a satellite system are 6/4 GHz and 14/12 GHz, Where 6 and 14 GHz represent uplink frequencies and 4 and 12 GHz represent downlink frequencies.

19. What is Satellite Transponder?

The transponder is an RF-to RF repeater. A Satellite Transponder consists of

- o a Band Pass Filter(BPF)
- o An input low noise amplifier(LNA)
- o A frequency translator and
- o a high power amplifier

20. Write few Communication Satellites

INTELSAT (USA based International Telecommunication Satellite Organization which provide services to 119 member countries)

INSAT (Indian Satellites which provide services to Indian region)

21. Write the advantages of satellite system

It provides Point to multipoint communication

Offers telecommunication links which includes telephone, TV, telegraphy, telex, FAX, video conferencing, video text, digital transmission services, etc.,

Global Positioning System (GPS)

22. Write the disadvantages of satellite system

There is a large time-delay of 250 millisecs between the transmission and reception of a signal.

The malfunctions in the satellite are highly difficult to correct.

The initial cost involved is quite large.

Advantages of Optical fiber Communication:

Enormous potential bandwidth

Small size less weight

Free from Electromagnetic Interference

Provides high degree of signal security and cross talk between parallel fibers is avoided.

Highly reliable and easy to maintain.

Flexible, compact and extremely rugged.

Low cost line communication.

High tolerance to temperature

24. Disadvantages of Optical fiber Communication:

Connections and taps are more difficult to make than copper wire

Fiber is not as flexible as a copper wire.

25. ***Applications of Optical fiber Communication:***

Computers

LAN

Industrial Electronics

Telecommunications

26. ***What is ISDN?***

ISDN - *Integrated Services Digital Network* – “A network that provides end to end digital connectivity, used for supporting a wide range of services which includes
Telephony (voice and music)

Data (telemetry, E-mail and alarm)

Text (telex, teletext and videotext)

Image (Facsimile, TV conferencing, video phone)

27. ***Applications using ISDN Smart Networks***

Based on these kinds of networks, we can achieve different services for different customers for minimum investment:

Wide Area Network (WAN)

Private network interconnection

Telecommunication (work at home)

Videotext

Call waiting - automatic call in case of no answer

Caller's identity by different ring tone

Personal Mailbox

Electronic directory and PC to PC screen sharing